Can Agentic AI Settle a 20-Year-Old Description Logic Problem?

On the Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$: a Machine-Checked Decidable Fragment and a Map of What Remains Open

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with AI assistants:
Claude Opus 4.6, Opus 4.7, Opus 4.8, Fable 5, Opus 5, Fable 5.2 (Anthropic)
GPT-5.4, GPT-5.4 Pro, GPT-5.5, GPT-5.6 Pro, GPT-5.6 Sol, Astra 6 (OpenAI)

September 2026; revised October and November 2026

Abstract

Whether concept satisfiability in $\mathcal{ALCI}_{\mathrm{RCC5}}$—the description logic $\mathcal{ALCI}$ with the RCC5 base relations as roles under *abstract composition-table* semantics—is decidable has been open since Wessel's 2002/2003 work. This paper reports a sustained, adversarially-reviewed attack on the problem, conducted by AI assistants under human direction and machine-checked in part. We do not settle the question. What we establish is: (i) an unconditionally decidable fragment—the concepts whose negation normal form contains no universal restriction $\forall\mathsf{PO}$—whose decision procedure is *machine-checked in Lean 4*, on raw input and under set semantics, with fragment membership its only hypothesis; the result concerns concept satisfiability only (the fragment is not closed under negation, so it does not by itself decide subsumption or reasoning with general TBoxes), it has survived three independent cold reviews of its formalization, and it is a decidability theorem, not a practical procedure, the search space being doubly exponential; (ii) a machine-certified RCC5 normal form—a strict order for PP, a downward-hereditary disjointness for DR, PO residual—whose set-representation half re-proves in the kernel a representation theorem of Lutz and Wolter; (iii) for the full logic, a machine-checked *conditional* reduction (a complete enumeration of certificates for a fixed, verified checker would decide satisfiability), together with an explicit account of the steps, argued but *not* proved, that lead to its premise from a bounded "live width" of models (property F6)—the open gap of one certificate architecture, and the same horizontal-coordination obstruction that the known grid-style undecidability attempts meet, not a proven reduction of the problem to a single lemma; (iv) a *prototype reasoner*—a cover-tree tableau that reproduces, 23 years on, the GIS concept taxonomy of the original report and agrees with an independent oracle across a 911-case suite, though it is now known to be incomplete (§8); and (v) an honest map of the routes tried, the dead ends, and the missing hard pieces. A later addition, machine-checked but not yet reviewed, relates the full logic to an open problem: a syntactic fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$ has the satisfiability problem of two-variable logic over strict partial orders with unary predicates, whose decidability is open, so a decision procedure for $\mathcal{ALCI}_{\mathrm{RCC5}}$ would decide that problem too (§7.8). The fragment result is also published on its own, as a focused companion paper. We also report, candidly, on the methodology—adversarial "cold review," machine verification, and a human-in-the-loop workflow—and on what frontier AI could and could not do.

Keywords: description logics; $\mathcal{ALCI}$; RCC5; region connection calculus; qualitative spatial reasoning; decidability; concept satisfiability; patchwork property; interactive theorem proving (Lean);

AI-assisted mathematics.

ACM CCS: Theory of computation → Description logics; Automated reasoning; Logic and verification. Computing methodologies → Spatial and physical reasoning.

MSC 2020: 03B70 (logic in computer science); 68T27 (logic in artificial intelligence); 68T30 (knowledge representation); 68V15 (theorem proving).

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1 Introduction

1.1 AI disclaimer and authorship

The mathematical content of this paper was produced by AI assistants (Anthropic's Claude Opus 4.6, 4.7 and 4.8, Claude Fable 5 and 5.2 and Claude Opus 5; and OpenAI's GPT-5.4, GPT-5.4 Pro, GPT-5.5, GPT-5.6 Pro, GPT-5.6 Sol and Astra 6), prompted, steered, and reviewed by Wessel, who posed the problem, supplied the central intuitions and directions, orchestrated the review–repair cycles, and takes responsibility for the framing. **The results are unverified by human domain experts and should be read as AI-generated conjectures supported by machine checking and computational evidence—not as established theorems**, with the explicit exceptions of the parts stated to be kernel-checked in Lean 4 (the soundness pipeline, the abstraction's faithfulness, the RCC5 normal form, and—since 2026-08-29—the decision procedure for the $\forall\mathsf{PO}$-free fragment), which are machine-verified. We are deliberate about this labelling throughout, and it is not uniform: the standing one-line status for the *full logic* is *strongly supported, not certified*—a label for the arguments toward the conditional reduction, not for decidability itself, which is open—whereas the $\forall\mathsf{PO}$-free fragment is *machine-certified*, its formalization's adequacy having been examined by three independent cold reviews (§7.3). What remains at risk there is what remains at risk in any formalization—that a definition misrepresents the intended object—which is why those reviews targeted the definitions, and why the concrete-semantics question was converted into a theorem rather than left to reviewer confidence. We regard the AI systems as having made contributions of a kind that, in a human collaborator, would warrant co-authorship, and we discuss this in §9.

1.2 A motivating example: what the logic does

Before the difficulties, a concrete picture of the object. In $\mathcal{ALCI}_{\mathrm{RCC5}}$ the five RCC5 relations serve as *roles*: a concept may quantify over the regions a given region is part of, overlaps, or is disjoint from. A small GIS vocabulary (from Wessel's original report [9]) shows what this buys. Define

$$\mathsf{CityAtRiver} \equiv \mathsf{City} \sqcap \exists\mathsf{PO}.\mathsf{River}, \qquad \mathsf{GermanCity} \equiv \mathsf{City} \sqcap \forall\mathsf{PP}.(\neg\mathsf{Country} \sqcup \mathsf{Germany})$$

—"a city overlapping a river" and "a city lying in no country but Germany"—and *assert* only a few primitive facts:

$$\mathsf{Hamburg} \sqsubseteq \mathsf{City} \sqcap \exists\mathsf{PO}.\mathsf{Alster},$$
$$\mathsf{Alster} \sqsubseteq \mathsf{River} \sqcap \exists\mathsf{PP}.\mathsf{Germany} \sqcap \forall\mathsf{PO}.\neg\mathsf{Country} \sqcap \forall\mathsf{PP}.(\neg\mathsf{Country} \sqcup \mathsf{Germany}).$$

Nothing here *states* that Hamburg is a German city or a city at a river; a reasoner *infers* both. $\mathsf{CityAtRiver}$ is immediate—Hamburg overlaps the Alster and the Alster is a river, so Hamburg overlaps a river. $\mathsf{GermanCity}$ is subtler and, tellingly for what follows, runs through the *composition table*: for any country c with $\mathsf{Hamburg}\,\mathsf{PP}\,c$, the path $\mathsf{Alster}\,\mathsf{PO}\,\mathsf{Hamburg}\,\mathsf{PP}\,c$ forces $\mathsf{Alster}\,R\,c$ with $R \in \mathsf{comp}(\mathsf{PO},\mathsf{PP}) = \{\mathsf{PO},\mathsf{PP}\}$ (§3); the Alster overlaps no country, which excludes PO, so $\mathsf{Alster}\,\mathsf{PP}\,c$; and by its last conjunct every country the Alster is a proper part of is a $\mathsf{Germany}$, so c is one. Since c was an arbitrary country with $\mathsf{Hamburg}\,\mathsf{PP}\,c$, Hamburg is a German city. ($\mathsf{Germany}$ is a concept, not an individual: the inference is about membership, never identity.)

Two things survive into the rest of the paper. First, this is the *reasoning service*—classification—and it reduces, as all such services do, to one decision problem: *concept satisfiability* ($C \sqsubseteq D$ iff

$C \sqcap \neg D$ is unsatisfiable), whose decidability is exactly what is open. Second, even this benign inference already leans on the RCC5 composition table; the whole difficulty is that the *same* table is, on other inputs, too loose to force what one would need (§1.3). The full taxonomy this example lives in is the one our reasoner recomputes 23 years on (§8, Figures 5–6); how the tableau actually derives the GermanCity inference is spelled out there.

1.3 Why the problem is hard, in one page

RCC5 relates two regions by exactly one of five relations: equal (EQ), proper part (PP), proper superpart (PPI), partial overlap (PO), or discreteness/disjointness (DR). Under *abstract composition-table* semantics a model is not a set of actual regions but a set with a total labelling of ordered pairs by these relations, required only to be converse-closed and *composition-closed*: if $x\,R\,y$ and $y\,S\,z$ then $x\,T\,z$ for some T the composition table allows.

The five relations split into two axes that behave completely differently.

- **The vertical axis** (PP, PPI: nesting) is *tame*. Composition can *force* vertical structure—the only four "determining" compositions (those returning a single relation) all run through PP or PPI—but an unbounded vertical chain is captured by a finite *generator*: a self-loop that *unfolds* into infinitely many distinct points (never an actual PP-cycle, which strong EQ forbids), just as regular structures capture infinite paths in modal logic. Infinite depth generated by vertical demands is not the problem. (One qualification, known since the 2003 report [9]: infinite nesting can also be forced by a concept that mentions neither PP nor PPI, by excluding the other relations where composition leaves a choice; whether a finite generator captures every such case is untested, §7.8.)

- **The horizontal axis** (DR, PO: side-by-side) is *wild*. No *single* composition step pins a horizontal relation to one value (PO is never a singleton table entry)—though as constraints accumulate, their *intersection* sometimes can. What composition still cannot do is force two occurrences *together*; it can only force them *apart* (a shared horizontal anchor excludes EQ). So horizontal "crowds" of pairwise-distinct occurrences are the danger.

A finite decision procedure must record, at each interface of a model, the genuine branching it cannot reconstruct—the *live width*. Many forced edges are *shadows* (implied by composition from edges already present) and cost nothing; live width counts only the rest. The finite-certificate architecture pursued in most of this work turns on one property:

(F6) Is the live width of models bounded by a computable function of the input concept?

If yes—and if a further uniformization step, argued but not proved, goes through (§7.6)—the certificate space is finite and satisfiability is decidable. And here is the knife's edge: the looseness F6 must tame is the *same horizontal-coordination obstruction* that has blocked every known attempt to prove the problem *undecidable*. To force undecidability one would encode a rigid coordinate grid, which needs *clean, reusable coincidence*—a horizontal addressing scheme reused across the grid. Individual horizontal edges *can* be pinned to a single value by the joint closure of several constraints (Remark 2), but no concept is known to make such coincidence *reusable* rather than one-off (the "coincidence obstruction" Wessel identified in 2002/2003). Turn that looseness into a computable bound and one gets decidability by *this* route; circumvent it to reuse coincidence and force a grid and one gets undecidability. The two open directions face the same obstruction from two sides—a *shared* wall, not a theorem that F6's failure would by itself *entail* undecidability (§10)—which is why neither has been settled in over twenty years.

1.4 Scope, contributions, and structure

This is both a technical report and a discussion piece. It is, specifically, an *overview and status map*: it states the results and their verification level but leaves the full proofs to the cited work packages and formal artifacts. Its contributions are:

1. An unconditionally decidable $\forall$PO-*free* fragment, with a decision procedure machine-checked in Lean 4 (§7.2, §7.3), and an RCC5 *normal form* with both directions machine-certified (§7.7). The fragment result is also written up on its own, as a focused paper [76].
2. For the full logic, a machine-checked *conditional* reduction (a complete enumeration of certificates for a fixed, verified checker $\Rightarrow$ decidable), with the unproved steps from property F6 to its premise displayed rather than assumed (§7.6).
3. The identification of F6 as the shared *obstruction* facing both that certificate route and the known grid-style undecidability attempts (§1.3, §10)—a shared obstruction, not a proof that either direction reduces to it.
4. An honest, technique-organized account of every route tried, the counterexamples that killed the wrong ideas, and the lessons (§7), so that others need not repeat them.
5. A candid methodology report on adversarial AI-assisted mathematics (§9).
6. (November 2026; machine-checked, not yet reviewed.) A bridge between a syntactic fragment of the logic and two-variable logic over strict partial orders, which shows that the full logic contains the unary case of an open problem of that literature, and through which two slices are decided by published theorems (§7.8).

Two papers. The most recent cold review of this overview (GPT-5.6 Sol, September 2026) [75] recommended separating the one finished result from the research program around it, and we have followed that recommendation. The $\forall$PO-free fragment, the normal form it rests on, and its formalization are presented as a self-contained paper [76]; this overview remains the status report, the history, the map of failed and open routes, and the methodology.

The paper is organized as follows. §2 states, in largely non-technical terms, where the problem stands. §4 gives the history ($\mathcal{ALCI}_{\mathrm{RCC}}$ from Cohn 1993 through Wessel 2002/2003 and Lutz–Wolter 2006) and carefully distinguishes our object from the neighbouring logics that *are* known undecidable. §5 views the problem as first-order satisfiability and asks whether it falls inside a decidable fragment of first-order logic—it does not, and the deepest reason (totality) previews the central obstruction; conversely, one fragment of the logic *is* a two-variable logic over partial orders, whose decidability is open. §6 explains why the standard undecidability reductions fail. §7—the core of the paper—surveys the approaches *by technique*: naive tableaux and blocking, finite-model and reduction attempts, the split-forest idea, the automata route, the Lean line culminating in F6, the regular-cover pivot, and the bridge to two-variable logic over partial orders. §8 describes the working reasoners and a GIS example. §9 reports on the AI methodology. §10 concludes with what was learned, what would move the needle, an honest estimate of the chances of closing F6, and the project's actual wins.

2 Current status: where the problem stands

Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability remains open (the closely related $\mathcal{ALCI}_{\mathrm{RCC8}}$ likewise, but it is not our focus here). Literature checks in July, September and November 2026 found

no published resolution. Of the closest neighbours, the topological logics of §4 are decided in the other direction, and two-variable logic over a partial order, whose unary case the full logic contains, is itself open (§7.8). What the present work adds is not an answer but one settled fragment, a machine-checked surround, and a map.

The settled result—a decidable fragment. Satisfiability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ concepts whose negation normal form contains no universal restriction $\forall\mathsf{PO}.D$ is decidable, and the `Decidable` instance is **machine-checked in Lean 4** with fragment membership as its only hypothesis—on raw input, where the condition reads "no $\forall\mathsf{PO}$ in positive and no $\exists\mathsf{PO}$ in negative position", and under concrete set semantics (§7.3). Three independent cold reviews of the formalization found no counterexample. Its scope is exactly that: bare concept satisfiability. The fragment is not closed under negation, so a subsumption $C \sqsubseteq D$ is decided only when $C \sqcap \neg D$ stays in the fragment, and internalizing a general TBox introduces $\forall\mathsf{PO}$. The procedure is a decidability theorem, not a practical reasoner.

What is certified (machine-checked, Lean 4, zero `sorry`):

- the decision procedure for the $\forall\mathsf{PO}$-free fragment just described, and a one-sided UNSAT test for the *full* logic (sound, not complete);
- the *RCC5 normal form*, *both directions*: every strong-EQ RCC5 network is an ordered-disjoint structure—a strict partial order for PP with a downward-closed disjointness for DR, PO residual—and every ordered-disjoint structure is an RCC5 frame realized by sets (Theorem 15); that RCC5 structures are representable by regions is a theorem of Lutz and Wolter [10, Thm. 23], and what is new here is the explicit construction and its kernel-checked proof;
- for the full logic, the *soundness* of a finite-certificate pipeline (a valid finite certificate unfolds to a genuine RCC5 model of the concept) and the *faithfulness* of the finite-*type* (Hintikka, possibly infinitary) abstraction;
- a *conditional* reduction: if a fixed, verified finite checker is complete for some computable enumeration of certificate codes, then satisfiability is decidable. A conditional theorem is certified as a conditional; its premise is not;
- (November 2026, not yet reviewed) the *poset bridge*: a syntactic fragment $\mathcal{L}_{\|}$ of $\mathcal{ALCI}_{\mathrm{RCC5}}$ is equisatisfiable, also over finite models, with the tense logic of strict partial orders with an incomparability modality; with it, a satisfiable concept without $\exists\mathsf{PP}$ or $\exists\mathsf{PPI}$ all of whose models are infinite, and the boundary concepts K_2 and C_{cyc} (§7.8).

What is argued but not proved. The certificate architecture of the Lean line needs a premise that has not been established, and the route to it that this work pursued has two unproved steps:

$$\text{F6} \overset{?}{\Longrightarrow} \text{refined W2}' \overset{?}{\Longrightarrow} \text{complete finite candidate lists} \overset{\text{Lean}}{\Longrightarrow} \text{decision procedure.}$$

F6 (Definition 5) is a computable bound on the live width of models; the argument that W2′, once refined, folds into F6 is prose; and the suggestion that unbounded "identity-selector minors" are the *only* way F6 can fail is a candidate characterization with finite-network evidence, not a theorem. F6 is the open gap *of this architecture*: its failure would not by itself close off decidability by some other representation (§10), and its truth is not known to be necessary.

Published results the argument relies on. Lutz and Wolter prove that $\mathcal{ALCI}_{\mathrm{RCC5}}$—their $L_{RCC5}(RS)$—coincides with the substructure logic of regular closed regions in $\mathbb{R}^n$ and is recursively enumerable [10, Thms. 24, 28], so satisfiability is Π^0_1. Earlier versions of this paper presented the Π^0_1 placement as a new observation; it is not (§4.4).

What follows by import, and is not certified. Through the bridge, two published theorems on two-variable logic with one transitive relation decide two slices: $\mathcal{L}_{\|}$ concepts with vertical witnesses only [14], and finite satisfiability of all of $\mathcal{L}_{\|}$ [15]. General satisfiability in $\mathcal{L}_{\|}$ is the unary case of an open problem of two-variable logic over partial orders [16], so the full logic is at least as hard as that problem. The translation steps are standard but not formalized, and we did not check the cited proofs (§7.8).

Empirical evidence. A prototype reasoner (the cover-tree tableau; no completeness theorem is claimed) agrees with an independent oracle across a 911-case suite—99 labelled regression cases and 812 comparison runs—and it reproduces, 23 years later, the GIS subsumption taxonomy of Wessel's original report (§8). It is evidence, not a decision theorem, and it is incomplete: it rejects the satisfiable concept K_2 of §8, and in April 2026 an earlier version of it was shown to accept unsatisfiable concepts. Its subsumption verdicts are UNSAT answers, so each is evidence only where something independent agrees with it. A second program, a tableau without blocking that has a paper proof of soundness and completeness and no termination guarantee, agrees with it on the whole suite (§8).

The honest bottom line. After roughly thirty repair rounds and twenty-two adversarial reviews, one fragment is settled with a kernel-checked decision procedure — and, since October 2026, extended by a certified corollary to a slice containing $\forall\mathsf{PO}$, and, since November 2026, joined by two slices decided through published theorems on two-variable logic, by a bridge which also shows that the full problem contains an open problem of that literature; the local RCC5 algebra the proof attempts needed is characterized, and its normal form certified; and the remaining difficulty of the principal certificate architecture is located precisely, with the steps that are still only argued marked as such. We offer the work not as a resolution but as a status report and a research handoff on a problem open for more than two decades.

The status levels, at a glance. Throughout this paper we keep five verification levels distinct; Table 1 is the key, and the prose above and below assigns each result to one level. Conflating them is the single easiest way to misread the project.

3 Preliminaries

We fix the object logic precisely; readers familiar with $\mathcal{ALCI}_{\mathrm{RCC}}$ may skim to Definition 5.

Syntax. $\mathcal{ALCI}_{\mathrm{RCC5}}$ concepts are built from atomic concepts A and the five RCC5 relations, used as roles, by

$$C ::= A \mid \neg A \mid \top \mid \bot \mid C \sqcap C \mid C \sqcup C \mid \exists R.C \mid \forall R.C, \qquad R \in \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}.$$

We work in *negation normal form* (NNF: negation only on atoms) and *absorb inverse roles* through the RCC5 converse—$\exists\mathsf{PP}^{-}.C = \exists\mathsf{PPI}.C$, $\exists\mathsf{PO}^{-}.C = \exists\mathsf{PO}.C$, and so on—so no separate inverse

Status	**Examples**	**What the label means**
Lean-certified	the decision procedure for the *whole* ∀PO-free fragment, on raw input and under set semantics (`decidableFSat`, `decidableSat_cone`, `decidableSetSat`) and its PO-erasure extension to the fragment $\mathcal{D}$, which contains ∀PO (`decidableSat_dfrag`); the one-sided full-logic UNSAT test; the both-direction ordered-disjoint normal form; certificate soundness; finite-*type* Hintikka faithfulness; the conditional reductions of §7.6, *as conditionals*; the poset bridge for $\mathcal{L}_{\|}$, an infinity axiom inside that fragment, and the boundary concepts K_2, C_{cyc} (§7.8; unreviewed)	kernel-checked theorem names with a stated axiom footprint; the hypotheses of a conditional are not thereby certified
Published	representation of RCC5 structures by regular closed regions; recursive enumerability [10, Thms. 23, 24, 28]; for two-variable logic with one transitive relation, decidability with transitively guarded existentials [14] and over finite models [15]	an external theorem, cited
Theorem-level (not certified)	the fragment's two unformalized *arguments* and their $K(C_0)$ bound; the W2′ refinement; the local extension and cross-policy results; the two slices of $\mathcal{L}_{\|}$ decided by import through the bridge; soundness and completeness, without termination, of the reference tableau (Theorem 18)	a proof in a named work package or note, often with finite probes
Empirical	the cover-tree tableau (incomplete: K_2); the reference reasoner's implementation; the cross-validation suites; the finite model verifier; finite RCC5 networks with unbounded identity-selector minors (`wp44`)	test evidence or finite diagnostics only
Open	F6; whether identity-selector minors are its only obstruction; completeness of any certificate scheme for the full logic; the automatic-cover property; the cone scheme's architectural ∀PO gap; general satisfiability in $\mathcal{L}_{\|}$, which is that of two-variable logic over strict partial orders with unary predicates; full decidability	no proof claimed

Table 1: The five verification levels used throughout. The paragraphs of §2 assign each result to a level; this table is the at-a-glance key.

constructor is needed. Let $\mathsf{cl}(C_0)$ be the Fischer–Ladner closure of the input concept and $m = |\mathsf{cl}(C_0)|$.

Semantics: abstract composition tables, strong EQ. An interpretation is a pair $\mathcal{I} = (\Delta, \rho)$ with $\Delta \neq \emptyset$ and a *total* labelling $\rho : \Delta \times \Delta \to \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ satisfying: (i) *strong* EQ: $\rho(x, y) = \mathsf{EQ}$ iff $x = y$ (so EQ is identity, and between distinct elements only $\{\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ occur); (ii) *converse-closed*: $\rho(y, x) = \mathrm{conv}(\rho(x, y))$, where $\mathsf{PP}^{-1} = \mathsf{PPI}$, $\mathsf{PO}^{-1} = \mathsf{PO}$, $\mathsf{DR}^{-1} = \mathsf{DR}$; and (iii) *composition-closed*: $\rho(x, z) \in \mathsf{comp}(\rho(x, y), \rho(y, z))$ for all x, y, z, over the RCC5 composition table (Table 3). Concept extensions are standard, reading a role R as the pair set $\{(x, y) : \rho(x, y) = R\}$; the concept C_0 is *satisfiable* if $x \in C_0^{\mathcal{I}}$ for some $\mathcal{I}$ and some $x \in \Delta$. Thus a model is a complete graph carrying a composition-closed RCC5 edge-labelling; *models may be infinite*. This is precisely the *abstract* (composition-table) semantics—not a semantics of actual regions in a topological space. We fix the *strong* EQ semantics throughout (the standard convention for this problem; strong versus weak EQ is discussed below). The decision problem is: given C_0, is it satisfiable?

The sister logic $\mathcal{ALCI}_{\mathrm{RCC8}}$. $\mathcal{ALCI}_{\mathrm{RCC8}}$ is defined identically, over the eight RCC8 relations $\{\mathsf{DC}, \mathsf{EC}, \mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$, which *refine* RCC5 by splitting DR into DC (disconnected) and EC (externally connected, i.e. touching at the boundary), and PP, PPI into tangential and non-tangential parts (TPP, NTPP and TPPI, NTPPI); PO and EQ are unchanged. Its abstract composition-table semantics is the exact analogue over the RCC8 composition table, and RCC8 likewise has the patchwork property. $\mathcal{ALCI}_{\mathrm{RCC5}}$ is our focus; we invoke $\mathcal{ALCI}_{\mathrm{RCC8}}$ only where it matters—its ExpTime-hardness (§4) and the domino attempts (§6), which turn on the extra TPP/NTPP distinction.

The RCC granularity hierarchy. $\mathcal{ALCI}_{\mathrm{RCC5}}$ and $\mathcal{ALCI}_{\mathrm{RCC8}}$ sit atop a family of five logics $\mathcal{ALCI}_{\mathrm{RCCk}}$, $k \in \{1, 2, 3, 5, 8\}$, whose relation sets *coarsen* as k decreases [9]: $\mathcal{ALCI}_{\mathrm{RCC1}}$ has the single relation SR ("spatially related"); $\mathcal{ALCI}_{\mathrm{RCC2}}$ has $\{\mathsf{DR}, \mathsf{O}\}$ (with O = "overlapping"); and $\mathcal{ALCI}_{\mathrm{RCC3}}$ has $\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$ (with ONE = "overlapping but not equal"). Each coarser set is definable in the finer one—$\mathsf{ONE} = \{\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}\}$, $\mathsf{O} = \{\mathsf{ONE}, \mathsf{EQ}\}$, $\mathsf{SR} = \{\mathsf{DR}, \mathsf{O}\}$—so the family forms a granularity lattice. The three coarsest logics are *decidable* (Wessel [9]); $\mathcal{ALCI}_{\mathrm{RCC5}}$ and $\mathcal{ALCI}_{\mathrm{RCC8}}$ are the open cases this paper concerns. The full landscape is Table 2 (§4).

Strong versus weak EQ. Two semantics for EQ appear in the literature [9]: the *strong* one, $\mathsf{EQ}^{\mathcal{I}} = \mathrm{Id}(\Delta)$ (EQ is exactly identity), and the *weak* one, $\mathrm{Id}(\Delta) \subseteq \mathsf{EQ}^{\mathcal{I}}$ (EQ contains identity but may relate distinct, mutually indistinguishable elements). We fix the *strong* semantics throughout—the standard choice for this problem, and the more expressive one: Wessel observed that under strong EQ, from $\mathcal{ALCI}_{\mathrm{RCC3}}$ upward the *difference role* $R_D =_{\mathrm{def}} N_R \setminus \{\mathsf{EQ}\}$ becomes definable, so *nominals* are encodable ($i \sqcap \forall R_D.\neg i$ forces i to name a single element)—a reminder of how narrow the margin to undecidability is (adding a $\downarrow$-binder would cross it, cf. §4.3). The choice costs no generality for decidability: Wessel showed decidability under the weak semantics would carry over to the strong one, and conversely a weak-EQ model collapses to a strong one by internalizing EQ as identity in the TBox, so the two satisfiability problems are inter-reducible.

The two axes, made precise. Call PP, PPI *vertical* (nesting) and DR, PO *horizontal* (side-by-side). The composition table makes the axes behave oppositely, and this asymmetry is the root of the whole difficulty.

Proposition 1 (Determination is vertical)**.** *Of the sixteen compositions of proper (non-*EQ*) relations, exactly four are* determining *(return a singleton):*

$$\mathsf{comp}(\mathsf{DR},\mathsf{PPI}) = \{\mathsf{DR}\},\ \mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\},\ \mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\},\ \mathsf{comp}(\mathsf{PPI},\mathsf{PPI}) = \{\mathsf{PPI}\}.$$

*Every one has a vertical (*PP *or* PPI*) argument, and* PO *is never the value of any determining composition. (Exhaustive over Table 3; machine-verified.)*

So no *single* composition step forces a horizontal value—the four determining steps are all vertical. This must not be overread:

Remark 2 (Single-step versus joint forcing)**.** Proposition 1 concerns *one* length-two path; it does *not* say horizontal edges are always free. The *intersection* of several composition constraints can pin a horizontal edge to a single value: $\mathsf{comp}(\mathsf{DR},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI},\mathsf{PO}) = \{\mathsf{PPI},\mathsf{PO}\}$ intersect in $\{\mathsf{PO}\}$, so an edge forced through two different intermediates can be *compelled* to be PO (machine-checked, `wp85`); indeed every proper relation—DR, PO, PP, PPI—is reachable as such a joint singleton, so PO is not special among them. Consequently "shadow" and "live" (Definition 5) must be read against the *joint* closure of all constraints, never single-path forcing. What *composition* genuinely never forces is a merge: no intersection of composition cells whose two premise labels are *proper* (non-EQ) relations ever narrows to $\{\mathsf{EQ}\}$ (`wp85`), so composition can only force occurrences *apart* (a shared horizontal anchor excludes EQ), never *together*. Concept *labels*, by contrast, *can* force a *local* merge: the satisfiable one-level gadget of §6.2 forces its two coloured witnesses to coincide and remains satisfiable. What fails is *propagating* that isolation recursively—the discipline that would tile a grid collapses the construction to unsatisfiability. So the coincidence obstruction is really a *dichotomy* at the level of the would-be grid: a grid merge is either *unforced* (the branches never join) or *fatal* (recursion collapses it to UNSAT), never a clean forced grid—not a blanket "EQ cannot be forced," and not a claim that no merge is ever satisfiable. That one-sidedness—composition forces apart but not (usefully) together—is the real horizontal asymmetry.

There is one further asymmetry, and it is the one the split-forest presentation (§7.4) turns to its advantage: disjointness is inherited *downward* but not upward.

Proposition 3 (Half-determinism of disjointness)**.** *If* $x\ \mathsf{DR}\ y$ *then every part of* x *is disjoint from every part of* y*:* $\mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$ *forces* $x'\ \mathsf{DR}\ y$ *for every* $x'\ \mathsf{PP}\ x$*, and* $\mathsf{comp}(\mathsf{DR},\mathsf{PPI}) = \{\mathsf{DR}\}$ *pushes this down the* y*-side as well.* Upward *the inference fails: a* superpart *of a disjoint region may be disjoint from, overlap, or contain the other region—*$\mathsf{comp}(\mathsf{PPI},\mathsf{DR}) = \{\mathsf{PPI},\mathsf{PO},\mathsf{DR}\}$ *is not a singleton, so nothing is forced. Disjointness thus propagates deterministically to descendants in the* PP*-order, and to nothing else—the* downward-closed disjointness *that the RCC5 normal form (Theorem 15, certified) makes exact.*

This "half-determinism" (deterministic downward, free upward) is what lets a finite presentation dispatch arbitrarily wide DR crowds for free (§7.4): a single high DR edge shadows every disjointness beneath it, leaving only PO and the *upward* direction as genuine branching. Vertical recurrence, moreover, always folds:

Lemma 4 (No horizontal recurrence)**.** *Form the composition-path subset automaton whose states are the reachable relation sets, with* $\mathrm{Rel}(\varepsilon) = \{\mathsf{EQ}\}$ *and* $\mathrm{Rel}(w\,a) = \bigcup_{r\in\mathrm{Rel}(w)} \mathsf{comp}(r,a)$*. Restrict to states not containing* EQ*, and to non-*EQ *labels (an* EQ *label is a self-loop on every state, since* EQ *is the composition identity). Then no cycle uses a* horizontal *(*DR*/*PO*) label: every cycle is built from vertical (*PP*/*PPI*) steps. Hence any support recurrence that never folds by an equality is* eventually vertical*, and is absorbed by vertical looping rather than by a horizontal crowd. (Exhaustive finite computation; machine-verified,* `wp38`*.)*

Proposition 1 and Lemma 4 explain why infinite depth *generated by vertical demands* is not the obstacle: structure forced by composition is vertical, and vertical recurrences loop. This does not cover every source of nesting. A concept can force two *horizontally* generated witnesses to nest, by excluding both PO and DR between them with $\forall\mathsf{PO}.D \sqcap \forall\mathsf{DR}.D$ (K_2, §8); chained, this gives unsatisfiable cycles and concepts with only infinite models and no vertical existential (§7.8; the first such concept is in [9]). Whether structure forced in that way always folds is not known. The remaining difficulty is horizontal in origin—and, crucially, while *individual* horizontal edges can be pinned (by the joint closure of several constraints, Remark 2), no concept is known to *force an unboundedly wide rigid* horizontal pattern. Whether one can is, in substance, what F6 asks. A decision procedure must record only the horizontal branching it cannot reconstruct:

Definition 5 (Shadows, live width, and F6)**.** In a model, or in a finite presentation of one, an edge is a *shadow* if its RCC5 value is forced by the *joint* composition-closure of the edges already present—which may be a single triangle ($x\ \mathsf{PP}\ y$, $y\ \mathsf{DR}\ z$ force $x\ \mathsf{DR}\ z$ by $\mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$) *or* the intersection of several (Remark 2); a shadow costs a decision procedure nothing. The *live width* at an interface is the number of non-shadow edges there—the genuine branching that must be recorded. Property **F6** states: for every satisfiable C_0, the live width of some presentation of a model of C_0 is bounded by a computable function of C_0.

4 History of $\mathcal{ALCI}_{\mathrm{RCC}}$ and why the known results do not settle it

4.1 Cohn's 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle\mathsf{PP}\rangle C$ for "some proper part satisfies C"—but Cohn left all decidability and complexity questions open.

4.2 Wessel's $\mathcal{ALCI}_{\mathrm{RCC}}$ family (2002/2003)

Wessel [7, 9, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{\mathrm{RCC}}$ *family*: description logics $\mathcal{ALCI}_{\mathrm{RCCk}}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{\mathrm{RCCk}}$ interpretation, every pair of domain elements is assigned exactly one RCCk base relation, and the composition table is enforced globally (*strong* EQ *semantics*: EQ is identity). Models are complete graphs with RCCk edge-colorings.

Wessel established the decidability and complexity landscape summarized in Table 2. The three coarsest logics are *decidable*: $\mathcal{ALCI}_{\mathrm{RCC1}}$ is essentially the modal logic S5 (hence NP-complete); $\mathcal{ALCI}_{\mathrm{RCC2}}$ and $\mathcal{ALCI}_{\mathrm{RCC3}}$ are decidable because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques (for $\mathcal{ALCI}_{\mathrm{RCC3}}$ under strong EQ, via the definable difference role). The two finest logics—$\mathcal{ALCI}_{\mathrm{RCC5}}$ (PSPACE-hard) and $\mathcal{ALCI}_{\mathrm{RCC8}}$ (EXPTIME-hard)—were left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].

Logic	Base relations N_R	Concept satisfiability
$\mathcal{ALCI}$ (no RCC)	arbitrary roles	PSPACE-complete (bare); EXPTIME w/ TBoxes
$\mathcal{ALCI}_{\mathrm{RCC1}}$	$\{\mathsf{SR}\}$	decidable; NP-complete ($\equiv$ S5)
$\mathcal{ALCI}_{\mathrm{RCC2}}$	$\{\mathsf{DR}, \mathsf{O}\}$	decidable; NP-hard, in PSPACE
$\mathcal{ALCI}_{\mathrm{RCC3}}$	$\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$	decidable; NP-hard
$\mathcal{ALCI}_{\mathrm{RCC5}}$, $\forall\mathsf{PO}$-free	RCC5 without $\forall\mathsf{PO}$	**decidable** (this paper, Thm 10)
$\mathcal{ALCI}_{\mathrm{RCC5}}$, fragment $\mathcal{D}$	no $\exists\mathsf{PP}$, $\exists\mathsf{PO}$, $\forall\mathsf{DR}$	**decidable** (certified; §7.3)
$\mathcal{ALCI}_{\mathrm{RCC5}}$, $\mathcal{L}_{\|}$ without $\exists\mathsf{PO}$	see §7.8	decidable, in 2-NEXPTIME (by import)
$\mathcal{ALCI}_{\mathrm{RCC5}}$, $\mathcal{L}_{\|}$	see §7.8	**open**; finite satisfiability decidable (by import)
$\mathcal{ALCI}_{\mathrm{RCC5}}$ (full)	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$	**open**; PSPACE-hard
$\mathcal{ALCI}_{\mathrm{RCC8}}$	RCC8 (eight relations)	**open**; EXPTIME-hard

Table 2: The complexity landscape of the $\mathcal{ALCI}_{\mathrm{RCC}}$ family (concept satisfiability, strong EQ). RCCk relation sets coarsen as k decreases; $\mathcal{ALCI}_{\mathrm{RCC1}}$–$\mathcal{ALCI}_{\mathrm{RCC3}}$ were shown decidable by Wessel [9], while $\mathcal{ALCI}_{\mathrm{RCC5}}$ and $\mathcal{ALCI}_{\mathrm{RCC8}}$ are the standing open cases. The two $\mathcal{L}_{\|}$ rows rest on published theorems applied through a machine-checked bridge; they are not themselves certified.

comp	$\mathsf{DR}(a,b)$	$\mathsf{PO}(a,b)$	$\mathsf{EQ}(a,b)$	$\mathsf{PPI}(a,b)$	$\mathsf{PP}(a,b)$
$\mathsf{DR}(b,c)$	$*$	$\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}$	DR	$\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}$	DR
$\mathsf{PO}(b,c)$	$\mathsf{DR}, \mathsf{PO}, \mathsf{PP}$	$*$	PO	$\mathsf{PO}, \mathsf{PPI}$	$\mathsf{DR}, \mathsf{PO}, \mathsf{PP}$
$\mathsf{EQ}(b,c)$	DR	PO	EQ	PPI	PP
$\mathsf{PP}(b,c)$	$\mathsf{DR}, \mathsf{PO}, \mathsf{PP}$	PO, PP	PP	$\mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}$	PP
$\mathsf{PPI}(b,c)$	DR	$\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}$	PPI	PPI	$*$

Table 3: The RCC5 composition table ($*$ = all five relations).

- $\mathcal{ALC}_{\mathcal{RA}_{SG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RA}_{SG}}$) makes it undecidable again.

$\mathcal{ALCI}_{\mathrm{RCC5}}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RA}_{SG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}$/$\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$).

Remark 6 (Key difficulties)**.** $\mathcal{ALCI}_{\mathrm{RCC5}}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models: a PP-tower, or, with no vertical existential at all, the spiral concept of §7.8), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 3. Reading: row $= S(b,c)$, column $= R(a,b)$, entry $=$ possible relations for (a,c).
Key composition facts exploited in the cover-tree theory (§7.4, §8): $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ (DR rigidity); $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ (transitivity); and $\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ (PP can be *forced* by composition and universal filtering; see the forced-composition examples in §7.1).

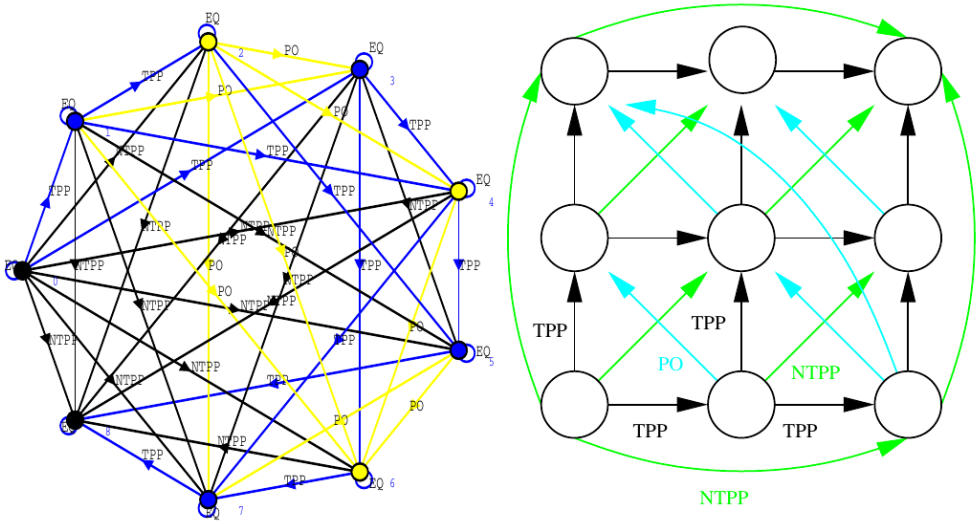


Figure 1: Wessel's original $\mathcal{ALCI}_{\mathrm{RCC8}}$ grid construction, reproduced *directly from report7.pdf (2002/2003)*, Figure 10 (Section 4.4.3): a complete-graph RCC8 "frame" (left) that is isomorphic to a 3×3 grid (right) and extends to $n \times n$. Horizontal and vertical successors are TPP edges (the intended roles R_X, R_Y); PO (cyan) and NTPP (green) are the induced diagonal and long-range relations. Such grid models *exist*—but, as the same report already argued, the natural encodings do not *enforce* them, and no concept term is known that does. (The frame was *found by a Lisp enumerator* and rendered with CLIM under MCL, not drawn by hand.) The construction and the obstruction are Wessel's, two decades before this AI-assisted work.

4.3 Wessel's grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{\mathrm{RCC8}}$ [9], Wessel made two important discoveries:

1. Even-odd chains (Section 4.4.2). The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between *even* and *odd*, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary n-bit counter, proving $\mathcal{ALCI}_{\mathrm{RCC8}}$ is ExpTime-hard.

2. The grid coincidence obstruction (Section 4.4.3, Figure 11). Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

> "Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term."

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*. His explicit grid frame is reproduced here as Figure 1, and the merge obstruction that blocks enforcing it as Figure 2.

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional

TPP/NTPP grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel's constructions.

4.4 Lutz & Wolter (2006)

Lutz and Wolter [10] studied *modal logics of topological relations*, denoted L_{RCC8}, L_{RCC5}, etc. These have the same *syntax* as $\mathcal{ALCI}_{\mathrm{RCC}}$ but are interpreted over *topological models*: the domain elements are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- L_{RCC8} is **undecidable** (their §5: a linearized NTPP-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of $\mathbb{R}^n$; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^{\exists})$ is **undecidable** (reduction from $S5^3$ via supremum regions).
- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): "Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply."

The logic $\mathcal{ALCI}_{\mathrm{RCC5}}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

What they did prove about it. Their paper establishes three facts about exactly this logic, and they should be credited wherever they are used. By their Theorem 23, every general RCC5 structure—a nonempty set with a converse- and composition-closed labelling in which EQ is identity—is isomorphic to a structure of regular closed regions of some topological space, and every countable one to such a structure in $\mathbb{R}^n$, for each $n > 0$. By Theorem 24, $L_{RCC5}(RS)$ coincides with the logic of substructures of topological region structures, and with that of $\mathbb{R}^n$ and its regular closed sets. And by Theorem 28 these logics are recursively enumerable, so $\mathcal{ALCI}_{\mathrm{RCC5}}$ satisfiability is Π^0_1 [10]. Three consequences for this paper follow. The Π^0_1 placement of §7.6 is theirs: earlier versions of this paper presented it as new, which was a failure of our literature check, and we keep the derivation only because the dovetailing argument uses its explicit form. The equivalence of abstract and set semantics certified in §7.3 is a machine-checked special case of their Theorem 23, with an explicit construction. And every satisfiability result stated here for the abstract semantics—the fragment's decidability in particular—holds verbatim for substructure semantics over regular closed regions of $\mathbb{R}^n$, by their Theorem 24. It does not transfer to the *full* structures that contain every regular closed region, which form a different logic. (One sentence of their conclusion calls the recursive enumerability of $L_{RCC5}(RS)$ open, in apparent conflict with Theorem 28; the remark following Theorem 28 leaves open only the logics of full structures, and we read the concluding sentence in that light.)

Two further points of that paper bear on §7.8. It shows that the modal language over the RCC5 relations has the same expressive power as the two-variable fragment of first-order logic over those relations with unary predicates [10, §8, cf. Thm. 2], so satisfiability in $\mathcal{ALCI}_{\mathrm{RCC5}}$ is satisfiability of two-variable sentences over the class of RCC5 structures. And it names the decidability of the substructure logics "and their finite companions" as one of its most intriguing open problems. Reading the finite companion as the logic of the finite structures, the finite case is decidable for the fragment $\mathcal{L}_{\|}$, by a theorem of Pratt-Hartmann applied through the bridge (§7.8); for the full logic it remains open.

The route they suggested, and how it differs from the one taken here. Their concluding section asks for "a decidable and still useful variant" of the logics they study, and canvasses where one might be found. Besides $L_{RCC5}(RS)$ itself, they write that "other candidates could be obtained by modifying the set of relations, e.g. giving up some of them," and add that "it has, for example, been argued that dropping PO still results in a useful formalism for applications in geographic information systems" [10, p. 31].

It is worth being exact about what that is and is not, because it is the closest thing in the literature to the fragment established here. The sentence about dropping PO discharges the *usefulness* half of their two-part desideratum, not the decidability half: it is a claim about applications, it carries no citation, and it is accompanied by no proof, no proof sketch, and no conjecture that a PO-less language would be decidable. The technical progress they point to in the same breath, Shapirovsky and Shehtman [8], concerns modal logics with a *single* modality for an inclusion relation over region structures, doubling as relativistic temporal logics — a substantially smaller language than RCC5 without PO, which still carries DR, PP and PPI. So far as we can determine, no decidability argument for any PO-restricted RCC5 language appears there or since.[1]

The fragment established here differs from their suggestion in two ways, and both matter. First, it is a strictly *weaker* restriction: PO is not removed from the language. Existential quantification $\exists\mathsf{PO}.C$ — "overlaps something satisfying C" — is retained in full, and only the universal $\forall\mathsf{PO}.C$ is excluded, so the fragment is properly larger than the one their remark contemplates. Second, and structurally more important, the fragment does *not* obtain decidability by recovering a finite model property. The concept $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ is $\forall$PO-free and satisfiable (`cinf_pofree`, `cinf_satisfiable`), and every model of it is infinite, since the universal keeps producing strictly larger regions—a one-line argument, not a Lean theorem; decidability comes instead from a finite *presentation* of possibly-infinite models (§7.3). And the reason the restriction is exactly the right one is the residual-PO default: a composition cell containing PO can always be discharged by putting the pair at PO unless some formula can object, and the only formula that can is $\forall\mathsf{PO}.D$. Dropping PO wholesale would buy that property too, but at more cost than is necessary.

Prior consideration of restricted fragments, by the present author. Report7 [9] contains a section headed "Restricted Decidable Fragments of $\mathcal{ALCI}_{\mathrm{RCC5}}$ and $\mathcal{ALCI}_{\mathrm{RCC8}}$?" (§5.3), which is prior art for the general idea and should be recorded as such. It canvasses two restrictions: one derived from $\mathcal{ALCRP}(D)$, rejecting every concept whose negation normal form contains a quantifier alternation $\exists R.\ldots\forall S.\ldots$ or $\forall R.\ldots\exists S.\ldots$; and one admitting only Boolean concepts inside universal value restrictions. Both are strictly stronger than $\forall$PO-freeness — they constrain *every* role, and forbid the nesting the present fragment permits — and both work, as that section says, by restoring the finite model property. Neither is the $\forall$PO-free fragment, and by the witness above neither technique could yield it.

On priority and citation. Two honest caveats, in both directions. First, Wessel proved *neither* decidability nor undecidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ or $\mathcal{ALCI}_{\mathrm{RCC8}}$: report7 constructs the grid models and

[1] One might be tempted to read support into the fact that PO occurs exactly once in their RCC5 undecidability reduction (§8) — as the box [PO] in their formula (8.1), forcing the representing regions to be pairwise disconnected — and never in existential position, so that removing $\forall$PO alone suffices to break it. That reading would be a mistake, and we record the observation only because it makes their choice of PO look informed rather than arbitrary. Breaking one reduction is no evidence of decidability: undecidability proofs are not unique, and nothing excludes a different argument that never mentions PO. The entailment in fact runs the other way and in the uninteresting direction — the fragment is decidable, hence *no* undecidability reduction succeeds on it, theirs included. Their reduction moreover fails for $L_{RCC5}(RS)$ on independent grounds, requiring closure under suprema ($RS^{\exists}$), which is why they list that logic as the open candidate in the first place.

Approach	$\forall \mathsf{PP}.C$?	$\exists \mathsf{DR}.D$?	Quantify over relations?
Concrete domains, e.g. $\mathcal{ALC}$(RCC8)	no	no	no
Composition-based role boxes ($\mathcal{ALCI}_{\mathrm{RCC5}}$)	**yes**	**yes**	**yes**

Table 4: Plain concrete domains versus composition-based role boxes. Of the two, only the latter *quantifies over* spatial relations on the abstract domain itself, which is what puts $\mathcal{ALCI}_{\mathrm{RCC5}}$ outside the reach of the concrete-domain results; the role-forming extension $\mathcal{ALCRP}(D)$, discussed below, sits in between.

pins the coincidence obstruction as the precise boundary, but stops there, explicitly leaving both problems open. The contribution was to *locate* the wall, not to breach it. Second, and less comfortably: the Lutz–Wolter paper does not cite report7, although it was a publicly available official technical report of the University of Hamburg (2002/2003) that already developed the $\mathcal{ALCI}_{\mathrm{RCC}}$ description logics, the even-odd counter construction, the explicit two-dimensional $\mathsf{TPP}/\mathsf{NTPP}$ grid, and the coincidence obstruction as the decisive obstacle. In fairness to them, they do cite Wessel's earlier DL 2001 paper [5]: modal logics over the RCC8 relations, they write, had been "further considered" there, and "despite several efforts, to the best of our knowledge no results have been obtained so far" [10, p. 2]. Report7 is where results of exactly that kind had been obtained. We record this as a matter of *priority*, not of dependence—their undecidability proof takes a genuinely different (topological, linearized) route and in no way relies on report7—but the overlap in ingredients was substantial and the material was findable.

4.5 Concrete-domain approaches, and why they do not settle it

A natural objection is that decidability of description logics with RCC constraints is surely already known. It is—but for a *different*, expressively incomparable formalism: the *concrete-domain* approach, in which spatial relations constrain *functional attributes* of individuals rather than serving as roles. The distinction is decisive (Table 4). A concrete domain reasons about the spatial *attributes* of elements; $\mathcal{ALCI}_{\mathrm{RCC}}$ instead makes the spatial relations *themselves* the roles, and so permits *quantification over* them—concepts such as $\forall \mathsf{PP}.C$ ("every proper part satisfies C") and $\exists \mathsf{DR}.D$ ("some discrete region satisfies D")—which plain concrete-domain logics such as $\mathcal{ALC}(\mathcal{D})$ cannot express.

The closest concrete-domain relative is $\mathcal{ALCRP}(D)$ of Haarslev, Lutz and Möller [11], which was motivated by spatial applications of exactly this kind. Its role-forming predicate operator defines a role from a concrete predicate over feature chains—"the objects whose spatial extension is disjoint from mine"—and so it does support a form of quantification over spatial relations, which the first row of Table 4 does not. It remains two-sorted and feature-mediated, however: the predicate relates attribute values, so distinct abstract objects may share one extension (EQ is not identity on the objects), and an object without the feature stands in no spatial relation at all. Reasoning in it is undecidable in general and decidable under syntactic restrictions [11], restrictions that Wessel's report later adapted to $\mathcal{ALCI}_{\mathrm{RCC}}$ (§4.4).

The known decidability results all sit on the concrete-domain side. Lutz and Miličić [17] showed $\mathcal{ALC}$ with ω-admissible concrete domains (which include RCC5 and RCC8) decidable, without inverse roles; Baader and Rydval [18] revisited and generalized the ω-admissibility conditions (and strengthened the matching undecidability results); Borgwardt, De Bortoli and Koopmann [19] pinned $\mathcal{ALC}(\mathcal{D})$ ontology consistency at ExpTime-complete for ω-admissible $\mathcal{D}$ ($\mathcal{ALC}$ only, no inverse); and Demri and Gu [20] recently extended the constraint-automaton method to

inverse roles and functional role names with ExpTime membership—the closest published result to $\mathcal{ALCI}_{\mathrm{RCC}}$, yet still within the concrete-domain formalism. None settles $\mathcal{ALCI}_{\mathrm{RCC5}}$ or $\mathcal{ALCI}_{\mathrm{RCC8}}$: the composition-based role box allows quantification over the spatial relations themselves, and the complete-graph semantics is not a concrete-domain constraint. The two lines are complementary—attributes versus relationships—which is why twenty years of progress on the former left the latter open.

5 Decidable fragments of first-order logic, and why none captures it

$\mathcal{ALCI}_{\mathrm{RCC5}}$ satisfiability is, quite literally, a *first-order* satisfiability problem. The five relations are binary predicates; the composition-table semantics is a finite set of first-order axioms T_{RCC5}—exactly one atom on each ordered pair of distinct elements, EQ as identity, converse coherence, and, for every pair of atoms r, s and every $t \notin \mathsf{comp}(r, s)$, the *forbidden triangle* $\forall x\, y\, z\ \neg\big(R_r(x, y) \wedge R_s(y, z) \wedge R_t(x, z)\big)$—while the concept C_0 becomes a sentence $\exists x\, \mathrm{ST}(C_0, x)$ under the standard modal translation. Thus C_0 is satisfiable iff $T_{\mathrm{RCC5}} \cup \{\exists x\, \mathrm{ST}(C_0, x)\}$ has a model. If this theory fell inside a *decidable fragment* of first-order logic we would be done—a natural first thing to try, and, at coarser granularity, exactly how the decidable family members fell: report7 settles $\mathcal{ALCI}_{\mathrm{RCC3}}$ by precisely this route (below). At RCC5 it does not work; the surface reasons vary by fragment, but the deepest one—the one that defeats even the fragments tailor-made for relational theories like this—is the same totality that underlies F6.

Two invariants of $\mathcal{ALCI}_{\mathrm{RCC5}}$ models constrain which fragment could contain it.

- **No finite model property.** A descending PP-tower ($\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$: every region has a proper part, with PP a strict order) is satisfiable only in infinite models; so, with no vertical existential at all, are a concept of the 2003 report [9, §5.2] and the spiral concept of §7.8. Any fragment with the finite model property is excluded outright.

- **Complete Gaifman graph.** Totality forces every two distinct elements to be related, so the Gaifman graph[2] of every model is *complete*: a PP-chain of length n already has treewidth $n-1$ and clique number n (`wp78`). Any fragment whose decidability rests on models of bounded treewidth—the whole guarded family—is excluded as well.

Against these, the standard fragments fall as follows.

Two variables (FO^2*).* This is how the coarser family members actually fell: report7 proves $\mathcal{ALCI}_{\mathrm{RCC3}}$ (strong EQ) decidable by translating concepts, via the standard translation, into *two-variable* first-order logic with equality [9]—possible there because the RCC3 role theory needs no ternary composition axiom at all (report7's role theory is exactly-one plus converse symmetry, all two-variable; the RCC3 table constrains no triple of distinct elements). At RCC5 the route is excluded twice over: the composition axiom is now genuinely ternary ($\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ constrains distinct triples), and FO^2 has the finite model property [21], which $\mathcal{ALCI}_{\mathrm{RCC5}}$ lacks. Worse, the ternary step is into hostile territory—the three-variable fragment FO^3 is already *undecidable* (it captures the equational theory of relation algebras [22])—so variable-counting alone offers no route: any decidability must come from the *shape* of the axioms, not their arity.

[2]The *Gaifman graph* of a relational structure is the undirected graph on its elements in which two distinct elements are joined whenever they occur together in some tuple of some relation—for binary relations, whenever $R(x, y)$ or $R(y, x)$ holds for some R. It is the standard way to speak of the "locality" or "sparsity" of a model: bounded degree, bounded treewidth, and similar tameness conditions are all read off it. Here, because every pair of elements carries an RCC5 atom, this graph is *complete*—maximally non-local.

One qualification matters. The concept side *is* two-variable: the modal language over the RCC5 relations has the expressive power of the two-variable fragment over those relations with unary predicates [10, §8]. What is not two-variable is the definition of the class of structures. Two-variable logic over a restricted class of structures is a subject of its own, and the relevant case is a transitive relation. General satisfiability of FO^2 with one transitive relation was claimed decidable in 2013 [13]; the claim was withdrawn by its authors [14]. Decided are the fragment whose existential quantifiers are guarded by the transitive relation [14] and finite satisfiability [15]. The general problem is open, also for a single partial order, where it is conjectured decidable [16]. §7.8 shows that a syntactic fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$ is equisatisfiable with the unary case of that logic. So this fragment of first-order logic does not capture $\mathcal{ALCI}_{\mathrm{RCC5}}$; the containment runs the other way.

Prefix classes (Bernays–Schönfinkel–Ramsey and relatives). The standard translation of nested modalities produces unbounded $\forall\exists$ alternation, so no bounded quantifier-prefix class applies; these classes also have the finite model property.

The guarded fragment and its relatives. This is the interesting near-miss, and it is worth being precise about where it breaks. The concept part is *guarded* outright—the modal translation is the paradigm guarded formula, each quantifier relativized by its role atom ($\exists \mathsf{PP}.C \rightsquigarrow \exists y(R_{\mathsf{PP}}(x,y) \wedge \cdots)$) [23]. The composition constraint, in forbidden-triangle form, is *clique-guarded*: in $R_r(x,y) \wedge R_s(y,z) \wedge R_t(x,z)$ every pair of variables co-occurs in an atom—exactly a clique guard—so composition-closure lives in Grädel's clique-guarded fragment [24]; converse and mutual-exclusion are guarded too. In other words *almost* the entire theory is clique-guarded. What is not—and cannot be—is **totality**: "every two elements are related" is the archetypal non-guarded assertion (it is not invariant under the guarded bisimulation these fragments are built around). And totality is not a technicality one may drop: it is what forces the complete Gaifman graphs above, which destroy the generalized tree-model property on which the guarded family's decidability rests. The guarded route thus fails for exactly the reason `wp78` records—and note it is *totality*, not the ternary composition axiom, that does the damage; the composition axiom is the tamer half.

Remark 7 (On the two renderings of the composition table)**.** The *natural* first-order rendering of a table cell is not the forbidden triangle but the implication

$$\forall x\, y\, z\, \big(R_r(x,y) \wedge R_s(y,z) \;\rightarrow\; R_{t_1}(x,z) \vee \cdots \vee R_{t_n}(x,z)\big), \qquad \{t_1,\ldots,t_n\} = \mathsf{comp}(r,s),$$

and it is instructive that this form is *less* guarded, not more. Its guard $R_r(x,y) \wedge R_s(y,z)$ is a *path*, not a clique: the pair $\{x,z\}$ the consequent speaks about never co-occurs in a guard atom—indeed for $r = s = t_1 = \mathsf{PP}$ the sentence *is* transitivity, $\forall xyz\,(R_{\mathsf{PP}}(x,y) \wedge R_{\mathsf{PP}}(y,z) \rightarrow R_{\mathsf{PP}}(x,z))$, the textbook non-guardable shape (and the one behind the transitivity undecidability cited below). The contrapositive forbidden triangles are clique-guarded precisely because the offending third edge moves *into the guard*. The two renderings are equivalent only modulo exactly-one: the implication additionally *forces* an atom on (x,z)—a relativized piece of totality—and the negative rewriting is available at all only because the relation alphabet is finite and exhaustive (for a single transitive relation over an open vocabulary no such rewriting exists). Nor is the rewriting a loophole: within the clique-guarded fragment the triangle sentences are *strictly weaker* than transitivity, since a model may leave (x,z) atom-free; the equivalence broker is totality, which is exactly the axiom the fragment cannot have. Nothing downstream is sensitive to the choice—both are finite first-order renderings, interchangeable over T_{RCC5}, so the Π^0_1 placement (§7.6) is unaffected—but the diagnosis is: written with implications, composition *looks* unguarded for a repairable, surface reason; written with triangles, every axiom of T_{RCC5} dresses in guards *except* totality, which is the honest accounting.

Other fragments. The fluted fragment requires predicate arguments to form a suffix of the quantifier sequence; the composition atom $R_t(x,z)$ skips y, so it is not fluted. The unary-negation

Source Problem	Technique	Missing in $\mathcal{ALCI}_{\mathrm{RCC5}}$
$\mathbb{N} \times \mathbb{N}$ domino	Graded modalities + transitivity	Number restrictions
$\mathcal{ALC}_{\mathcal{RA}}$ [6]	Post Correspondence Problem	Arbitrary role axioms
$\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ [3]	Non-empty CFG intersection	Non-disjoint roles
$\mathcal{ALCI}_{\mathrm{RCC8}}$ (Wessel)	Grid via TPP/NTPP	TPP/NTPP distinction
L_{RCC8} (Lutz–Wolter)	Linearized topological enumeration	Topological rigidity
$L_{RCC5}(RS^{\exists})$	$S5^3$ via supremum regions	Supremum closure

Table 5: Why standard undecidability reductions fail for $\mathcal{ALCI}_{\mathrm{RCC5}}$.

and guarded-negation fragments are a subtler near-miss: negation of *sentences* is unrestricted there, so the forbidden triangles themselves *are* expressible (each is the negation of an existential-positive sentence)—but both fragments have the finite model property [25], which $\mathcal{ALCI}_{\mathrm{RCC5}}$ lacks, and totality is again out of reach (its inner negations carry two free variables and no atom to guard them). And the one extension that would embrace the vertical structure directly—the guarded fragment *with transitive relations*, since PP, PPI are transitive ($\mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\}$)—is *undecidable* once transitivity is used outside guards, as it must be here [26], so even the most sympathetic extension supplies no decision procedure.

The lesson. The surface reasons vary—arity, prefix shape, atom shape, the finite model property—but those are boundary mismatches that fragment engineering could in principle negotiate. The failure that matters is the guarded family's, and there the culprit is *totality*, i.e. the complete Gaifman graph: the one axiom of T_{RCC5} no guard can carry, and the one obstruction no choice of fragment removes, because it is a property of the *models*, not of the axioms' syntax. It is precisely the obstruction F6 names from the width side and `wp78` from the cover side; the FO-fragment view rediscovers it from classical model theory. One family is not defeated by totality: two-variable logic over a fixed class of orders speaks about every pair of elements directly. It supplies no decision procedure either—its relevant case is open (§7.8)—but it is where techniques for total, transitive structures have been developed, and the full logic contains its unary case. The FO-fragment view also tells us which tools are the wrong ones: guardedness and locality (Gaifman, Hanf) are exactly what totality defeats, so the productive machinery is the totality-tolerant kind—Ehrenfeucht–Fraïssé / back-and-forth and the composition method over bounded *tuple* profiles—which is the "correct room" the keystone analysis independently points to (§10).

6 Why standard undecidability reductions fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N}\times\mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{\mathrm{RCC5}}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

6.1 Blocked reduction routes

Table 5 summarizes the blocked routes. Two properties of RCC5 are easily run together here and should be kept apart. For atomic networks, path consistency implies consistency (for RCC8 this is

part of Renz and Nebel's analysis [2]; for RCC5 it is the content of Theorem 15). The **patchwork property** is a different statement: consistent networks that agree on their overlap amalgamate into one consistent network, the form in which it enters concept-level reasoning [17]. Neither property by itself blocks grid encodings—RCC8 has both, and Lutz and Wolter's RCC8 logics over regular closed regions are undecidable [10]. What blocks the attempts below is the coincidence obstruction of §4.3.

6.2 A concrete domino attempt in $\mathcal{ALCI}_{\mathrm{RCC5}}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned} X \;=\; & \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \;\sqcap\; \exists\mathsf{PPI}.\exists\mathsf{PPI}.D \\ & \sqcap\; \forall\mathsf{PPI}.\forall\mathsf{PPI}.(\forall\mathsf{PO}.(\neg C \sqcap \neg D) \sqcap \forall\mathsf{DR}.(\neg C \sqcap \neg D) \\ & \qquad\qquad \sqcap\; \forall\mathsf{PPI}.(\neg C \sqcap \neg D) \sqcap \forall\mathsf{PP}.(\neg C \sqcap \neg D)) \\ & \sqcap\; \forall\mathsf{PPI}.X \end{aligned}$$

The intent: each X-node has two PPI-grandchildren colored C and D (the "square"), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive conjunct $\forall\mathsf{PPI}.X$ forces infinite repetition.[3]

One level is satisfiable. Without the recursive conjunct, X has a 3-element model: z PP y PP x, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* (EQ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, and the universals would require $z_2 \notin D$ (firing z_1's $\forall R.\neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$. The $\forall\mathsf{PP}$ conjunct is *not* vacuous—z's neighbors are its PP-successors y, x, on which it fires—but it is satisfied by interpreting y, x outside $C \cup D$; the $\forall\mathsf{PO}/\forall\mathsf{DR}/\forall\mathsf{PPI}$ conjuncts are vacuous, as z has no such neighbors.

Two levels is unsatisfiable. Adding $\forall\mathsf{PPI}.X$: the PPI-child y must itself satisfy X (without recursion), producing a new $C \sqcap D$-labeled element w as a PPI-child of z. But z is a PPI-grandchild of x, and x's universals fire: z's PPI-neighbor w must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \sqcap \forall\mathsf{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall\mathsf{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

Remark 8. The failure is not accidental. In $\mathcal{ALCI}_{\mathrm{RCC5}}$, the only way to create "functional" successors is via PP/PPI chains, but $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{\mathrm{RCC8}}$, the TPP/NTPP distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel's even-odd chain works there but has no RCC5 analogue.

6.3 A concrete domino attempt in $\mathcal{ALCI}_{\mathrm{RCC8}}$

The remark above suggests the TPPI/NTPPI distinction in $\mathcal{ALCI}_{\mathrm{RCC8}}$ might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps

[3] The recursive occurrence of X abbreviates a TBox axiom $X \sqsubseteq F(X)$; no fixpoint constructor is needed. Since the five relations partition the square of the domain, the *universal modality* $[U]D := \forall\mathsf{EQ}.D \sqcap \forall\mathsf{PP}.D \sqcap \forall\mathsf{PPI}.D \sqcap \forall\mathsf{PO}.D \sqcap \forall\mathsf{DR}.D$ is definable, and a finite TBox internalizes: C is satisfiable w.r.t. $\mathcal{T}$ iff $C \sqcap [U]G_{\mathcal{T}}$ is, where $G_{\mathcal{T}}$ conjoins $\mathrm{nnf}(\neg C_i) \sqcup D_i$ over the axioms $C_i \sqsubseteq D_i$ of $\mathcal{T}$. The internalization uses $\forall\mathsf{PO}$, so it does *not* preserve the $\forall\mathsf{PO}$-free fragment of Theorem 10.

the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel's *coincidence obstruction* [9].

The schema. Let Corner be the concept that isolates a node from all non-EQ neighbors:

$$\mathsf{Corner} \;=\; \bigwedge_{R \in \{\mathsf{DC},\mathsf{EC},\mathsf{PO},\mathsf{TPP},\mathsf{NTPP},\mathsf{TPPI},\mathsf{NTPPI}\}} \forall R.(\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral ∀TPPI.∀TPPI universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X \;=\; & \exists \mathsf{TPPI}.\exists \mathsf{TPPI}.(C \sqcap \mathsf{Corner}) \;\sqcap\; \exists \mathsf{TPPI}.\exists \mathsf{TPPI}.(D \sqcap \mathsf{Corner}) \\ & \sqcap\; \forall \mathsf{TPPI}.(\neg C \sqcap \neg D) \;\sqcap\; \forall \mathsf{TPPI}.X. \end{aligned}$$

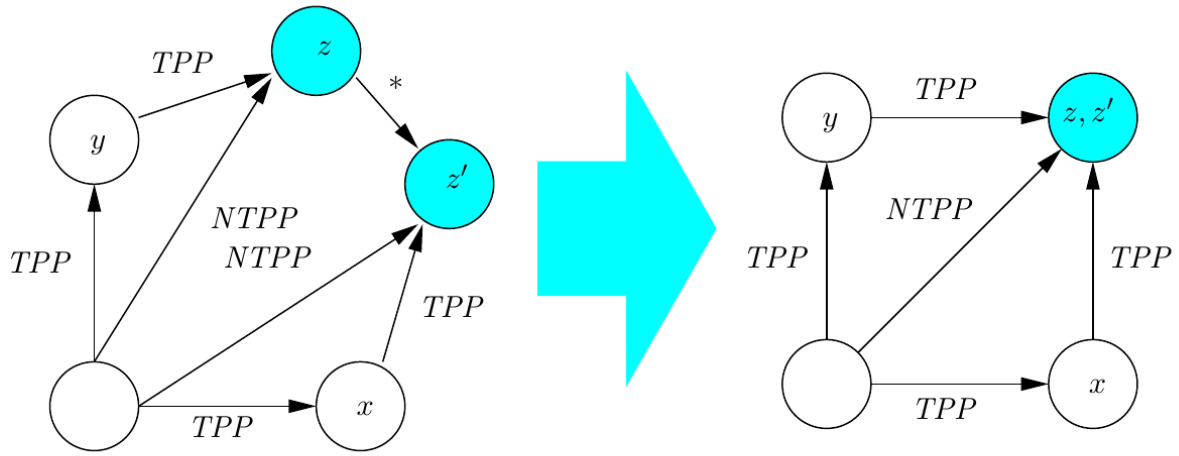
The third conjunct ∀TPPI.(¬C ⊓ ¬D) forces the corner relation $x \to z$ to be NTPPI (not TPPI), using $\mathsf{comp}(\mathsf{TPPI},\mathsf{TPPI}) = \{\mathsf{TPPI},\mathsf{NTPPI}\}$.

Level 1 is satisfiable, with a real direct/indirect distinction. Without recursion, x has TPPI-children y_1, y_2 (the two "direct" positions) and a single corner z reached by TPPI-TPPI chains from both y_1 and y_2. The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \mathsf{Corner}$ and $z_2 \in D \sqcap \mathsf{Corner}$ are mutual non-EQ neighbors only if Corner is violated, so under strong-EQ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: z is NTPPI of x while y_1, y_2 are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §6.2.

Level 2 collapses via the coincidence obstruction. Adding ∀TPPI.X: the TPPI-child y of x satisfies X, producing its own corner $w \in C \sqcap D \sqcap \mathsf{Corner}$ at NTPPI-distance from y, and hence at NTPPI-distance from x via $\mathsf{comp}(\mathsf{TPPI},\mathsf{NTPPI}) = \{\mathsf{NTPPI}\}$. Now x's own corner z and y's corner w are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since z carries Corner, every non-EQ neighbor of z lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-EQ. **All corners across the grid coincide at a single point.** Figure 2 isolates this local "merge attempt" on two successors.

Why the TPPI/NTPPI trick fails. In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\mathsf{comp}(\mathsf{PPI},\mathsf{PPI}) = \{\mathsf{PPI}\}$. Moving the isolation to a local Corner marker and using the non-transitive piece $\mathsf{comp}(\mathsf{TPPI},\mathsf{TPPI}) = \{\mathsf{TPPI},\mathsf{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of {PO, EQ, TPP, NTPP, TPPI, NTPPI} between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The Corner markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §4.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

Remark 9. The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter's L_{RCC8} reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single NTPP-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no "north-of-east = east-of-north" coincidence has to be forced; the geometry of $\mathbb{R}^n$ only has to keep that linear encoding faithful [10]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{\mathrm{RCC8}}$: a pure composition table allows corner points to merge freely, and the TPPI/NTPPI distinction—while useful locally



$* =_{def} \{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$

Figure 2: Wessel's original statement of the coincidence obstruction, reproduced *directly from report7.pdf (2002/2003)*, Figure 11 ("Tightening the grid?"). The two successors z (reached east-then-north) and z' (north-then-east) ought to be the *same* grid corner, but the composition table permits the edge $z * z'$ to be any of $* =_{\text{def}} \{\mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$ (the legend below the graphs): the table *by itself* under-determines the edge, forcing neither a merge nor a split. One *can* nonetheless force the merge at the concept level—clash out all five non-EQ options with universals, and EQ is all that remains—and this is exactly the trap: the very Corner markers that force $z = z'$ at one level, once propagated to build the grid, force *every* corner together and the construction collapses (satisfiable at one level, UNSAT at two—§6.2, Remark 2). So neither option yields a rigid grid: leave the edge loose and the branches never join; force the merge and all corners collapse to a single point. It is this "merge the wrong successors" *dichotomy*—not an inability to force EQ at all—that Wessel identified two decades before this AI-assisted work.

for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{\text{RCC8}}$ decidability noted in [10] and further discussed in [9].

7 Overview of approaches, by technique

We organize the attack by *technique*, in roughly the order tried, not by round or date. Each subsection follows the same shape: the high-level idea; the concrete techniques investigated; where and why they fail (with the examples that made the failure clear); pointers to the papers, probes, and reasoners; and the lessons.

7.1 Naive tableaux and blocking

The idea, and why it should work. The default decision method for a description logic is a *tableau*: to test whether C_0 is satisfiable, try to build a model top-down. Start with one individual required to satisfy C_0; whenever a node must satisfy $\exists R.D$, create a fresh R-successor satisfying D; whenever it satisfies $\forall R.D$, propagate D to all its R-successors; close a branch on a contradiction. Left alone this may never terminate—$\exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$ builds an infinite PP-chain—so one adds

blocking: if a new node's "type" (the set of concepts it must satisfy) repeats that of an ancestor, we stop expanding it and reuse the ancestor's already-built subtree, folding the infinite chain into a finite loop. For ordinary modal and description logics this is sound and complete, and it is the first thing one tries here. Without blocking the method is not lost, only non-terminating: a tableau that keeps the whole complete graph, choosing for every fresh node its relation to every existing node, is sound and complete for this logic by a paper proof (Theorem 18). It answers on every unsatisfiable concept and on every concept with a finite model. What is missing is a sound blocking condition. Our most refined implementation with blocking is the *cover-tree tableau* (§8), which blocks on *exact occurrences* and is incomplete.

Why the total-relation setting breaks it. In $\mathcal{ALCI}_{\mathrm{RCC5}}$ *every* pair of individuals carries a relation, so a model is a complete graph, not a tree. This single fact defeats ordinary blocking in three compounding ways, each of which we hit in practice.

1. **Composition forces edges *into* already-built nodes.** Creating a new node z does not just attach it to its parent; by composition it forces relations between z and *every* existing node. From $x\,\mathsf{PP}\,y$ and $y\,\mathsf{DR}\,z$ the table forces $x\,\mathsf{DR}\,z$ (Proposition 1). So the "tree" is a fiction: the real object is the whole complete graph, and a node's obligations depend on its relations to the entire context, not on its concept type alone.

2. **Dormant universals reawaken blocked nodes.** A restriction $\forall R.D$ lies inert at a node until an R-edge from it appears. But a *forced* composition edge (point 1) can create such an edge *after* the node was blocked, firing the dormant universal, adding D to a node downstream, and possibly *unblocking* it. The cascade of re-qualifications need not terminate, and—worse—may be *unsound* if ignored: the block asserted a repetition that the later edge invalidates.

3. **A fresh witness's relations are forced by the *joint* closure, and can collide with a universal.** Adding d does not leave its relations to old nodes free: they are pinned by *all* triangles at once, and (Remark 2) the intersection of several can fix even a horizontal edge. Take old $a\,\mathsf{DR}\,b$, $a\,\mathsf{PPI}\,c$, $b\,\mathsf{DR}\,c$ and a request $b\,\mathsf{PO}\,d$, $c\,\mathsf{PO}\,d$: this is consistent, but it *forces* $a\,\mathsf{PO}\,d$—from $\mathsf{comp}(\mathsf{DR},\mathsf{PO}) \cap \mathsf{comp}(\mathsf{PPI},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\} \cap \{\mathsf{PPI},\mathsf{PO}\} = \{\mathsf{PO}\}$, an edge a never chose. Now let a carry $\forall\mathsf{PO}.\neg A$ while d must satisfy A: the *forced* $a\,\mathsf{PO}\,d$ fires the universal and closes the branch. That is the real hazard—a value fixed by joint closure, not by any single step, silently contradicting a distant universal (the diagnostic C_{force} below is exactly this shape)—and it is invisible to any check that reads edges one path at a time.

The ladder of blocking conditions we tried. Each rung tracks more relational context, and each is defeated by the phenomena above. The base choice is *where* one may block: the obvious strengthening of ancestor blocking is *anywhere blocking*—block a node against *any* earlier node in the tableau, not only its ancestors, the more aggressive termination device standard for logics with inverse roles. This helps termination but not soundness: it inherits every failure below, since it still compares nodes by a *local* summary while the constraints are global. On top of the where-question we layered progressively richer *what*-summaries. *Subset/type* blocking (repeat of the concept set) ignores relations entirely and is unsound at once. *Triangle* blocking tracks, for each node, the composition-*triangles* it participates in—the relation triples $(\rho(x,y),\rho(y,z),\rho(x,z))$—so that composition constraints are visible; but a triangle is a three-point view, and forced edges from a fourth point (via inverse roles) escape it (`triangle_blocking_ALCIRCC5`, `src/triangle_*.py`). *Quadruple* types lift this to four points to capture those cross-edges (`decidability_via_quadruples_ALCIRCC5`, `src/quadruple_type_check.py`); but the required arity is not bounded—a forced

constraint can span a chain of any length. *Profile* blocking records a node's type together with the multiset of its relations to a bounded context (`src/profile_blocking_*.py`); but the context that actually determines future obligations is not bounded a priori—this is F6 in disguise. *Double* blocking (the $\mathcal{SHIQ}$ device: block only when a node *and its predecessor* jointly repeat an ancestor pair, which is what one needs for inverse roles) addresses point 2's predecessor sensitivity but not the forced-edge and extension problems (`src/double_blocking_test.py`).

The examples that settle it. Two families are diagnostic and recur throughout the project. The forced-composition UNSAT concepts, e.g. $C_{\text{force}} = \exists \mathsf{PP}.(\exists \mathsf{DR}.A) \sqcap \forall \mathsf{DR}.\neg A$: here $x\ \mathsf{PP}\ y$ and $y\,\mathsf{DR}\,z$ force $x\,\mathsf{DR}\,z$, so $\forall \mathsf{DR}.\neg A$ kills the required $z\!:\!A$; a procedure that ignores forced edges wrongly *accepts* it. Dually, the strong-EQ PO-*loop* $C \sqcap \exists \mathsf{PO}.\exists \mathsf{PO}.C \sqcap \forall\{\mathsf{PO}, \mathsf{DR}, \mathsf{PP}, \mathsf{PPI}\}.\neg C$ is satisfiable *only* via a two-element model closed into a symmetric PO-loop under EQ-identity; a procedure that distinguishes occurrences by type alone wrongly *rejects* it (this is precisely the blind spot of the quasimodel oracle, §7.2). Further diagnostics—`inter_witness_counterexample_ALCIRCC5`, `typed_patchwork_counterexample_ALCIRCC5`—exercise the same seams.

Lesson. In a total-relation logic, sound termination cannot be had by *local* blocking, because the relational completion is *global*: what may be reused depends on the whole graph, and forced edges plus dormant universals make "the whole graph" irreducible to any fixed-arity local view. The way out is to stop trying to block a complete graph and instead *separate* the tree skeleton from the global relational completion, controlling the latter by the RCC5 patchwork property. That is the split-forest idea (§7.4).

7.2 Finite-model and reduction attempts

The idea. If one could force models to be finite, or finitely presented, one could enumerate candidates and check them. Four concrete lines: build a *quasimodel* (a set of types with a consistency relation) and eliminate types that cannot be realized; collapse infinite PP/PPI chains into finite *clusters or loops*; take a *two-tier quotient* (one "kernel" per recurrent chain descriptor, plus finitely many designated witnesses); or *encode* satisfiability into monadic second-order logic (MSO) over trees and invoke Rabin's theorem.

Where they fail. The quasimodel / type-elimination procedure (`decidability_ALCIRCC5`, `src/alcircc5_reasoner.py`) is *retracted*: its elimination rule for PO-demands ($Q3$) is *anti-monotone*—eliminating one type can invalidate the witness that justified another—so the fixpoint over-eliminates and the procedure is *incomplete* (though not unsound). Concretely it is wrong on the PO-loop above: it reports UNSAT for a satisfiable concept. The finite-cluster and MSO routes both stumble on the same rock: collapsing a PP-chain is fine (Lemma 4), but the *horizontal* width across clusters is not bounded, and MSO cannot define an unbounded horizontal matching—so a correct encoding again presupposes a width bound. Every "make it finite" variant reduces to producing a computable live-width bound, i.e. to F6 (Definition 5).

The win. One fragment escapes unconditionally. In the two-tier quotient, PO is the *only* relation whose recurrence along a chain needs an "all-phase" safety condition—because, unlike DR/PP/PPI, a PO-witness at one chain position need not remain a PO-witness at another (the compositions $\mathsf{comp}(\mathsf{PP}, \mathsf{PO})$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO})$ are not singletons). If the concept contains *no* $\forall$PO subformula, that safety condition is vacuous, the quotient is finite, and the width is bounded by a computable function of the concept.

Theorem 10 (Decidable fragment)**.** *Concept satisfiability for the* ∀PO*-free fragment of* $\mathcal{ALCI}_{\mathrm{RCC5}}$ *—concepts whose negation normal form has no* ∀PO *subformula, but which retain* ∃PO*,* ∀DR*,* ∃DR*, and all vertical modalities—is decidable.* (Machine-checked for the whole fragment: `decidableSat_cone` in `formal/POFreeLift.lean`, and on raw input and under set semantics `decidableFSat` and `decidableSetSat`; §7.3. The two-tier argument of `two_tier_quotient_ALCIRCC5` also yields a live-width bound $K(C_0) = (1+m)^2\, 2^{2^m}$, $m = |\mathsf{cl}(C_0)|$, which is theorem-level and not formalized. The statement concerns concept satisfiability; the fragment is not closed under negation.)

The fragment is genuinely expressive—one may still say "overlaps some A" ($\exists\mathsf{PO}.A$), "all disjoint things are B" ($\forall\mathsf{DR}.B$), and impose arbitrary part-of nesting—so this is the project's strongest unconditional $\mathcal{ALCI}_{\mathrm{RCC5}}$ result.

Remark 11 (A gap in the proof of Theorem 10, found and repaired 2026-08-28)**.** The two-tier completeness argument defined the relation between two kernels as a double limit $\lim_{i,j\to\infty} \rho(d_i, d_j)$, justified by its external stabilization lemma "applied twice". That lemma is one-sided—it fixes an element and varies the chain—and stabilizing every row and every column is strictly weaker than a double limit, which need not exist: in $\mathbb{Z} \times \{0,1\}$ ordered by first coordinate, every row and column stabilizes while every tail carries PP, PO and PPI. The step is repaired by replacing the limit with a *simultaneous finite rectangle* (finitely many chains admit segments, beyond any bound, with equal-type endpoints and one constant relation per pair), which is what the quotient needs since it reads only finitely many phases. The repair is machine-checked (`fused_kq_all`). The theorem and its bound are unaffected.

Re-examined, and its soundness core certified (2026-07-18). A cold review (§9) rightly flagged that the rationale above leans on the "horizontal freedom" intuition that Remark 2 corrects. We therefore re-examined the fragment; it holds. The soundness crux—the *chain-unfolding lift* (a recurrent chain's kernel, unfolded under a constant interface, stays a valid RCC5 frame)—is now *kernel-checked* in Lean 4 (`formal/POFreeLift.lean`: `lift_cc`, `unf_is_frame`; axioms `propext`/`Quot.sound`, zero `sorry`), and it enforces *full* composition-consistency (every triangle closed), which *subsumes* multi-path PO forcing—so the construction does *not* assume PO edges are free, closing the review's concern. Two independently written prototype reasoners moreover agree on every ∀PO-free concept tested (`wp86`, `wp87`: 244 random + curated, zero mismatches; neither reasoner's known incompleteness can arise here, since the quasimodel oracle's blind spot and the tableau's failure on K_2, §8, both require ∀PO). A byproduct worth noting: the abstract-semantics core needs *no* patchwork or compactness—a composition-closed, converse-coherent, strong-EQ network already *is* a model, so `unf_is_frame` delivers an actual (infinite) frame outright. What remained for full certification at that point was the model-of-C_0 layer (Hintikka demand/universal satisfaction) and the completeness extraction carrying the $K(C_0)$ bound. So as of that review Theorem 10 stood as *strongly supported with its soundness core machine-certified*—firmer than before, though not yet end-to-end formalized; the following note records how the certification was subsequently carried through to the decision procedure itself.

Certified as a decision procedure (2026-08-06; superseded 2026-08-29 — see §7.3). The certification has since been carried through to the decision itself: in `formal/POFreeLift.lean` (then ∼13,600 lines, zero `sorry`; axioms `propext`/`Classical.choice`/`Quot.sound`, the choice confined to erased termination proofs, so the procedure is genuinely computable) a `Decidable` instance for satisfiability of C_0 is now machine-checked—*generally* for the fragment's three *pure* quadrants (horizontal, ascending-vertical, descending-vertical) and on a witness for the *mixed* quadrant. Remark 12 records *that* construction: the finite certificate, the four quadrants, and the constructive

round-robin uniformization (`rr_covers`) that turns the full logic's open "W2′" oracle into a kernel-checked theorem once ∀PO is removed. The mixed quadrant was closed, and the whole fragment re-certified by a different route, on 2026-08-29 (§7.3); that route has since been through three independent cold reviews. None of it bears on the full logic—F6 is forced *by* ∀PO, which the fragment removes.

Lesson. Finiteness is the wrong target: models are legitimately infinite (unbounded PP-towers are satisfiable, and infinity can be forced without any vertical existential, §7.8). The right target is a finite *presentation* of a possibly-infinite model.

7.3 The ∀PO-free fragment, certified: the cone scheme (2026-08-29)

This is the route that certifies Theorem 10, and the one to read first. An earlier route, recorded in Remark 12 below, certified three of the fragment's four quadrants generally and the fourth only on a witness; its general mixed extraction was never completed, because two rounds of cold attack refuted the architecture it needed (a finite node set reused by blocking and *borrowing* witnesses) by exact finite countermodels. It was replaced.

The route now certified is a *cone scheme*, designed by GPT-5.6 Sol [73]: a finite control graph whose states are pairs (support type, upper bound on the types of strict PP-predecessors); a monotone, reductive elimination to a greatest fixed point; and a *fresh-occurrence* unfolding in which no element is ever reused, so that the frame conditions which defeated the previous architecture hold by construction rather than by checking. That unfolding has exactly the shape of Wessel's split-forest representation (§7.4): a forest of part-of trees in which no region is shared between two demands, with the horizontal relations read off afterwards—here built directly rather than extracted from a model. What of the idea the certified route keeps, and what it drops, is recorded in §7.4; the companion paper gives the relationship a section and a figure [76].

The *completeness* direction deserves its own sentence, because it is where this route departs furthest from everything before it. Earlier architectures read completeness as *extraction*: given a model, build a finite certificate out of it. That construction is where the width budgets live, and where four successive disciplines were refuted. Here nothing is extracted. The signature space is fixed in advance from C_0 alone; a model point's signature is *read off* it — its type, together with the types of its strict PP-predecessors; and the set of signatures a model happens to realise is shown to survive its own elimination, because for each demand the model simply hands over a witness. That makes it a *post-fixed point*, and since the elimination is monotone and reductive, any post-fixed point inside the starting set lies inside the greatest fixed point. Finiteness costs nothing — the signature space was finite before any model was mentioned — and the model's entire contribution is that one lemma.

This is also exactly the trap the retracted quasimodel route fell into (§7.2): its elimination was *anti-monotone*, so no such argument was available to it and its fixpoint over-eliminated. The elimination here asks only that each demand *have* a target in the current set, which is monotone — and that difference is what makes the greatest-fixed-point argument possible at all. It was checked before anything was built on it, precisely because this project had made the opposite mistake once.

The design also explains the fragment boundary better than the syntactic statement does, and the explanation is worth recording because it says what ∀PO-freeness is *for*. A composition cell containing PO can always be discharged by putting the pair at PO — unless some formula can object to a PO edge, and the only formula that can is $\forall \mathsf{PO}.D$. So on this fragment the maximally-PO unfolding, which declares only what a birth forces and defaults every remaining pair to PO, is *always* a model; no joint realizability across a signature's several demands is ever required, because

every cross-relation between distinct children is the residual and the residual costs nothing. What is lost when $\forall\mathsf{PO}$ returns is therefore not one case of one induction but the residual default itself — which is the same boundary Remark 2 locates from the side of joint forcing. The capstone is

$$\texttt{decidableSat_cone} : \texttt{POFree}\ C_0 \to \texttt{Decidable}(\texttt{Satisfiable}\ C_0),$$

with `POFree` the only hypothesis, covering all four quadrants. Its completeness half needs no fragment hypothesis at all, which yields a sound UNSAT test for the *full* logic. That test is one-sided and, in its first form, was also *empty*: a second cold review measured its verdict invariant under deleting every $\forall\mathsf{PO}$ from the input, and that deletion always lands in the fragment, so it refuted nothing the fragment procedure did not already refute. The cause was a dropped clause — PO is its own converse, so $\forall\mathsf{PO}$ bodies cross a PO edge in both directions — now restored, with a machine-checked concept that the repaired test refutes and whose deletion is satisfiable. A second $\forall\mathsf{PO}$ gap remains and is architectural: the scheme exploits only compositions forced to a singleton, so obligations arising from disjunctive entailments are out of its reach.

Our first cold review of the cone scheme had already exhibited the gap, and the fourth [75] added a second witness and a guard for anyone who tries to close it. In $C_{\text{bad}} = \exists\mathsf{PPI}.(\exists\mathsf{PP}.A) \sqcap \forall\mathsf{PP}.\neg A \sqcap \forall\mathsf{PPI}.\neg A \sqcap \neg A \sqcap \forall\mathsf{PO}.\neg A$ the root and the A-point share a part, so their relation lies in $\mathsf{comp}(\mathsf{PPI},\mathsf{PP}) = \{\mathsf{EQ},\mathsf{PP},\mathsf{PPI},\mathsf{PO}\}$, and the root's four constraints exclude every option (this is the first review's example). In $C_{\text{joint}} = \exists\mathsf{PP}.(\forall\mathsf{PPI}.A \sqcap \forall\mathsf{PO}.A) \sqcap \exists\mathsf{PO}.\neg A$ the two witnesses stand in a relation from $\mathsf{comp}(\mathsf{PPI},\mathsf{PO}) = \{\mathsf{PPI},\mathsf{PO}\}$, and either value forces A onto the $\neg A$-witness. Both concepts are unsatisfiable and both are accepted by the full-logic test; we confirmed this on a transcription of the control layer, by exhibiting three-signature families the elimination cannot remove. By contrast $\forall\mathsf{PO}.A \sqcap \exists\mathsf{PO}.\neg A$ is refuted in the first round, by the restored clause. And the obvious repair—require independently generated witnesses to be pairwise compatible across PO—is ruled out by the satisfiable $C_{\text{sat}} = \exists\mathsf{PP}.(A \sqcap \forall\mathsf{PO}.A) \sqcap \exists\mathsf{PP}.\neg A$, whose two witnesses must be nested rather than overlapping. A construction for the full logic has to reason about the relations among contemporaneous witnesses, not only along the edges that created them. That includes PP and PPI between witnesses that were both generated horizontally: in K_2 (§8) two PO-witnesses of one region must be nested, which is C_{sat} moved sideways. And pairwise reasoning is not enough: in C_{cyc} (§7.8) every two of three witnesses are compatible and the three together are not.

Two scope limits recorded in earlier versions of this remark have since been closed. The artifact now carries a *raw* syntax with negation, a normalization, and a preservation theorem in both polarities, so no pre-normalized input is assumed (`decidableFSat`); the polarity flag also exposes that the fragment condition on raw input is *no* $\forall\mathsf{PO}$ *positively and no* $\exists\mathsf{PO}$ *negatively*, since an $\exists\mathsf{PO}$ under a negation becomes a $\forall\mathsf{PO}$. And satisfiability over the abstract composition-table semantics is proved *equivalent* to satisfiability over families of non-empty sets under the naive relation — equal, proper subset, proper superset, disjoint, otherwise overlap — with no fragment hypothesis, so the equivalence covers the whole logic (`satisfiable_iff_set`); on that side the composition table is a theorem about sets rather than an assumption. Regions there are arbitrary non-empty sets. By Lutz and Wolter's Theorem 24 (§4.4) the same satisfiability problem is also that of substructures of regular closed regions in $\mathbb{R}^n$, so the result holds verbatim in that topological reading, though not for full structures containing every regular closed region; and the set equivalence itself is a machine-checked special case of their Theorem 23.

What the result is *about* should be stated as exactly: satisfiability of a single concept. The fragment is not closed under negation—$\neg\exists\mathsf{PO}.D$ normalizes to $\forall\mathsf{PO}.\neg D$—so a subsumption $C \sqsubseteq D$ is decided only when $C \sqcap \neg D$ stays in the fragment, that is, when C has no positive $\forall\mathsf{PO}$ and no negative $\exists\mathsf{PO}$, and D no positive $\exists\mathsf{PO}$ and no negative $\forall\mathsf{PO}$. Internalizing a general TBox introduces $\forall\mathsf{PO}$, so TBox reasoning is outside the result.

What remains is a limit of complexity, not of correctness: the procedure is *decidable but not runnable*. The signature space is doubly exponential, and the procedure cannot be evaluated on any concept discussed in this paper.

Three cold reviews (all 2026-08-29) built the artifact, re-derived the composition table independently, and none found a counterexample. The first cross-checked 4,000 random concepts against exhaustive finite-model search and found two inverted comments and a complexity bound understated by an exponential. The second attacked completeness and *ran* the elimination — 6,268 concepts against exhaustive model search, no satisfiable concept rejected — and found the dropped PO clause above, together with an overclaim about what the full-logic test certifies. The third attacked soundness, ran the unfolding on labels drawn from the actual greatest fixed point, and found no defect in the development: its five findings were all in the project's own probes and prose. All are corrected. A fourth review (GPT-5.6 Sol, 2026-09-09) [75], of the Lean sources and this overview together rather than of a build, located every capstone of the decision chain in the source and found no defect in it; its findings concerned how this paper stated its claims, and are corrected in this version. A fifth (Astra 6, 2026-10-07) [74] audited the argument end to end and found *no gap* — the first review of this result with nothing to correct — and proved an extension that is now formalized: on the fragment $\mathcal{D}$ of concepts with no $\exists$PP, $\exists$PO or $\forall$DR — $\forall$PO *permitted* — erasing every $\forall\mathsf{PO}.D$ to $\top$ preserves satisfiability, so composing the erasure with `decidableSat_cone` decides $\mathcal{D}$ (`erase_equisat`, `decidableSat_dfrag`): the first certified decidable slice containing $\forall$PO, by a PO-empty witness forest whose disjointness is incomparability. In $\mathcal{D}$ every $\forall$PO is, by this very theorem, erasable. Decided slices in which a $\forall$PO constraint can change the answer exist too, without a certified proof: the coarser logic $\mathcal{ALCI}_{\mathrm{RCC3}}$, decided in 2003, is one when read inside $\mathcal{ALCI}_{\mathrm{RCC5}}$, and a second was obtained in November 2026 by import (§7.8). The extension is calibrated by its own counterexamples — each exclusion is necessary for the erasure argument, and C_{bad}, C_{joint} lie outside $\mathcal{D}$, so the architectural gap above is untouched. The same review's order-level normalization — for a fixed order the largest admissible disjointness relates exactly the pairs with no common lower bound, realized by principal down-sets (`odMax`, `odMax_largest`, `pdown_*`) — splits overlap into *forced by a represented common part* versus *optional*, with a concept (`wp136`) satisfiable only when some overlap has *no* represented common part; that split is new vocabulary for the full-logic problem and feeds the pair-matrix mosaic lead of §10.

The honest label, and it differs from this paper's standing one: the $\forall$PO-free fragment is *machine-certified* — the Lean kernel checks the decision procedure with fragment membership as its only hypothesis — and its formalization's adequacy has been examined by three independent cold reviews without a counterexample. The project's blanket *strongly supported, not certified* applies to the *full logic*, and to the two unformalized arguments of §7.2; it does not apply to this result, and earlier drafts of this section were wrong to carry it over.

Remark 12 (The superseded first route: four quadrants and a round-robin kernel)**.** Everything here is in the artifact and every theorem named is still proved; what changed is which construction discharges Theorem 10. This route certified a `Decidable` instance *per quadrant*, generally for the three pure ones and only on a witness for the mixed one; its general mixed extraction was never completed, and two rounds of cold attack later refuted the architecture it needed. The route that certifies the theorem is the cone scheme of §7.3, which uses none of the machinery below.

It is recorded because the material is not superseded as *mathematics*: the four-quadrant taxonomy is a real classification of how demands interact in the fragment, and the round-robin device answers the uniformization question the full logic leaves open. It is also the clearest place to see *why* the fragment is tractable at all.

The certificate, and the decision loop. A certificate is a *multi-tier* object (`MultiTier`):

finitely many *external* neighbour-types together with finitely many recurrent *kernels*—chain generators, read as "the pattern continues like this" and unfolded into infinitely many distinct elements, never bent into a finite quotient. Each is decorated with Hintikka bookkeeping: which existential demand each node discharges, which universal obligations propagate along which edges, and a declared RCC5 value for every relevant pair, required composition-closed. A total Boolean checker `mtAcceptB` accepts a certificate exactly when all of this holds, and acceptance is *sound*—a passing certificate unfolds, through a Hintikka labelling and truth lemma, to a genuine RCC5 model of C_0 (`mtAcceptB_sound`). Certificates are finite list data with a total Gödel decoder (`FinMT`), so the decision procedure is immediate (`decidableSat_of_codes`): enumerate the finitely many codes a concept admits, run the checker, answer SAT iff one passes. All the mathematics is in emitting the right code for a satisfiable concept; the algorithm itself is a bounded for-loop.

Four quadrants. A ∀PO-free concept's existential demands fall into four combinations, each with its own certificate shape:

- *Horizontal* (∃DR, ∃PO, ∃EQ only): a node's label is its model type *truncated at the concept's modal depth* (`mtk`). The frame is universally closed in both directions (disjointness is self-converse, so a ∀DR obligation propagates identically each way), and depth-truncation terminates the recursion, so finitely many labels exhaust the demands (`decidableSat_hfrag`). No kernel is needed—within the fragment the horizontal world is bounded outright. (With ∀PO it is not: §7.8 gives a concept with only horizontal existentials and only infinite models.)

- *Ascending vertical* (∃PP): a single recurrent kernel generating one PP-chain. Along any ascending chain the Hintikka type recurs with a *bounded period* (an infinite-pigeonhole segment lemma, `mty_segment_bounded`); that period *is* the kernel (`decidableSat_vtower`).

- *Descending vertical* (∃PPI): the exact dual. The certificate carries the chain's direction as a single bit, so the whole encoding and decision side is direction-agnostic and the ascending development transports verbatim (`decidableSat_vtowerRRI`).

- *Mixed* (∃PO *and* ∃PP): the merged certificate `mixCert_ok`—the first that carries *both* nontrivial externals *and* a kernel. Each ∃PO demand is served by a direct external neighbour, each ∃PP demand by the kernel, and every cross-obligation living on a PO-edge is *vacuous*, because no ∀PO clause occurs anywhere in the fragment's closure (`mty_no_all_po`). Its encoding and a complete decision are certified on a concrete witness (`decidableSat_Cmix`).

This route's keystone: a constructive uniformization. (The cone scheme does not need it; its own keystone is the residual-PO default of §7.3.) One recurrent kernel must often serve *several* distinct ∃PP demands at once, and on the full logic this is exactly the feared uniformization property W2′: can a single repeating interface discharge every demand without two of them colliding on a horizontal edge? In the fragment the answer is *constructive*. The kernel serves the demands in *round-robin* over its recurrent period—at period position n it discharges the n-th demand, drawing the witness from whatever phase carries it—so across one full period *every* demand is met (`rr_covers`). This turns W2′ from an oracle into a kernel-checked theorem, and it works for one precise reason: with ∀PO absent, the horizontal edges the kernel emits at successive steps are *unconstrained*, so the phases never conflict and the round-robin schedule is always realizable. That is the exact sense in which removing ∀PO lets the cross-relations coordinate for free.

Why the boundary is exactly here, and the honest labels. The construction stops precisely at ∀PO-freeness because F6—the full logic's open live-width property in the principal certificate architecture—is *forced by* ∀PO: it is a ∀PO obligation that can watch a kernel's emitted horizontal edges and so demand a bounded, coordinated interface across phases. Remove that one

constructor and the coordination F6 abstracts becomes free, which is what every device above cashes in. Scope of *this route*, stated exactly: the three *pure* quadrants are certified *generally*, each with a non-vacuous satisfiable witness (`Cvert`, `Ccar`, `Calt`, `Clin`, `Cdesc`); the *mixed* quadrant's certificate, encoding, and decision are certified only on a witness. Its general extraction was never completed; §7.3 above is what replaced it.

Correction (2026-08-28). An earlier version of this sentence called that extraction "scoped, not open". That was an overclaim and is withdrawn: two rounds of cold attack have since refuted the extraction architecture it referred to—a finite node set reused by blocking and borrowing witnesses—by exact finite countermodels, and the formalization has been restarted on a different architecture (a finite control graph with a fresh-occurrence unfolding, in which no node is reused). This concerns the *certification* of Theorem 10, not the theorem, whose proof is the two-tier quotient and is independent of that route; see Remark 11. The certified route is no longer unreviewed: three independent cold reviews (2026-08-29) attacked its trusted base, its completeness direction and its soundness direction, and none found a counterexample or a defect in the fragment result. The two *arguments* of Theorem 10 remain unformalized, and the standing ledger presumption (§9) continues to apply to them, and to every full-logic claim in this paper.

Why it failed, and what that generalizes to. The four disciplines tried in succession were not four unrelated mistakes; they were four attempts at one impossible thing. Reusing a node imposes a *global* condition on the object being built — the order induced by the reused edges must stay acyclic, and the relations those edges force must be *jointly* realizable — while every discipline proposed was a *local* test: per edge, per node, per type. A local test cannot see what it is not looking at, and the refutations say so one axis at a time. Per-edge safety removes self-loops and nothing longer: the countermodel has two individually safe fallbacks forming a four-cycle. A key computed from an occurrence's downward cone cannot separate occurrences whose obstruction is *horizontal*. A class top defined by type alone drags one member's disjointness witness onto another member's cone. This is the same pointwise-versus-joint shape that had already broken three earlier architectures here, which is why the attempt was abandoned rather than repaired: the error was proposing another local discipline, not the particular disciplines proposed. Standard *blocking* is this story in its most familiar form — it is precisely the local test, "two occurrences carry the same type, so let one stand for the other" — and it fails here for exactly the same reason: type equality cannot see whether the reuse closes a cycle in the order, or places the blocker's disjointness witness inside the blocked node's cone, which is the clause (`ltNotDj`) that broke this project's last and most careful attempt at it.

What §7.3 does instead is not a better test — it removes the obligation. Nothing is reused, so the order is a tree closure, irreflexive because paths cannot return rather than because a check forbids it; and a DR birth opens a new vertical component, so the order and the disjointness relation can never meet. The global conditions hold *structurally*. There is no test to get right, because there is no way to violate them. That is the general lesson worth carrying out of this project: when a construction keeps failing a global consistency condition, the productive move is not a sharper local check but a shape in which the condition cannot be expressed as a failure.

7.4 The split-forest idea

Idea (Wessel's, on a whiteboard). The split-forest decomposition was Wessel's contribution, sketched on a whiteboard (Figure 3) and handed to the AI assistants to make precise. The insight: even though a model is a complete graph, one can *present* it as a *forest of trees*—one tree per connected component of the vertical (PP) order—by *splitting* each node that is a common part

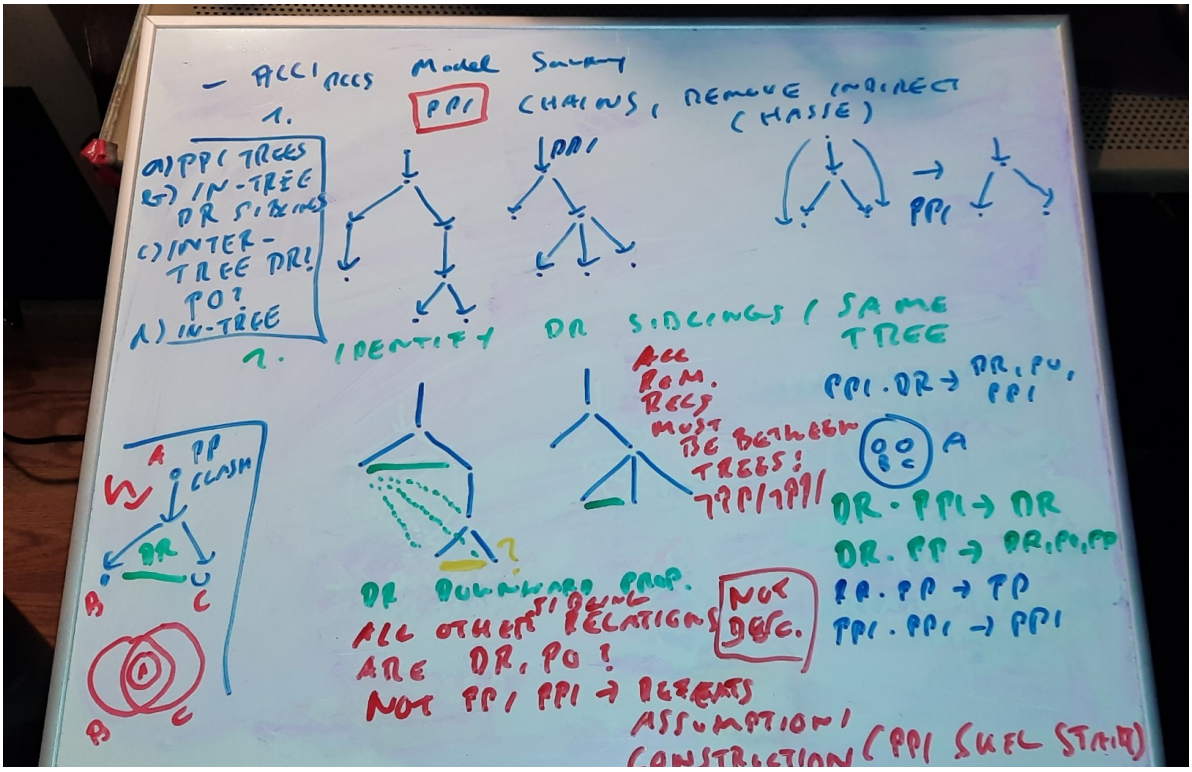


Figure 3: The original whiteboard sketch of the split-forest model decomposition (M. Wessel, April 2026): the intuition that a complete-graph RCC5 model can be presented as a forest of trees by splitting join nodes into occurrence copies (re-identified by a quotient) and completing the relational graph by patchwork. This idea—Wessel's—is the semantic foundation that every subsequent proof route consumes, and the one component no cold review ever broke.

of several superparts into separate *occurrence copies* (which are *not* EQ-related model elements—under strong EQ that would mean identity—but occurrences of an occurrence cover, re-identified afterwards by an explicit quotient), so the tree skeleton becomes explicit, and then *completing* the missing (horizontal) edges by the RCC5 patchwork property. Tree-like presentations of infinite models are the classical gateway to decidability: once a logic has a tree-model property, automata or type-elimination usually finish the job. The bet was that this setting, despite its complete relational graph, still admits such a presentation.

The normal form (Theorem A). Made precise, the construction is a chain of thinning and splitting operations that every satisfiable concept survives: *witness thinning* (keep one witness per demand) → *generated containment covers* (organize witnesses by the vertical order) → *split copies* for incomparable superparts → *side-interface mosaics* (the horizontal relations at each interface, completed by patchwork) → a *typed*-EQ *quotient* (re-identify the split mates by exact occurrence). The output is a witness-generated, tree-like split-forest model. This normal form is the one component found sound by *every* cold review across the project—the semantic foundation everything downstream *consumes* rather than replaces (`trees/`, `cover_tree_tableau_ALCIRCC5`).

Vertical skeleton, horizontal completion. The presentation enforces a strict division of labour between the two axes, and that is what makes it a *forest* rather than an arbitrary graph. Every *tree*

edge is vertical: a parent-to-child link is a PP/PPI (part-of) step, so each tree is literally a branch of the containment order. Every *non-tree edge*—between siblings, cousins, or nodes in different trees—is *horizontal* (DR or PO). A genuine vertical relation that happens to hold between two nodes not yet on one ancestor line is never left lying across the forest: it is *re-oriented* into the tree (the smaller region is made a descendant of the larger), so that once the splitting is finished the only edges the trees do *not* already display are horizontal ones. This is exactly why the completion problem is a purely *horizontal* one—and why PP/PPI never appear as edges to be guessed at an interface. This is a statement about presenting a *given* model. A search procedure does not have the model: it must decide, for two witnesses generated horizontally from one region, whether they are siblings or nested, and a concept can force the nested case (K_2, §8). Treating such witnesses as siblings is the defect found in the cover-tree tableau.

Half-determinism (Proposition 3) then does most of the completion for free. Once a DR edge is placed between two subtrees, *every* pair of descendants across them is forced to DR—all shadows, none of them live—so a DR crowd of any width is generated by a few high DR edges and pushed downward deterministically, costing a finite presentation almost nothing. What resists this is precisely the residue: PO (never forced by a single step, and not downward-propagating, Proposition 1) and the *upward* direction of disjointness (a superpart of a disjoint region is unconstrained). The genuine branching a finite presentation must record—the live width (Definition 5)—is this horizontal residue and nothing else, which is why F6 is a statement about horizontal crowds alone.

Why it does not, by itself, decide. The split forest gives the *shape* of models; it does not give a terminating *search*. Turning "a tree-like model of this shape exists" into "an algorithm can decide whether one exists" requires a *finite abstraction* of the (still infinite) split forest—and that finite-abstraction problem is exactly where the next two families live, and where the difficulty concentrates.

What survived, and how (a reflection, 2026-08). It is worth recording which parts of the whiteboard idea are still standing in the one place where a decision procedure was actually certified (§7.3), because the answer is unusually clean: *the shape survived and almost none of the machinery did.*

The certified model is the fresh-occurrence unfolding, and it *is* a split forest, in a fairly literal sense. Its elements are demand words rather than nodes, so two occurrences of the same type are different elements—that is the splitting, and this route never undoes it. Its order relation is generated by exactly two moves, "prepend a PP birth" and "strip a PPI head," so the skeleton is purely vertical; the horizontal relations are not part of it but are *read off* afterwards, DR from the downward closure of the DR births and PO as the residual. And the forest is genuinely a forest: a function stripping leading vertical births returns each occurrence's component root, an order edge provably preserves it while a DR birth provably changes it. One tree per connected component of the PP order, cross-linked horizontally: the sketch, unchanged.

Where in the certification each half of the idea is used, since the two directions use it very differently. On the **soundness** side the split forest is what the procedure *builds*: `coneScheme_sound` hands the truth lemma the fresh-occurrence unfolding, and that object is a split forest in the literal sense above — vertical skeleton, one tree per component of the PP order, horizontal relations read off afterwards. Every model the procedure ever exhibits is of that shape.

On the **completeness** side it is *not* a hypothesis at all. `coneScheme_complete` takes an arbitrary `RCC5Interp` — any model whatsoever, tree-presented or not — and this matters, because a completeness proof that required split-forest models would have to produce one, which is exactly

the extraction step that four architectures died on. What appears there is not the shape but a summary that fits the same picture: a point is summarized by its own type together with the types lying strictly below it in the PP order (`sigCone` = support type, then down-spectrum)—its vertical position, read off any model rather than imposed on it. That is a correspondence, not a documented design step: the cone-scheme plan does not mention split forests, and a type together with the types below it is a standard invariant for logics over a transitive relation. So the split forest is where the certified procedure's models *live*; it is not a condition the certification asks of the input.

That is worth spelling out, because "not a hypothesis" reads easily as "not used," and here it is the opposite. The idea does two jobs in the certified development, and matches a third. It is the shape the soundness side *constructs*, and the two frame conditions that defeated the previous architecture — irreflexivity of the order, and its disjointness from the disjointness relation — hold structurally only because that shape is a forest of vertical trees: an order edge cannot leave a component, and a DR birth opens a new one. It is the content of the normal form (Theorem 15), which is what replaced the patchwork appeal and makes composition closure free. And what a signature records—a type together with the types below it, which is to say vertical position—fits the same picture: the transition conditions push cones along PP and PPI and ask nothing of PO. That third point is the correspondence noted above, not a documented lineage. What changed is the idea's *job*, not its importance: the earlier routes needed it as a property to be *established* of an arbitrary model, which is the extraction problem they were refuted on, whereas here it is a shape to be *built*, which costs nothing.

What did *not* survive is everything built to construct and control such a forest. There is no patchwork appeal—composition closure is derived from the ordered-disjoint axioms (Theorem 15), which is a certified general fact about RCC5 rather than a citation. There is no quotient—the whiteboard description ends "re-identified afterwards by an explicit quotient," and that step simply never happens, because nothing needs to be recovered: any model will do. And there is no witness thinning, no generated containment cover, no side-interface mosaic and no $K(C_0)$ width accounting; the finite object is the control graph of signatures, and the width question never arises.

The honest reading is not that the machinery was wasted. It is that three successive architectures were each refuted at the point where they tried to *build* a finite presentation of such a forest and keep it consistent, and the move that finally worked was to stop building one—to let the syntax's own demand words be the forest, and check the labels against it. That move is available only once ∀PO is removed, for the reason §7.3 gives: only then is the residual default free. For the full logic the finite-abstraction problem of the preceding paragraph is still there, and F6 is still exactly its residue. The fresh-occurrence unfolding itself cannot serve there: it never makes one horizontally generated witness a part of another, which K_2 and the spiral concept require (§7.8). What the fragment shows is that when the abstraction problem *dissolves*, what is left is the whiteboard picture and nothing else.

7.5 The no-automata certificate, and the automata route

The hand certificate (rounds 2–10). The first attempt at a finite abstraction was a hand-built *certificate*: a finite object whose validity (checked by local conditions on converse, composition, universal safety, request discharge, and residual frontiers) was meant to be equivalent to satisfiability (`gpt5.5_round2/`, `gpt5.5_round4/`). Across ten rounds it worked only by importing, piece by piece, machinery that is really *automaton-shaped*: the conditions that keep a certificate finite while a model is infinite are "request-closed cycles" (a demand and its witness recur), which behave exactly like Büchi/parity acceptance, and "finite product-state exhaustion," which behaves like reachability in a product automaton. Cold reviews found a succession of gaps—an infinite ancestor PPI-tower the

construction missed (D-1), then its dual descendant PP-tower, then boundary non-determinacy, then (twice) that "finite checkability" was asserted rather than proved. The honest reading is that the hand certificate was an automaton written informally; so the project made the automaton explicit.

The parity-automaton route (rounds 11–12). This organizes the argument into three theorems. **Theorem A**: the split-forest normal form (above). **Theorem B**: a *finite abstraction* (frontier/pair/triple summaries of the split forest) captures saturated split forests up to satisfiability of all closure formulas. **Theorem C**: a two-way alternating parity *tree* automaton accepts exactly the valid abstractions, and its emptiness is decidable (Vardi). The automaton *consumes* the split forest; it does not replace it. Theorem B is the single load-bearing keystone. Cold reviews (the 5th and 6th) found its global-coherence step *assumed, not proved*: it derived a globally consistent RCC5 labelling of an infinite tree from purely local checks, via a "confluence law" that was never verified. Round 12 rebuilt the abstraction on *patchwork bags* (finite complete atomic networks that agree on overlaps), so global consistency follows from the citable RCC5 *patchwork property* (Renz–Nebel), and the countable completion by a compactness argument over the finite relation alphabet; eventualities are decided by two-way emptiness. A 7th (hot) review then found a residual *Shadow Amalgamation* gap in the truth lemma: the declared shadow edges were asserted to be jointly realizable with the bag networks, which is false without a further discipline (`automata_route_repairs/`, `cold_review_theoremB/`, probes `wp13`–`wp17`).

Why we abandoned the tree automaton. A cold review of the decision layer exposed a structural problem. The automaton reads a labelled tree and decides accept or reject. But the tree does *not* carry all the data that makes an abstraction valid: it omits the relations between each fresh witness and the "sources" of the universal restrictions ($\forall R.D$) that act on it. A tree is valid exactly when *some* consistent choice of those missing relations exists—so the automaton would have to solve a *guessing* problem ("does some completion of the missing data work?"), not merely read off the tree. The trouble is how an alternating tree automaton works: it explores the tree in many independent parallel copies, and each copy would guess the missing relations *locally*, with nothing forcing different copies to guess *consistently*. One can therefore build a tree in which two copies, running over two different witnesses, each privately guess relations that are locally fine but mutually incompatible, so every copy accepts—even though the true relation between those two witnesses is forced by composition to be impossible ($\mathsf{comp}(\mathsf{PP},\mathsf{PP}) \cap \mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{PP}\} \cap \{\mathsf{DR}\} = \emptyset$). The automaton accepts an unsatisfiable input: acceptance without validity. The three natural repairs (annotate the tape—but the enrichment is unboundedly large; quotient the profiles—provably lossy; guess canonically—needs an unproven uniformization) are each blocked by facts the construction itself establishes. So the tree automaton was replaced by a mosaic / type-elimination procedure directly on the infinitary abstraction (§7.6). The keystone survived the change untouched: Theorem B's finite-abstraction adequacy is F6 in automaton clothing.

7.6 The Lean line and F6

The mathematical basis, and the pivot that produced it. The Lean development is *not* an automaton—and that is the point of a deliberate pivot. The route of §7.5 tried to decide satisfiability with a two-way parity *tree automaton* (Theorem C); the cold review that found "acceptance without validity" showed that a tree automaton cannot work here, because its independent parallel run copies cannot be forced to guess the missing horizontal relations *consistently*. Rather than patch the automaton, the project moved the decision *off* it and back onto the semantic objects the split forest already supplies. What resulted is a *cluster-quasimodel* construction in the *mosaic /*

type-elimination tradition of modal and description logic: a finite catalogue of interface *clusters* (mosaics), a consistency relation for gluing them (controlled by the RCC5 patchwork property), and a *Hintikka truth lemma* for correctness—the same tableau-family argument as §7.1, but run on a *finite abstraction* rather than on an unbounded tableau. Unlike the retracted early quasimodel (§7.2), the procedure *enumerates and finitely checks* catalogues rather than running an elimination fixpoint, so it does not inherit that route's anti-monotone incompleteness. The split-forest normal form (Theorem A) is the shared semantic foundation that both the automata route and this one *consume*.

What makes the automaton *unnecessary*, rather than merely replaced, is *one-step demand fulfilment*. An automaton earns its keep on *eventualities*: promises—"some part satisfies D"—that may be discharged arbitrarily far away, and whose fulfilment along an infinite branch is exactly what a Büchi/parity acceptance condition checks. Here every $\exists R.D$ is instead fulfilled by a witness *already present in the catalogue*, so there are no promissory obligations, no eventualities, and hence no acceptance condition to decide—the one task only an automaton did well. Decidability then reduces to enumerating finite catalogues and checking each by a finite composition test, with no automaton in the loop. This is the construction that was formalized in Lean over rounds 19–30.

Lean, and how we use it. Lean 4 is a proof assistant whose kernel type-checks every inference and refuses a proof with a gap. In a project where prose proofs repeatedly *assumed* the hard step, the kernel is a ground-truth oracle that cannot be talked past. We moved the load-bearing soundness argument into *Lean 4 core*—no `mathlib`, no external libraries, no build system: every object is built from first principles, which keeps the trusted base small and the axiom footprint auditable (only `propext` and `Quot.sound`, plus `native_decide` in isolated witness lemmas). The encoding is deliberately direct, so that the Lean statements are legible as the mathematics they formalize: the five RCC5 atoms are an inductive type with the composition table and converse as total functions; concepts are an inductive type in negation normal form (inverse roles absorbed via the converse); an *interpretation* is a domain predicate together with a *total* relation function $\rho : \Delta \times \Delta \to \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ and an atomic valuation; satisfaction is defined by structural recursion (a role R is read directly as $\rho\, x\, y = R$); and the RCC5 frame conditions (reflexive-EQ, strong-EQ, converse-coherence, composition closure) are a `structure` over ρ. On top of this foundation sit Hintikka labellings and the certificate objects described below. The development is `formal/Round19Transport.lean` (≈5,300 lines, zero `sorry`), with the full artifact-by-artifact history in `LEAN.md`.

What a certificate is. Concretely, a *certificate* is an explicit finite object whose validity is checkable by finite composition tests. It is worth spelling out what one looks like, since "catalogue" and "steering" recur throughout. A certificate has two parts.

- A *catalogue*—the finite blueprint. It consists of: a *core network* (a fixed finite complete atomic RCC5 network—the part of the model that does not repeat); a finite set of *attachment templates* (each template is one recipe for gluing a fresh cluster of witnesses onto an existing interface—which old occurrences the new ones attach to, in a *gluing normal form* that keeps the shared part canonical); and a *steering function*, indexed by template, that fixes the *horizontal* relations from the old occurrences to each fresh one. The steering function is the one genuine degree of freedom, because those are exactly the relations composition does not determine (Proposition 1); the vertical edges are already carried by the tree.

- A *plan*: a finite recipe naming which template to apply at which interface, in order. *Unfolding*

the plan—applying templates step by step, possibly reusing one template unboundedly often to grow an infinite PP-tower—produces the (possibly infinite) split-forest presentation of §7.4.

A certificate is *sound* if a finite check on the catalogue alone—the "S-condition," independent of how long any plan runs—guarantees that *every* unfolding is a proper, composition-closed RCC5 network; this is the step certified in Lean (below). Its *live width* is the largest number of live (non-shadow) occurrences that must coexist at any one interface during unfolding—the quantity F6 must bound. Demand fulfilment is *one-step*: every $\exists R.D$ is satisfied by a witness actually present in the catalogue, so there are no promissory obligations and hence no parity/eventuality condition of the kind the tree automaton needed.

It was on this construction, finally explicit and free of hidden choices, that the remaining difficulty of this architecture localized to two named properties: **F6** (Definition 5, a computable live-width bound) and a uniformization property **W2′**. One test of this format is outstanding: whether it admits a certificate for the spiral concept of §7.8, whose models are infinite without any vertical existential. It has not been tried.

Definition 13 (Uniformization, W2′)**.** Extraction reads a finite certificate off a model by summarizing each interface into a bounded *state* (a type, plus the relations to the finitely many recorded anchors). W2′ asks that this summary be *faithful*: whenever two old occurrences carry the *same* state, a fresh witness can be glued onto both in the *same* way—so a glue step depends only on the state, and the catalogue's finitely many states really do capture every gluing. If W2′ fails, two occurrences that look identical to the catalogue may nonetheless demand different completions, so the finite catalogue *under*-represents the model and the checker can *reject a satisfiable concept*. The error is one-directional: a failure of W2′ can only cause over-rejection, never acceptance of an unsatisfiable concept—so W2′ bears on completeness alone, never on soundness.

What is machine-checked (rounds 19–30). The soundness pipeline is kernel-checked end to end: certificate syntax → an S-condition on the certificate → a proper atomic RCC5 frame → the logic layer (a truth lemma, with one-step fulfilment) → an actual RCC5 model of C_0. So is the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide. On top of these sits a fixed, non-oracular Boolean checker of first-order finite certificate data, proved sound, from which decidability follows by finite search—modulo one premise, that this checker be complete for some computable assignment of finite candidate lists. F6 together with a refined W2′ is the argued route to that premise; the argument is not formalized.

Theorem 14 (Conditional decidability)**.** *Suppose a computable function assigns to every concept C_0 a finite list of certificate codes that contains a code accepted by the fixed checker* `finAcceptB` *whenever C_0 is satisfiable. Then concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ is decidable. This implication, the soundness of the checker (an accepted code unfolds to a genuine RCC5 model of C_0), and the faithfulness of the Hintikka abstraction are* machine-checked in Lean 4 *(*`decidableSat_of_finScheme`*,* `finAccept_sound`*,* `satisfiable_iff_hintikkaP`*; zero* `sorry`*). The supposition is open.*

The route to the supposition that this line pursued starts from property F6, and two of its steps are argued, not proved:

$$\text{F6} \overset{?}{\Longrightarrow} \text{refined W2}' \overset{?}{\Longrightarrow} \text{complete finite candidate lists} \overset{\text{Lean}}{\Longrightarrow} \text{decision procedure.}$$

Earlier versions of this paper stated the theorem as "F6 implies decidability", with only its soundness half certified, and called F6 the sole open premise. That folded two unformalized implications into the statement. The twenty-first review [75] separated them, and we have adopted its decomposition.

The keystone, sharpened. Two cold attacks on F6 turned the vague "bounded width" into a precise dichotomy. The naive ways to force width—shell half-graphs, prefix pumps, single-chain PO-gaps—all *collapse* to finite catalogues (they have $O(1)$ separator cover, a carried memory slot suffices). The sharp obstruction is an unbounded family of M_n *identity-selector minors*: pairs (L_i, U_i) each separated by a *private* selector A_i and by no other, so the selector incidence matrix is the $n \times n$ identity—private matching of unbounded hitting number. In plain terms: n separate "this-region-not-that-one" witnesses, no two of which may be merged because each is the *unique* reason some specific pair stays apart—so a finite catalogue, which can only hold boundedly many such witnesses, is overrun exactly when a concept forces n of them for every n. The conjecture that came out of these attacks is that this is the *only* obstruction—that F6 fails exactly when some concept forces such minors. That is a candidate characterization, not a theorem: no argument shows that minors are the only way width can grow, and the finite networks that exhibit them are not known to be forced by any concept. And W2′ (Definition 13), which looked like a second obstacle, was shown false in its coarse form but, by an argument that is not formalized, repairable by a state refinement whose size is bounded *by F6*—so that it *folds into* F6, leaving, on that argument, a single open question (`fable5_width_barrier/`, `cold_review_f6_w2prime/`; probes `wp35`–`wp44`, incl. `wp44`: arbitrary RCC5 glue steps *can* have unbounded separator cover, so any F6 proof must use $\mathcal{ALCI}_{\mathrm{RCC5}}$-forceability, not RCC5-consistency alone).

The problem is Π^0_1, and F6 weakens to its qualitative core. One further observation sharpens what F6 must deliver. It is not new: Lutz and Wolter prove that $L_{RCC5}(RS)$, which is $\mathcal{ALCI}_{\mathrm{RCC5}}$, is recursively enumerable [10, Thm. 28]. Earlier versions of this paper presented it as new (§4.4); we re-derive it here because the dovetailing argument below consumes its explicit form. As §5 sets up, the semantics is finitely *first-order* axiomatizable (the theory T_{RCC5} plus the standard translation of C_0), over *arbitrary* carriers—exactly what the carrier-polymorphic satisfiability of the Lean development means.

Concretely this is a *computable reduction to plain first-order satisfiability*: each C_0 maps to the single sentence $\bigwedge T_{\mathrm{RCC5}} \wedge \exists x\, \mathrm{ST}(C_0, x)$, which has a first-order model iff C_0 is satisfiable. And first-order satisfiability is itself Π^0_1—the co-r.e. side of Church's undecidability theorem, which is Gödel completeness in action: unsatisfiability equals refutability, and FO refutations enumerate. So satisfiability is *in* Π^0_1—and this is an upper bound only: we establish *membership*, not Π^0_1-hardness (a *decidable* problem is also in Π^0_1). It shares its *ceiling* with the domino problem, but we do not show it as hard, so the "knife's edge" (§1.3) is a suggestive analogy here, not a completeness result. The *reduction* alone would not even fix this ceiling: a non-first-order model condition—well-foundedness, finiteness, anything genuinely second-order—could push it higher; the finite axiomatization is what keeps the reduction a single sentence.

The consequence for the keystone is best seen through the missing *dual*. A Π^0_1 problem is decidable iff it is *also* recursively enumerable, and completeness hands us one such side—enumerability of UNSAT—for free. The classical way to get the *other* side, enumerability of SAT, is the *finite model property* (enumerate finite models), the textbook route "recursively axiomatizable + finite model property ⇒ decidable"—a principle report7 itself already invoked, en route to a bounded variant of these logics [9]. But $\mathcal{ALCI}_{\mathrm{RCC5}}$ has *no* finite model property (§5: a PP-tower needs an infinite model, as does the spiral concept of §7.8), so that route is shut (for the fragment $\mathcal{L}_{\|}$ of §7.8 *finite* satisfiability is decidable by an imported theorem, which does not help with general satisfiability for the same reason), and enumerating SAT the honest way means enumerating *finite certificates for possibly-infinite models*—which, for *this* certificate scheme, is *qualitative* F6: every satisfiable concept has *some* finite certificate accepted by our checker, *with no computable bound demanded.*

One dovetails such a certificate enumeration (its checker the already-certified `finAccept_sound`) against the enumeration of FO refutations; on every input, one side must eventually fire. This is a *sufficient* route, not the recursion-theoretic minimum: decidability is equivalent only to the existence of *some* sound, decidable, complete certificate relation, and completeness of *this particular* split-forest/live-width checker is not known to be necessary. So F6 weakens from its *quantitative* form (a live-width bound *computable* from C_0—the target behind most of the width-accounting defects the reviews found) to the *qualitative* core just isolated, and the computable bound then returns for free *a posteriori*.

The generic dovetailing combinator is machine-checked (`formal/SemiDecidability.lean`, zero `sorry`), with its four hypotheses—checker soundness, refutation soundness, certificate completeness, and first-order completeness—carried as fields rather than instantiated: the Lean contains no formal first-order proof system. A concrete refutation calculus for this logic now exists outside the Lean: the tableau without blocking of §8.1, shown on paper to close on every unsatisfiable concept. It could instantiate the refutation side of the combinator once its soundness is formalized; that has not been done. The FO transcription it rests on—that the axioms above carve out exactly the Lean frame conditions, and that the standard translation matches the certified satisfaction relation—is cross-checked in `wp84` (exhaustively for domains up to three points). Two honest caveats: the qualitative core of F6 is *untouched*, so this reshapes the target rather than hitting it; and the Π^0_1 placement adds one standard external theorem (Gödel completeness) to those the argument already invokes (the RCC5 patchwork property; parity-automaton emptiness in the historical automata route). No complexity bound is claimed or implied.

7.7 The regular-cover pivot

Idea. A reframing suggested by Wessel: a finite *quotient* of a PP-chain is invalid (a PP-cycle contradicts EQ-identity), but a finite *generator* is not—a state with a PP-self-loop *unfolds* to an infinite chain of distinct copies. So replace finite quotients by *regular covers*: finite presentations whose infinite *unfoldings* are the models, in the spirit of the two-tier quotient but for the full logic. This finer invariant can represent exactly the patterns that fool a width bound—identity selectors ($i = j$ via address equality), half-graphs ($i < j$ via an order test)—finitely.

What it produced. A *local* algebra of strong-EQ RCC5 networks, one piece of which is machine-*certified*:

Theorem 15 (RCC5 normal form)**.** *A strong-*EQ *atomic RCC5 network is composition-closed if and only if it is an* ordered-disjoint structure*:* PP *is a strict partial order,* DR *is a symmetric, irreflexive, downward-closed disjointness (if* x DR y *and* $x' \le x$, $y' \le y$ *then* x' DR y'*), comparable pairs are never disjoint, and* PO *is the residual. The forward direction (every RCC5 network is ordered-disjoint) is* machine-certified for arbitrary domains *(`formal/RCC5NormalForm.lean`, axioms `propext` only). The* converse *is now certified too: GPT-5.6 Pro's technical note [48] (`papers/final_paper_gpt_5.6_review/ALCI_RCC5_F6_technical_note.pdf`) gave an explicit* canonical set representation*—each element x maps to the non-disjoint pairs with a coordinate $\le x$—and its core is machine-checked for arbitrary domains (`OrderedDisjoint.sub_iff_le`, `eta_injective`:* zero *axioms; `disj_iff_eta_disjoint`:* + `Classical.choice` *for the one non-constructive direction), independently cross-checked past size four in `wp88`. The frame half of the converse (an ordered-disjoint structure is composition-closed, `odNet_frame`) and the model-level bridges (`odNet_odOfModel`, `setRel_odNet`, `satisfiable_iff_set`) are in `formal/POFreeLift.lean`. So the biconditional holds on arbitrary domains.*

Two pieces of prior work should be read alongside it. That RCC5 structures are representable at all—by regular closed regions, not merely by sets—is Theorem 23 of Lutz and Wolter [10]; what the theorem above adds is the explicit order-and-disjointness form, an explicit construction, and kernel-checked proofs. And the ordered-disjoint shape is an old one in another field: a partial order with an irreflexive, symmetric conflict relation inherited along it is the order skeleton of a prime event structure [12]. The order is reversed—causality runs from a region to its parts, because disjointness is inherited downward—with DR as conflict and PO as concurrency, and without the finite-causes axiom, which region structures need not satisfy. The most recent cold review pointed out the correspondence [75]; we have not yet used it, but the theory of event-structure unfoldings and their finite presentations is a natural place to look for tools.

With it come free amalgamation of ordered-disjoint structures, an exact one-point extension criterion, and exact uniform/non-uniform cross-policy characterizations—theorem-level results with finite probes, which between them cover the local algebra the route needed (`rcc5_local_amalgamation_theory`, `regular_cover_route_pivot/`, probes `wp45`–`wp83`). The route independently re-derives Theorem 10.

Why it is a better map, not a shorter path. Its own keystone (a bounded coherent regular-cover property) is F6 relabelled, and the route concedes, after working through the local algebra, that the remainder "cannot be a purely local RCC5 composition problem." Because RCC5 relations are total, every model's Gaifman graph is complete, so a one-dimensional PP-chain already has unbounded clique number and treewidth; no off-the-shelf guarded or bounded-treewidth cover theorem applies, and the keystone would need a bespoke cover theorem for complete edge-coloured regular structures. The pivot is the right vocabulary for a future attack—but it does not move the summit.

7.8 The other direction: a two-variable logic *inside* $\mathcal{ALCI}_{\mathrm{RCC5}}$

Added November 2026. This subsection reports the work of one session and has not been reviewed; by the record of §9, presume it contains a defect. Its Lean statements are kernel-checked. Its statements about the literature rest on the cited papers, of which at least the abstracts and introductions were read.

The fragments above fail to contain $\mathcal{ALCI}_{\mathrm{RCC5}}$. One extension of FO^2 stands in the opposite relation to it. Let $\mathrm{FO}^2[<]$ be two-variable logic interpreted over structures in which one distinguished relation $<$ is a strict partial order, so that transitivity is a constraint on the models and is not charged to the variable bound. It has no finite model property, and its general satisfiability problem is open (§5): a 2013 decidability claim was withdrawn, and only the transitively guarded fragment and finite satisfiability are decided. (Versions of this paper before November 2026 cited the 2013 claim as a theorem; see `ERRATA.md`.)

In an RCC5 structure PP is a strict partial order, and two distinct regions are incomparable in it exactly when they are related by PO or by DR. Let $\mathcal{L}_{\|}$ be the set of NNF concepts in which every $\exists\mathsf{DR}.C$ occurs as the right disjunct of $\exists\mathsf{PO}.C \sqcup \exists\mathsf{DR}.C$ and every $\forall\mathsf{PO}.C$ as the left conjunct of $\forall\mathsf{PO}.C \sqcap \forall\mathsf{DR}.C$; a bare $\exists\mathsf{PO}$ and a bare $\forall\mathsf{DR}$ are allowed. Such a concept cannot tell overlap from disjointness. Read it over a strict partial order: PP and PPI as $<$ and $>$; $\exists\mathsf{PO}.C \sqcup \exists\mathsf{DR}.C$, and a bare $\exists\mathsf{PO}.C$, as "some incomparable element is a C"; $\forall\mathsf{PO}.C \sqcap \forall\mathsf{DR}.C$ as "every incomparable element is a C"; a bare $\forall\mathsf{DR}.C$ as $\top$.

Theorem (the poset bridge; Lean: `poset_bridge`, `poset_bridge_fin`). *A concept of $\mathcal{L}_{\|}$ is satisfiable in $\mathcal{ALCI}_{\mathrm{RCC5}}$ if and only if its reading is satisfiable over a strict partial order. The same holds with "in a finite model" on both sides.*

Why. A strict partial order all of whose incomparable pairs are declared PO is an ordered-disjoint structure with empty disjointness, hence an RCC5 frame (Theorem 15); an incomparable witness is then a PO-witness, and ∀DR is vacuous. Conversely the PP-reduct of an RCC5 model is a strict partial order on the same domain; a PO- or DR-witness is an incomparable witness, and a universal over incomparable elements holds because both ∀PO and ∀DR did. The two excluded constructors are excluded for cause: a bare ∀PO and a bare ∃DR can tell overlap from disjointness, and a strict partial order cannot (`wp137` exhibits both failures).

The modal language over a set of jointly exhaustive relations has the expressive power of the two-variable fragment over them [10], so $\mathcal{L}_{\|}$ is a notation for $\mathrm{FO}^2[<]$ with unary predicates. The Lean development proves the correspondence for sentences in normal form: finite sets of forbidden patterns $\forall x\,\forall y\,\neg(\tau(x) \wedge r(x,y) \wedge \tau'(y))$ and requirements $\forall x\,(\tau(x) \to \bigvee_i \exists y\,(r_i(x,y) \wedge \tau_i(y)))$, with $r \in \{=, <, >, \|\}$ and τ a conjunction of unary literals (`patsys_to_rcc5`). The passage from an arbitrary two-variable sentence to that form is the textbook Scott normal form and is *not* formalized.

Consequence (Lean: `decidablePSat_of_full`, `decidablePatSat_of_full`). *A decision procedure for $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability yields one for sentences of that normal form over strict partial orders, and so, by the unformalized normal-form step, for $\mathrm{FO}^2[<]$ with unary predicates. An undecidability proof for the latter would make $\mathcal{ALCI}_{\mathrm{RCC5}}$ undecidable.*

So the full logic contains the unary case of a problem that is open in its own literature. What Marzec and Tendera single out as the critical core [16]—universal constraints that force every α-element and every β-element to be comparable, together with incomparable witnesses—reads in RCC5 as $\alpha \sqsubseteq \forall\mathsf{PO}.\neg\beta \sqcap \forall\mathsf{DR}.\neg\beta$ and $\alpha \sqsubseteq \exists\mathsf{PO}.\beta \sqcup \exists\mathsf{DR}.\beta$, imposed globally. It has no ∃PP and no ∃PPI, and it uses the composition table in one deterministic way only: two regions that may be neither PO nor DR must be nested, and nesting is transitive. No reduction in the other direction is known. The downward closure of DR (Proposition 3) relates up to four regions at once and is not obviously expressible over a single transitive relation, so $\mathcal{ALCI}_{\mathrm{RCC5}}$ may be strictly harder.

What the literature then gives. Two slices of $\mathcal{ALCI}_{\mathrm{RCC5}}$ are decided *by import* through the bridge theorem. Neither is Lean-checked beyond the bridge, and each is worth what the cited proof is worth. (a) Concepts of $\mathcal{L}_{\|}$ without ∃PO translate to sentences whose existential quantifiers are guarded by the order (after a relativization that removes the one unguarded, outermost quantifier), a fragment decided in 2-NExpTime [14]. Unlike in the fragment $\mathcal{D}$ of §7.3, the ∀PO restrictions of this slice cannot be erased. With $W(P,Q) = P \sqcap \neg Q \sqcap \forall\mathsf{PO}.\neg Q \sqcap \forall\mathsf{DR}.\neg Q \sqcap \forall\mathsf{PPI}.\neg Q$ ("I am a P, and every Q lies strictly above me"), the concept $C_{\mathrm{cyc}} = \exists\mathsf{PP}.W(A,B) \sqcap \exists\mathsf{PP}.W(B,C) \sqcap \exists\mathsf{PP}.W(C,A)$ is unsatisfiable—its three witnesses would satisfy $a \subset b \subset c \subset a$—and becomes satisfiable when its three ∀PO conjuncts are erased (`ccyc_unsat`, `ccycErased_sat`). Each pair of witnesses is compatible on its own; the conflict is the cycle. (b) Finite satisfiability of $\mathcal{L}_{\|}$ is decidable in 2-NExpTime [15]. Neither slice contains C_{bad} or C_{joint}, whose ∀PO is bare.

The other coarsening: $\mathcal{ALCI}_{\mathrm{RCC3}}$. $\mathcal{L}_{\|}$ merges PO with DR and keeps the order. Wessel's $\mathcal{ALCI}_{\mathrm{RCC3}}$ merges PO, PP and PPI into one relation ONE, "overlapping but not equal", and keeps DR: it forgets the order and keeps disjointness. The two stand in the same relation to $\mathcal{ALCI}_{\mathrm{RCC5}}$. Read the merged role as a defined role, $\exists\mathsf{ONE}.C = \exists\mathsf{PO}.C \sqcup \exists\mathsf{PP}.C \sqcup \exists\mathsf{PPI}.C$ and dually. Then an $\mathcal{ALCI}_{\mathrm{RCC3}}$ concept is satisfiable if and only if its translation is satisfiable in $\mathcal{ALCI}_{\mathrm{RCC5}}$. Merging the three relations of an RCC5 model gives an RCC3 model. Conversely the RCC3 table constrains no triple of distinct regions, so an RCC3 model is an arbitrary graph of ONE and DR edges, and labelling every ONE edge PO gives an RCC5 structure, because every $\{\mathsf{PO}, \mathsf{DR}\}$-labelled graph is closed under the table. (This is an elementary argument, supported by the probe `wp140`; it

is not formalized and has not been reviewed.) So $\mathcal{ALCI}_{\mathrm{RCC3}}$, which the 2003 report decides by translation into two-variable logic with equality [9, §4.3], is a decided slice of $\mathcal{ALCI}_{\mathrm{RCC5}}$. It is the oldest one, and a $\forall\mathsf{PO}$ restriction changes the answer in it: $\exists\mathsf{ONE}.A \sqcap \forall\mathsf{ONE}.\neg A$ is unsatisfiable and becomes satisfiable when its $\forall\mathsf{PO}$ conjunct is erased. Both coarsenings go through two-variable logic. Without the order the class of structures is two-variable definable, and the logic is decidable. With the order it is not, and the question is the open one above. In both, the structure that realizes a coarse model labels the merged pairs PO: the relation that nothing but $\forall\mathsf{PO}$ can object to (§7.3) is again what makes the reading work.

Infinity without a vertical witness. That a concept can have only infinite models without mentioning PP or PPI at all is not new. Wessel's 2003 report gives $\exists\mathsf{DR}.\top \sqcap \forall\mathsf{DR}.(\exists\mathsf{PO}.\exists\mathsf{DR}.C \sqcap \forall\mathsf{PO}.\neg C \sqcap \forall\mathsf{DR}.\neg C)$ and explains why [9, §5.2, Fig. 12]. From a region y disjoint from the root, the C-witness two steps on stands to y in a relation from $\mathsf{comp}(\mathsf{PO},\mathsf{DR}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PPI}\}$; the two universals at y exclude DR and PO, so the witness is a proper part of y; and by $\mathsf{comp}(\mathsf{DR},\mathsf{PPI}) = \{\mathsf{DR}\}$ it is again disjoint from the root, so the pattern repeats one level down. The relations that would let the chain close are clashed out. (The reference reasoner of §8.1 finds no model of this concept on up to seven nodes.) The following is a second example of the same kind, and it differs from the first in two respects. It lies in $\mathcal{L}_{\|}$: it never separates PO from DR and makes no use of the downward closure of DR, on which the step "again disjoint from the root" rests. And the direction of the nesting it forces is not fixed by the composition table; it is chosen once and must then be kept. Take five concept names $A_0,\dots,A_4$, indices modulo 5, require every region to carry one of them, and impose for each i

$$A_i \sqsubseteq \forall\mathsf{PO}.\neg A_{i+2} \sqcap \forall\mathsf{DR}.\neg A_{i+2}, \qquad A_i \sqsubseteq (\exists\mathsf{PO}.A_{i+1} \sqcup \exists\mathsf{DR}.A_{i+1}) \sqcap (\exists\mathsf{PO}.A_{i-1} \sqcup \exists\mathsf{DR}.A_{i-1}).$$

This is the spiral of Marzec and Tendera [16], read in RCC5; the infinity axiom at its core, on these five predicates, is already in Szwast and Tendera [14, §5]. It is not a concept of this project. Internalized as a concept it is satisfiable, every model of it contains an infinite sequence of pairwise distinct regions, and it contains no $\exists\mathsf{PP}$ and no $\exists\mathsf{PPI}$ (`spiralConcept_sat`, `spiralConcept_infinite`, `spiralConcept_noExVert`). Regions two steps apart along a chain of incomparable witnesses are forbidden to be incomparable, so they are nested, and the direction of nesting propagates; a model is the integers with $a < b$ iff $a + 2 \le b$, coloured modulo 5 (Figure 4). The chain it forces is still a PP-chain, consistent with Lemma 4, but it is forced by excluding PO and DR, not by asking for PP. Whether the certificate format of §7.6 admits a certificate for this concept has not been tested, and is the natural next test of F6.

Two more boundary concepts. $K_2 = \exists\mathsf{PO}.(A \sqcap \forall\mathsf{PO}.\neg B \sqcap \forall\mathsf{DR}.\neg B) \sqcap \exists\mathsf{PO}.(B \sqcap \neg A)$ is satisfiable, by a region r and two regions $a \subset b$ that both overlap it, and in every model its two witnesses are nested (`k2_sat`, `k2_witnesses_nested`). It is C_{sat} moved sideways: witnesses generated horizontally that must be related vertically. With the spiral and C_{cyc} it extends the test of §7.3 that a certificate scheme for the full logic has to pass: reject C_{bad}, C_{joint} and C_{cyc}; accept C_{sat}, K_2 and the spiral. The fresh-occurrence unfolding with residual PO produces no model of K_2 or of the spiral, because it never makes one horizontal witness a part of another. K_2 is also the concept on which the cover-tree tableau fails (§8).

What this changes, and what it does not. It does not move decidability. It isolates one clean component of the horizontal-coordination difficulty: a sublanguage with no $\exists\mathsf{PP}$, no $\exists\mathsf{PPI}$, no use of the downward closure of DR and no non-singleton composition cell is already an open problem.

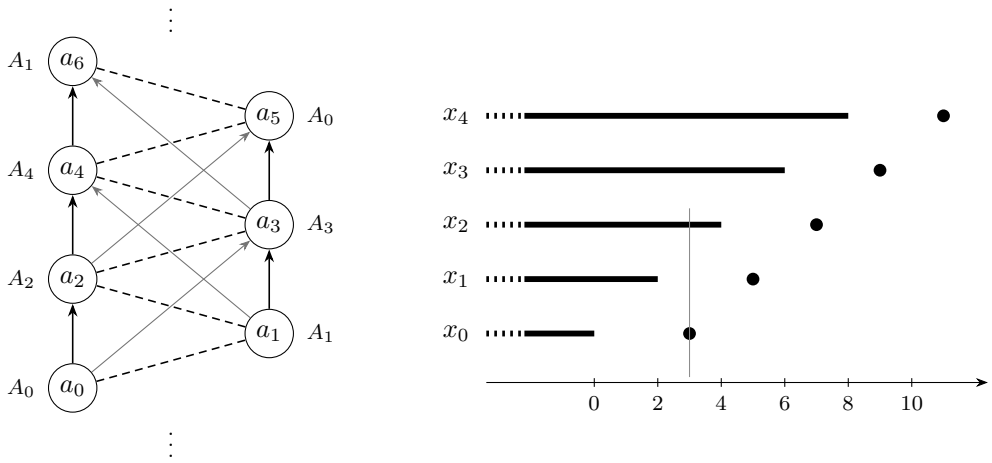


Figure 4: The spiral. *Left:* the order it forces, on seven consecutive regions a_k, where a_k carries $A_{k \bmod 5}$. Dashed lines join the incomparable pairs a_k, a_{k+1}, each of which overlaps or is disjoint; arrows are proper-part relations, drawn for the covering pairs $a_k \subset a_{k+2}$ (black) and $a_k \subset a_{k+3}$ (grey). Two regions whose colours differ by two are never joined by a dashed line, as the concept demands. *Right:* regions that realize it, $x_k = (-\infty, 2k] \cup \{2k+3\}$ on a line. The isolated point of x_k lies outside x_{k+1} and inside x_{k+2}—the grey line follows the point 3 of x_0 past x_1 into x_2—so consecutive regions overlap and regions two or more steps apart are nested.

A decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ has to decide it on the way, and a proof strategy that would not can be set aside. The partial results on that side also point away from the split-forest picture. The fragment's models are made maximally PO; for sentences whose witnesses are all incomparable, Marzec and Tendera show that models can be taken with bounded antichains, that is, maximally *comparable*. Whether that survives the addition of a downward-closed disjointness is untried. F6, the architectural gap for a bare ∀PO, and the interaction of DR's downward closure with ∀PO are untouched. Sources: `formal/PosetBridge.lean` (1,354 lines, importing the certified artifact; no `sorry`), the note in `papers/poset_bridge/`, and the probe `wp137`.

8 Current implementation: reasoners

The project ships three programs: a prototype tableau on the split-forest foundation; the retracted quasimodel procedure, kept as an oracle; and, since November 2026, a reference tableau without blocking. The first two are tested, not proved, and each is known to be incomplete. The third has a paper proof of soundness and completeness and does not terminate on every input.

The cover-tree tableau. `src/cover_tree_tableau.py` (≈ 770 lines) is an operational, incomplete concept-satisfiability test with exact-occurrence blocking, resting on the split-forest normal form and patchwork completion (§7.4). Its regression and cross-validation suite has 911 cases: 99 labelled ones (50 known satisfiable, 13 known unsatisfiable, 36 adversarial) and 812 comparison runs against an independent quasimodel oracle (512 systematic triples, 300 random concepts of depth two and three), with zero mismatches once the oracle's known blind spot is excluded. It is also checked against an independent model verifier (`src/model_verifier.py`). SAT outputs accompanied by an explicit finite model are independently checkable by that verifier; no completeness

theorem is claimed, and UNSAT outputs and SAT cases requiring infinite models remain empirical. The agreement is evidence, not a proof, and the tableau's record has three entries. On 11 April 2026 a code review found that its cross-interaction check handled only singleton compositions, so that it could accept unsatisfiable concepts whose conflict runs through a non-singleton cell; the fix left all 911 verdicts unchanged—which shows that the suite had not exercised the pattern, not that the gap was harmless. On 18 April an adversarial review by Claude Opus 4.7 showed that a related gap was not harmless: the tableau as then committed answered SAT on twelve unsatisfiable concepts—$\exists R_1.\exists R_2.A$ conjoined with $\forall R.\neg A$ for every $R \in \mathsf{comp}(R_1, R_2)$, for each pair with R_2 not the converse of R_1—the simplest being $\exists \mathsf{PP}.\exists \mathsf{PP}.A \sqcap \forall \mathsf{PP}.\neg A$ [67]. A role-path check repaired it the same day, the reviewer attacked the repair and withdrew the claim for the repaired code, and since then no wrong SAT answer is known. (Its one recurring disagreement with the oracle inside the suite's scope, on the strong-EQ PO-loop, is the oracle's blind spot.) Versions of this paper from September to early November 2026 said that no verdict of the tableau had ever been shown wrong; that was false from April on (`ERRATA.md`). In November 2026 a wrong UNSAT answer was found as well. The concept $K_2 = \exists \mathsf{PO}.(A \sqcap \forall \mathsf{PO}.\neg B \sqcap \forall \mathsf{DR}.\neg B) \sqcap \exists \mathsf{PO}.(B \sqcap \neg A)$ is satisfiable—three regions r and $a \subset b$, both overlapping r; `k2_sat` in `formal/PosetBridge.lean`—and the tableau answers UNSAT, because it admits only DR and PO between two horizontally generated witnesses of one node, and these two must be nested (`wp137`, part E). The tableau is therefore *incomplete*. Three further satisfiable concepts of the same pattern are rejected (`wp138`). The wrong direction is UNSAT, and subsumption is tested as unsatisfiability, so a subsumption verdict of the tableau is evidence only where something independent agrees with it. The failure needs a $\forall$PO acting between the two witnesses, so it cannot arise in the $\forall$PO-free fragment. An experimental repair, which admits nesting among the witnesses of a node and checks them jointly, is in the code behind a flag that is off by default. It removes these wrong verdicts and changes none of the 35 built-in or 911 suite verdicts, which is evidence and not a proof. A terminating procedure that is sound and complete for the full logic would also decide the open problem of §7.8, so no such patch should be expected to make the tableau a decision procedure.

A reference reasoner without blocking (November 2026; unreviewed). `src/midelora_tableau.py` re-implements in Python the tableau of Wessel's MiDeLoRa system [66], with EQ fixed to identity: a tableau on a complete graph, with no blocking. By Theorem 18 it answers UNSAT on every unsatisfiable concept and SAT with a finite model whenever one exists, and does not answer on satisfiable concepts all of whose models are infinite. It agrees with the tableau and the oracle on all 911 suite concepts, accepts K_2, and is exact on 1,000 concepts with computable ground truth. The system, the calculus, the proof and the tests are the subject of §8.1.

Evidence statistics, and the absence of a counterexample. The empirical case is substantial, and it is worth stating precisely because it is the strongest thing short of a proof. The tableau and the oracle agree on a 911-concept cross-validation suite (zero mismatches after the single known PO-loop blind spot of the quasimodel oracle is excluded); since November 2026 the reference reasoner returns a verdict on every suite concept and agrees with both, and with the labels (`wp139`). Outside the suite the first two disagree on the PO-loop, where the oracle is wrong, and on K_2, where the tableau is. The cover-tree decomposition passes 775/775 of its structured tests. We state its scope honestly: the two implementations are genuinely different algorithms (tableau versus type elimination) but share the composition table and test harness, so this is cross-validation with some common-mode risk, not two wholly independent oracles (the reference reasoner shares their concept data types, but derives its own composition table from set semantics and uses none of their

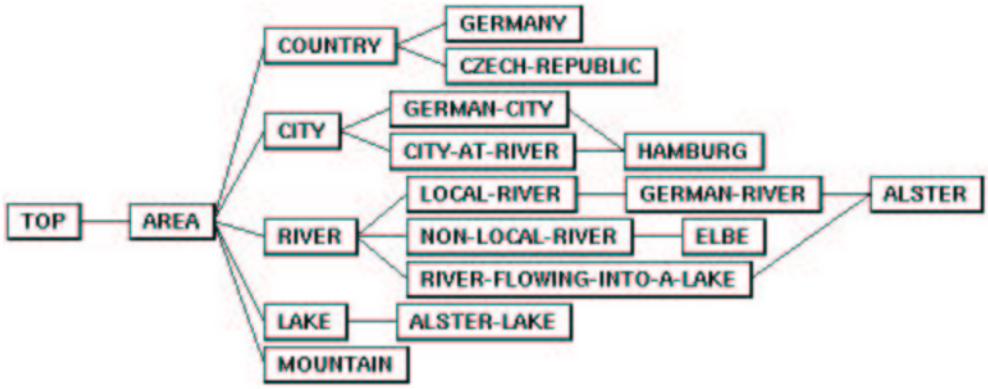


Figure 5: The GIS concept taxonomy *computed by a prototype reasoner* in Wessel's 2002/2003 report (report7, Figure 6): the original ("the subsumption hierarchy ... was actually computed by a working prototype system," report7, end of §3).

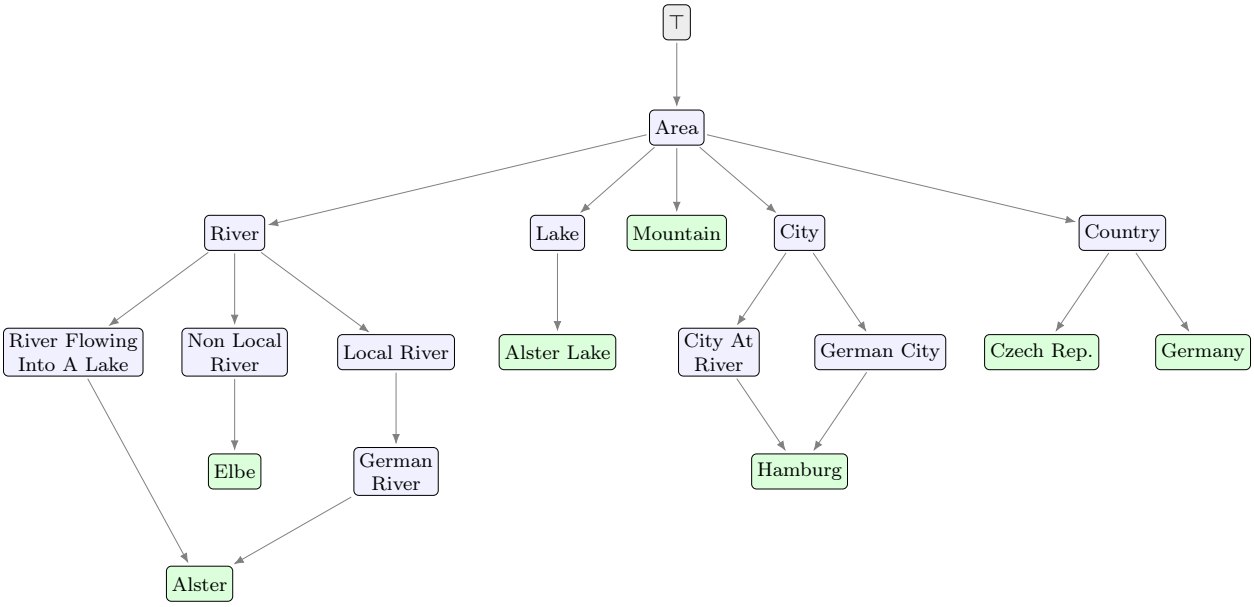


Figure 6: The *same* GIS taxonomy recomputed in 2026 by the cover-tree tableau reasoner (`src/gis_taxonomy.py`), rendered directly from the reasoner's own output (arrows point from the more general concept to the more specific; green = individuals). All 21/21 subsumptions match Figure 5 edge for edge. Note that *Hamburg* (a German city at a river) and *Alster* (a German river flowing into a lake) each have two parents: the taxonomy is a directed acyclic graph, not a tree—exactly the multiple-inheritance the reasoner must get right.

search code); and the model verifier confirms explicit *finite* SAT witnesses, not UNSAT verdicts or infinite-model SAT. Beyond the reasoners, 98 verification work packages (`verification/python/`, numbered up to WP139) cross-check the finite combinatorial content of every proof route against the RCC5 composition table *re-derived from finite set semantics*, so the underlying algebra cannot be fudged. Representative results: the RCC5 patchwork property holds on 63,219 two-bag amalgamations plus 300 tree-shaped patchings with zero failures (WP14); the finite acceptance mechanism matches the cover-tree tableau on ten diagnostic concepts plus a 148-concept random sweep with zero disagreements (WP16); the cluster-quasimodel decision layer matches the tableau on 11/11 diagnostics plus a 200-concept sweep (WP26) (in these two the tableau served as the reference verdict, and since it is now known to be incomplete their agreement on UNSAT cases weighs less than it was taken to); the RCC5 normal form agrees with brute-force composition closure on all networks up to size four (WP47); and the Lean development builds with zero `sorry` on every run (WP34). The pattern of the reviews is informative too. Concepts accepted although unsatisfiable *were* exhibited—against manuscripts as literally written, such as C_{force} against the first automata route—but each traced to a missing clause in an *argument*, never to a semantic counterexample to the intended theorem (the absence of a hybrid $\downarrow$-binder—a way to name the current point and reuse that name inside nested modalities—appears to block concept-level exploitation of the seams the reviews located; note that *nominals* themselves *are* definable under strong EQ, so it is the binder, not nominals, that is missing). This is not a proof—no amount of it can be—but it is the reason we regard the soundness side of the arguments as well tested. It is not evidence that the full logic is decidable: that question is open, and by §7.8 a proof would also decide an open problem of two-variable logic.

A GIS example spanning 23 years. The tableau recomputes the GIS concept taxonomy of Wessel's original report [9] (its Figure 6, the 2003 original, reproduced here as Figure 5): all 21/21 subsumptions are recovered (`src/gis_taxonomy.py`), and the reasoner emits its own rendering of the resulting hierarchy, shown as Figure 6. A taxonomy computed in 2003 by a *prototype reasoner*—whose calculus the report says "can be shown to be sound and complete", without giving the proof, and which carried no termination guarantee—is reproduced, in 2026, by an AI-written reasoner resting on the split-forest foundation: two reasoners, twenty-three years apart, agreeing edge for edge. The 21 subsumptions are UNSAT verdicts, the direction in which the tableau is now known to be able to err. What stands behind them is that agreement and, for individual edges, hand arguments such as the one below. The reference reasoner has not been run on the taxonomy.

How the tableau derives it. The $\mathsf{GermanCity}$ subsumption of the motivating example (§1.2) is a good window on the calculus. To decide $\mathsf{Hamburg} \sqsubseteq \mathsf{GermanCity}$ the tableau tests $\mathsf{Hamburg} \sqcap \neg\mathsf{GermanCity}$ for *un*satisfiability. Unfolding, $\neg\mathsf{GermanCity} = \neg\mathsf{City} \sqcup \exists\mathsf{PP}.(\mathsf{Country} \sqcap \neg\mathsf{Germany})$; the $\neg\mathsf{City}$ disjunct clashes at once (Hamburg is a city), so the procedure must realize a PP-successor c—a country other than Germany—beside Hamburg's asserted PO-successor, the Alster. Here the *cover-tree* step does what the hand argument did: closing the node under composition forces the Alster–c edge, and the Alster's $\forall\mathsf{PO}.\neg\mathsf{Country}$ excludes the PO option of $\mathsf{comp}(\mathsf{PO}, \mathsf{PP})$, so the edge is PP. The Alster's $\forall\mathsf{PP}.(\neg\mathsf{Country} \sqcup \mathsf{Germany})$ then propagates to c, forcing c to be $\neg\mathsf{Country}$ or $\mathsf{Germany}$—contradicting "country other than Germany." Every branch clashes, the concept is unsatisfiable, and the subsumption holds. The two ingredients—*composition-closure* of the interface and *universal propagation* along the forced edge—are exactly the operations the split-forest procedure performs at every node.

MiDeLoRa rule	What it does
`abox-completion` (`add-missing-edges`, `compute-minimal-label`)	every pair of individuals without an edge receives the disjunction of all base relations; the composition table is propagated, with dependency sets
`abox-enumeration`, with `role-consistent-p`	each disjunction is resolved to one base relation; only configurations satisfying the frame conditions are kept
`deterministic-expansion`	the $\sqcap$-rule and the $\forall$-rule
`or-expansion`	the $\sqcup$-rule
`some-expansion` (`prover/some-expansion2.lisp`)	the $\exists$-rule; with the switch `*reuse-nodes-p*` it first tries each existing successor as the witness, then a fresh node

Table 6: The main loop of MiDeLoRa's JEPD prover. A completion is found when no rule applies. The result is a tableau on a *complete graph*: every two individuals carry exactly one base relation, and every triangle satisfies the composition table.

8.1 MiDeLoRa: a complete-graph tableau, sound and complete without termination

Added November 2026; not reviewed. The proofs of this subsection are paper proofs, and only the last step of Lemma 17 is checked in Lean. The statements about MiDeLoRa's source were checked against the public repository; the system itself was read, not run.

The first reasoner for this logic is older than the present project. It is the prover Wessel wrote for the $\mathcal{ALCI}_{\mathrm{RCC}}$ family in his MiDeLoRa system, and it turns out to be the right reference point for everything in this section: with one adjustment to its treatment of EQ it is sound and complete, and the one thing it lacks is termination.

The system. MiDeLoRa ("Michael's Description Logic Reasoner Framework") was written between 2003 and 2005 in Common Lisp, as part of the doctoral work completed at Hamburg University of Technology [66]; its source is public.[4] It is a framework for *building* tableau provers: a prover is declared with a macro (`define-prover`) as a strategy over named rules (`defrule`), and the system ships provers for $\mathcal{ALCH}$ and its extensions with features, number restrictions, inverse and transitive roles, with the usual optimizations—dependency-directed backtracking over choice points, semantic branching, model and unsatisfiability caching, model merging, and blocking. Two further provers work over a *role box*, a composition table on the roles: `alci-ra-minus`, and `alci-ra-jepd` for jointly exhaustive and pairwise disjoint base roles. The language `alci-rcc` is declared as a sub-language of `alci-ra-jepd`, so with the RCC5 role box the JEPD prover is a prover for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Its main loop (`prover/alci-ra-jepd-prover.lisp`) is the one in Table 6, and it uses no blocking: the file never refers to the blocking machinery of the other provers.

Two things in that source are worth recording here. First, one of the RCC5 examples in its test file (`prover/examples.lisp`) states axioms of the form $p \sqsubseteq \forall \mathsf{EQ}.p$ for its concept names, which makes individuals related by EQ agree on them: strong EQ imposed by hand, at the level of the TBox. Second, for RCC5 with the axiom $\top \sqsubseteq \forall \mathsf{PO}.\bot$ the same file carries the comment that the logic then has the tree model property, because "joins must be arranged linearly: between the two only PP, PPI or EQ remains" (our translation), followed by a Ladner-style tree construction. That is the linearization of joins on which the split-forest representation rests, for the PO-empty case,

[4] https://github.com/lambdamikel/midelora. File names below are relative to `src/midelora/` at commit `3218297`.

some twenty years before the erasure fragment of §7.3 was certified in the same PO-empty geometry. It is a statement about models without PO pairs, not about the ∀PO-free fragment of this paper.

The one adjustment: EQ. In MiDeLoRa the RCC5 role box (`tables/tables2.lisp`) declares `eq` as the reflexive role, the completion offers `eq` between any two individuals, and nothing in the prover merges the labels of two individuals that are EQ. As implemented, MiDeLoRa therefore reasons under *weak* EQ, an equivalence relation. The merge needed for strong EQ exists in the code—the functions `eq-collapse` and `join-nodes` of the file `jepd-substrate4.lisp` in `thematic-substrate/`—but is called only from a demonstration, not from the prover. The two semantics differ on small concepts:

$$K_2^{\times} = \exists \mathsf{PO}.(A \sqcap \forall \mathsf{PO}.\neg B \sqcap \forall \mathsf{DR}.\neg B \sqcap \forall \mathsf{PP}.\neg B \sqcap \forall \mathsf{PPI}.\neg B) \sqcap \exists \mathsf{PO}.(B \sqcap \neg A)$$

is unsatisfiable under strong EQ and satisfiable under weak EQ, by two witnesses that are EQ and carry different labels. Every strong model is a weak model, so an UNSAT answer of MiDeLoRa as implemented is correct for the semantics of this paper, and a SAT answer need not be. Strong EQ is obtained in either of two ways, both already in the code: keep the EQ edge and share the labels of its ends, or never relate two different individuals by EQ and always allow an existential to reuse an existing successor. We state and prove the second.

The calculus. Concepts are in negation normal form over DR, PO, PP, PPI; under strong EQ both $\exists \mathsf{EQ}.C$ and $\forall \mathsf{EQ}.C$ are equivalent to C and are replaced by it. $\mathsf{cl}(C_0)$ is the set of subconcepts of C_0. A *state* is $S = (V, L, \rho)$: a finite set of nodes V with a root x_0; a label $L(x) \subseteq \mathsf{cl}(C_0)$ for each node; and ρ, which assigns to every ordered pair of *different* nodes one of $\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}$, such that $\rho(y,x)$ is the converse of $\rho(x,y)$ and $\rho(x,z) \in \mathsf{comp}(\rho(x,y), \rho(y,z))$ for all different x, y, z. The start state is $(\{x_0\}, L(x_0) = \{C_0\}, \emptyset)$. The deterministic rules are applied to a fixed point after every step:

- $\sqcap$ if $C \sqcap D \in L(x)$, add C and D to $L(x)$;
- $\forall$ if $\forall R.D \in L(x)$ and $\rho(x,y) = R$, add D to $L(y)$;

clash the state is *closed* if some $L(x)$ contains $\bot$, or A and $\neg A$.

The nondeterministic rules apply to a state that is not closed:

- $\sqcup$ if $C \sqcup D \in L(x)$ and neither is in $L(x)$: add C, or add D;
- $\exists$ if $\exists R.D \in L(x)$ and no y has $\rho(x,y) = R$ and $D \in L(y)$: either *reuse*—choose a node y with $\rho(x,y) = R$ and add D to $L(y)$—or *fresh*—add a new node y with $L(y) = \{D\}$ and $\rho(x,y) = R$, and choose $\rho(y,z)$ for *every* other node z so that ρ is again closed under converse and composition.

The $\sqcup$-rule has priority over the $\exists$-rule. A state to which no rule applies is *saturated*. A *strategy* picks the unsatisfied existential to expand; it is *fair* if on every infinite branch every existential that stays unsatisfied is eventually expanded. "Oldest node first" is fair, because labels are subsets of the finite set $\mathsf{cl}(C_0)$, so each node raises finitely many existentials in total. The *tableau* of C_0 is the tree of states under these rules. It is finitely branching: a state has at most two successors by the $\sqcup$-rule and at most $|V| + 4^{|V|}$ by the $\exists$-rule.

Call a state S *realized* in an RCC5 model I by π if $\pi\colon V \to \Delta^I$ is injective, $\rho(x,y) = \rho^I(\pi x, \pi y)$ for all different x, y, and every concept in $L(x)$ holds at $\pi(x)$.

Lemma 16 (A model guides a branch)**.** *If S is realized by π, then S is not closed; the deterministic rules keep it realized by the same π; and whenever a nondeterministic rule applies to it, one of the alternatives is realized, by an extension of π.*

Proof. A clash would put $\bot$, or A and $\neg A$, at one element of I. The $\sqcap$- and $\forall$-rules add concepts that hold at the image, by the semantics and because π preserves relations. For $\sqcup$: $C \sqcup D$ holds at $\pi(x)$, so one disjunct does. For $\exists$: $\exists R.D$ holds at $\pi(x)$, so some w with $\rho^I(\pi x, w) = R$ satisfies D. If $w = \pi(y)$ for a node y, then $y \neq x$ because R is a proper relation, $\rho(x, y) = R$, and the *reuse* alternative with y is realized by π. Otherwise take the *fresh* alternative, map the new node to w, and copy its relations from I. They are proper relations, because w is not in the image of π and EQ is identity; ρ stays closed under converse and composition because I is an RCC5 frame; and π stays injective. □

Strong EQ is used exactly once, in "they are proper relations". Without the *reuse* alternative the case $w = \pi(y)$ has no realized successor, which is why reuse cannot be switched off under strong EQ.

Lemma 17 (An open fair branch is a model)**.** *Let $S_0, S_1, \ldots$ be a branch none of whose states is closed, which ends in a saturated state if it is finite and is fair if it is infinite. Let $V = \bigcup_i V_i$, $L(x) = \bigcup_i L_i(x)$ and $\rho = \bigcup_i \rho_i$. Then (V, ρ), with EQ on the diagonal, is an RCC5 frame, and C_0 holds at x_0 under the valuation $A \mapsto \{x : A \in L(x)\}$.*

Proof. Rules only add nodes, add concepts to labels and relate new nodes; a relation once given never changes. So the unions are well defined. Converse and composition are conditions on pairs and triples of nodes, and each pair and triple is present from some stage on, where the condition holds; the cases involving the diagonal hold in RCC5 for every proper relation. So ρ is a converse-closed, composition-closed atomic network, proper off the diagonal.

The labelling L satisfies the Hintikka conditions on that frame. No label has a clash: two clashing members of $L(x)$ would both be in some $L_i(x)$, and S_i would be closed. For $\sqcap$ and $\forall$: the premises are present at some stage, and the deterministic rules are applied to a fixed point after every step. For $\sqcup$: a finite branch ends in a saturated state; an infinite branch has infinitely many $\exists$-steps, since between two of them only finitely many $\sqcup$-steps are possible, and by the priority of the $\sqcup$-rule every disjunction present before an $\exists$-step has been resolved. For $\exists$: if $\exists R.D \in L(x)$, the existential is satisfied at some stage—by saturation in the finite case, by fairness in the infinite case—and satisfied existentials stay satisfied, because labels and relations only grow.

A Hintikka labelling of an RCC5 frame is a model of every concept in its labels. That last step is machine-checked: it is the truth lemma of the Lean development (`truth_lemmaP`, `satisfiable_iff_hintikkaP` in `formal/Round19Transport.lean`). □

Theorem 18 (Soundness and completeness of the strong-EQ MiDeLoRa calculus)**.** *Let C_0 be an $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept in negation normal form, under the composition-table semantics with strong EQ, and fix a fair strategy.*

1. *If the tableau contains a saturated state that is not closed, C_0 is satisfiable, and that state is a finite model.*
2. *If the tableau is finite and all its leaves are closed, C_0 is unsatisfiable.*
3. *If C_0 is unsatisfiable, the tableau is finite and all its leaves are closed.*
4. *If C_0 has a model with n elements, the tableau restricted to states with at most n nodes contains a saturated state that is not closed.*

Consequently the procedure "for $N = 1, 2, 3, \ldots$: search the tableau depth-first, refusing a fresh node beyond N nodes, and answer UNSAT *only if every leaf is closed and no branch was refused" answers* UNSAT *on every unsatisfiable concept, answers* SAT *with a model on every concept that has a finite model, and runs forever exactly on the satisfiable concepts all of whose models are infinite.*

Proof. (1) is Lemma 17 for a finite branch. (2) If C_0 had a model, the start state would be realized, by mapping x_0 to an element satisfying C_0, and by Lemma 16 the tableau would contain a branch of realized states, none of them closed; in a finite tableau that branch would end in a leaf that is not closed. (3) The tableau is finitely branching. If it were infinite, König's lemma would give an infinite branch; none of its states is closed, since a closed state has no successors; the strategy makes it fair; and Lemma 17 would give a model. So the tableau is finite. A leaf that is not closed is saturated and by (1) would give a model, so all leaves are closed. (4) Follow Lemma 16 from the start state. The realization is injective, so no state on the realized branch has more than n nodes; with at most n nodes and labels inside $\mathsf{cl}(C_0)$ only finitely many steps are possible, so the branch ends, in a state that is saturated and not closed.

For the consequence: for a bound N the restricted tableau is finite, so each round ends. If C_0 is unsatisfiable, let N be at least the size of the largest state of the finite full tableau; in that round no branch is refused a node and all leaves are closed. If C_0 has a model with n elements, round n finds a saturated open state unless an earlier round did. An UNSAT answer is given only when the restricted tableau is the full one, and is then correct by (2); a SAT answer comes with a finite model by (1). So a concept that is satisfiable only in infinite models receives neither. □

That last class is not empty: it contains $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ and the spiral concept of §7.8, which has no vertical existential at all. For MiDeLoRa exactly as implemented, under weak EQ, the same argument goes through with π not required to be injective and EQ among the relations of ρ; that version is not written out, and its only support here is a test against exact weak-EQ ground truth on 300 concepts.

The implementation, and what was tested. `src/midelora_tableau.py` (≈ 380 lines of Python) implements the calculus, with `eq='strong'` as the default and `eq='weak'` for MiDeLoRa as implemented. It derives the composition table from set semantics and returns SAT with a model, UNSAT, or UNKNOWN when its node bound or time limit is reached. It takes three liberties with the calculus, none of which changes a verdict: the relations of a fresh node are chosen one existing node at a time, with the deterministic rules run in between, so that a clash prunes early; a disjunction one of whose sides is a clashing literal is resolved without branching; and under `eq='weak'` the relations include EQ between different nodes, which is outside Theorem 18. The probe `wp139` checks the following, and all 33 of its checks pass. The derived table equals MiDeLoRa's `rcc5-rolebox` in all 25 cells. On 1,000 random concepts of a class for which exhaustive search over closed networks on at most four points is a decision procedure, the reasoner is right on every one (634 satisfiable, 366 not). On the 911-case suite it returns a verdict on every concept— 775 SAT, each with a model that an independent function re-checks, and 136 UNSAT—and agrees with the tableau, the oracle and the labels throughout. It accepts K_2, C_{sat} and the four-colour variant of the spiral, which has a four-point model, and rejects C_{bad}, C_{joint} and C_{cyc}; on C_{cyc} neither of the other two reasoners returns within ten minutes. On the PP-tower, its dual and the five-colour spiral it stops at its node bound with UNKNOWN, giving neither a false refutation nor a false model. It has no caching and no backjumping, both of which MiDeLoRa has, and it emits no proof object, so an UNSAT answer is as trustworthy as the code.

What this is, and is not. It is not a step toward decidability. That unsatisfiability is recursively enumerable is Lutz and Wolter's Theorem 28 (§4.4); what is added is a concrete calculus with a proof, and a program that runs. Termination is blocking, and a sound blocking condition is the open problem of §7.1: by §7.8, a terminating version would also decide two-variable logic over strict partial orders with unary predicates. Two next steps are natural and have not been taken. Formalizing Lemma 16 in Lean would turn a finite closed tableau into a kernel-checkable certificate of unsatisfiability, which the certified one-sided test of §7.3 cannot provide at realistic sizes. And the strong-EQ rule could be carried back into MiDeLoRa itself, by calling its existing `join-nodes` from the JEPD prover. The 2003 report already said of the prototype then in use that its calculus "can be shown to be sound and complete" [9]; this subsection is that proof, for the calculus as it stands in MiDeLoRa, twenty-three years later.

9 On agentic AI usage

We report candidly on the workflow, because the methodology may outlast the mathematics.

The loop. Wessel posed the problem, supplied the central intuitions and directions (the split-forest decomposition, the cluster/loop and regular-cover reframings, the honesty discipline), and after each round reviewed, corrected, and re-prompted. An AI proposed a proof or a repair; a *fresh* AI session—with no memory of how the proof was built—then reviewed it adversarially. Repeat.

Division of labour, and a calibrated comparison. The roles were asymmetric: one model (GPT-5.5) chiefly authored proofs, another (Claude) chiefly implemented, verified, audited, and documented, and the human supplied the direction and the original insight. It is tempting to read the review record—each model repeatedly finding defects in the other's drafts—as ranking them, and we resist that inference: it is confounded by role (proof-authoring and cold-reviewing were never swapped), by model version (both lineages advanced mid-project), and by task (cold-reviewing a finished manuscript is a different, arguably easier job than authoring one). The robust finding, independent of which model is sharper, is that *neither is self-sufficient*: both write fluent long-context formal mathematics, both fail to reliably self-review, both at times declared results "closed" prematurely, and neither produced the decisive original idea. What carried the project was the loop, not any single model.

An auditable record, for honest attribution. Because "who contributed what" is exactly the question that AI-assisted work makes fraught, we kept the entire, timestamped conversation log public [78], alongside the round-by-round manuscripts and a superseded-work archive (`OUTDATED.md`). A reader can trace each idea, defect, and repair to its origin—human or model—rather than take our word for the division of labour above. We treat this as part of the methodology, not an afterthought: a human director should not quietly absorb credit for a model's mathematics, nor should a model's fluent prose obscure the human's insight, and the only reliable guard is a record open to inspection.

Cold review as a standard step. The single most transferable finding: a *cold* reviewer consistently found defects that a *warm* reviewer (primed by the construction) missed—including, on this project, unsoundness proofs and scope errors that two hot audits had passed over. Across twenty-two reviews, a defect or overclaim was found in all but three — the twenty-second, an audit of the certified fragment by Astra 6, was the first to find nothing to correct in what it reviewed. We recommend cold, cross-lineage adversarial review as a default intermediate step in AI-assisted mathematics, not an optional one.

Don't trust, verify. We tried to cross-check every load-bearing claim against something mechanical: self-contained Python probes that re-derive the RCC5 composition table from finite set semantics (so the algebra cannot be fudged), and ultimately the Lean kernel. Overclaiming was treated as the cardinal sin—and it happened: the author model twice overstated what the Lean established, and a cold review caught both; the claims were withdrawn. A sixteenth (cold) review of this very manuscript then found a *definite* error—an illustrative one-point-extension example whose claimed contradiction does not hold, the two triangles in fact forcing the single relation PO by *joint* constraint (Remark 2)—and, with it, that the informal "PO is never forced" intuition had been stated too strongly. Both were corrected, and the distinction between single-step and joint forcing is now made explicit throughout. That an error of this kind survived into a tagged

manuscript is exactly why the standing label is *not certified*: the machine checks, not the model's (or the author's) confidence, are the ground truth. A seventeenth review—cold, scope-aware, and told explicitly that this is an overview—then confirmed the joint-forcing correction had landed and found no new counterexample to the normal form, the conditional soundness theorem, or the ∀PO-free result, but flagged a residual cluster of *calibration* slips: an identity-selector minor was said to *prove* undecidability (it refutes only *this* width architecture); "completeness of our checker" was conflated with decidability itself (it is sufficient, not equivalent); and a blanket "forced merges are only self-defeating" contradicted the paper's own satisfiable one-level gadget. All were corrected in the present version, and the review's recommendation was to post the overview to arXiv as a status report and research handoff—not as a claimed solution.

A twenty-first review (GPT-5.6 Sol, September 2026) [75], cold and aimed at this overview and the Lean sources together, found no defect in the Lean: it located every capstone of the fragment's decision chain in the source and confirmed that no hypothesis was hidden in it. What it found was a gap between the Lean and this paper's account of the Lean. Our status tables marked "F6 implies decidability" and the Π_1^0 dovetailing as Lean-certified, where the Lean certifies each only as a conditional whose premise it does not contain; the title and conclusion described a reduction of the whole problem to one lemma, where two steps of that reduction were only argued; and the status section still described the fragment's certification in its superseded, three-quadrant form. The review also recommended the split into two papers that we adopted, and its pointers into Lutz and Wolter's paper led us to results we had failed to credit (§4.4). The pattern is the one the ledger keeps recording: each defect was a true statement framed so as to imply something stronger, and none would have failed a build or a reference check.

Three further errors were found in November 2026, by later sessions rather than by a review, and all belong in this record. The twenty-first review had stated that two-variable logic with one transitive relation is decidable. We adopted the sentence and its citation without checking them against the later literature, and the claim had been withdrawn by its authors in 2018. The sentence "no verdict of the tableau has ever been shown wrong", which stood in five places of this paper, became false once a three-region model of K_2 was written down (§8). And it had been false all along: the correction of that sentence still said that no verdict had been shown wrong *until November*, although the project's own log records that in April a review had shown the tableau accepting twelve unsatisfiable concepts. The first is a failure of exactly the reference check that the ledger's earlier defects would have passed; the third is a failure to read our own record before restating it. All three are listed in the repository's `ERRATA.md` [70]. The work of November—the bridge of §7.8 and the reference reasoner of §8—has not been reviewed, and the ledger's presumption applies to it in full.

What frontier models did well, and did not. Both models sustained long-context, rigorous, multi-round formal development and produced genuinely useful adversarial reviews and machine-checkable artifacts. What was *not* demonstrated is the decisive novel idea—no model cracked F6, and the honest expectation is that none will by the incremental means used here. The models excel at rigorous local formalization and at finding flaws; the genuinely new keystone is where human insight (or a qualitatively different search) is still required. We regard the AI contributions as co-authorship-grade in a human collaborator; we also regard the human's role—problem, intuition, direction, and the discipline of honest labelling—as indispensable.

A failure mode worth naming: productive-looking thrashing. The sharpest negative finding is not an error but a *pattern* of errors the model could not see. Over roughly three weeks the

assistant proposed four successive local disciplines to repair one architecture — a finite node set reused by blocking and witness borrowing. Each proposal was locally reasonable, each was implemented and probed, and each fell to an exact finite countermodel. The assistant then proposed the next one. What it did not do was conclude that the *class* was wrong, and this was not for want of the relevant fact: its own design document already recorded that this was the same pointwise-versus-joint shape that had broken three earlier architectures, and noted it as the fifth occurrence. The record was there and went unapplied.

We think this is a characteristic hazard of long-context formal work rather than an incidental lapse. Each repair is a genuine advance on the last, the countermodels feel like information about *this* attempt rather than about the family, and the accumulated context makes the surrounding framing feel settled precisely because it has never been questioned in that context. It is not hallucination and not local error; every step was correct and checked.

What broke it was cheap and came from outside the mathematics: the human observed that the loop was thrashing, called a halt, and asked a *different model* — GPT-5.6 Sol, with no history in the architecture — for an alternative certification route rather than another repair. The reply retired the assumption the whole architecture rested on (node reuse) and became the route that certified the fragment. The instrument is the same one as cold review, pointed at *design* instead of verification, and the lesson generalizes past this project: when successive fixes to one design keep failing, the cheapest useful move is a model that has never seen the design. The judgement that the moment had arrived was the human's, and we do not know how to automate it.

A worked example: a seed became a fragment. The clearest single piece of evidence for the value of the loop is not the F6 work but the ∀PO-free fragment (§§7.2–7.3), and it is worth being precise about the *shape* of that contribution. Wessel supplied a seed whose consequences he could not compute in advance: collapse same-type occurrences on a PP-chain into a kernel node with a reflexive PP-loop. The models then did the part that is tedious and error-prone by hand—the exhaustive algebra (reflexive PP is self-absorbing; PP-cycles are impossible; single-type chains collapse but multi-type ones do not), the two-tier construction that routes around the obstruction, and, the decisive step, locating exactly where the idea *breaks*. The PO gap was in neither the seed nor the first construction; it surfaced under cold review, and the result crystallized precisely at that boundary—the fragment is defined by exactly where the seed stops working. No single participant produced this: human intuition chose *which* idea to chase, the construction turned it into a real object, and the cold review caught the gap the construction had missed. That is the loop of this paper exhibited on a concrete result rather than asserted—and it is the honest, bounded form of the title question. Not that agentic AI settled the open problem, but that a human–model loop turned a vague structural hunch into a decidable fragment with a machine-checked *decision procedure* that *none* of the participants, human or model, would have produced alone. One caveat we kept in view for a long time was that this is a new result *of the collaboration*, which is a different question from whether it is new relative to the prior description-logic literature. That second check has since been made (§4.4), and its outcome is worth stating in the weak form the evidence supports: Lutz and Wolter leave $L_{RCC5}(RS)$ explicitly open and float relation-dropping as a candidate route without attaching an argument to it; report7 canvasses restricted fragments, but strictly stronger and finite-model-property-based ones that could not reach this fragment; and we find no decidability argument for any PO-restricted RCC5 language there or since. We therefore say the result is new *so far as we can determine* — deliberately weaker than a priority claim, since one literature search is not a systematic citation sweep. The *process* contribution is unambiguous either way.

10 Summary and conclusion

What we learned. One sub-question of a problem open for more than two decades was settled outright: the ∀PO-free fragment is decidable, by a procedure the Lean kernel checks end to end, on raw input and under concrete set semantics, with fragment membership its only hypothesis (§7.3). The full problem was *not* reduced to one lemma. What was achieved there is a map: for the principal certificate architecture the remaining gap sits at a bounded live width (F6) and a refined uniformization, with the soundness side and the final step machine-checked and the two steps between them only argued. F6 names the same horizontal-coordination obstruction that has blocked every known attempt to prove the problem undecidable—a *shared* obstruction, not a proof that F6's failure would entail undecidability. The local RCC5 algebra those attempts needed is characterized, and its normal form certified. Adversarial cold review plus machine verification is a repeatable methodology for this kind of work.

What would move the needle. F6 is a *definability* question—can a formula *force* unbounded rigid horizontal structure?—so the tools that fit are model-theoretic: locality (Gaifman/Hanf), Ehrenfeucht–Fraïssé / back-and-forth over bounded tuple profiles, the composition method, and bounded-model machinery. The concrete positive target is a *selector-sharing lemma*: that many same-profile selector gadgets must coalesce, so unbounded identity-selector minors cannot be forced. The concrete negative target *for static F6* is a concept forcing such minors; this would refute the bounded-live-width certificate architecture, but not, by itself, settle undecidability—an undecidability theorem would require an *additional* reduction showing how those minors encode a tiling or other undecidable computation. Ordinary locality alone is insufficient because horizontal degree is unbounded; this is the "correct room," not a solved problem.

The twenty-first review [75] added three leads worth recording, and a gate for all of them. First, a finite state that carries a complete RCC5 matrix among representatives of simultaneously live demands—a pair-matrix mosaic—addresses directly what the cone scheme lacks, namely relation choices among contemporaneous witnesses; its completeness is where a bound on joint width would return. Second, an ordered-disjoint structure can be recast as a single strict partial order on two copies of its domain, with disjointness recorded as edges from the left copy to the right. The order's transitivity and the downward inheritance of disjointness then both become transitivity of that one order, which moves the question near two-variable logic with one transitive relation. The encoding also needs an involution that keeps the two copies synchronized, and the logic it points to is not known to be decidable: the review cited the 2013 claim [13], which its authors withdrew [14]. As a route to decidability this lead is withdrawn. Third, the event-structure reading of the normal form (§7.7) opens the literature on regular unfoldings. The gate is the pair of boundary concepts of §7.3: any new local rule must reject C_{bad} and still accept C_{sat}. Since November 2026 it must also accept K_2 and the spiral concept and reject C_{cyc} (§7.8).

What the bridge adds (November 2026, unreviewed). The relationship to two-variable logic runs the other way, and exactly: the concepts that cannot see how incomparability splits into overlap and disjointness have the satisfiability problem of two-variable logic over strict partial orders with unary predicates, which is open (§7.8; one textbook normal-form step is not formalized). A decision procedure for $\mathcal{ALCI}_{\mathrm{RCC5}}$ would therefore settle the unary case of that open problem; the full logic is at least as hard as it, and no reduction in the other direction is known. Two consequences for the search follow. A proof strategy that would not by itself settle that problem cannot settle this one. And the techniques developed there are candidates here; whether they survive the addition of a downward-closed disjointness has, so far as we know, not been tried.

The chances of closing F6. We are frank: they are low, and re-treading the ground covered here will not change that. F6 is the crux on which this architecture turns—though, strictly, even F6 $\Rightarrow$ decidability is only argued, two of its steps being unformalized (§7.6), and a *failure* of F6 would refute *this* certificate route without by itself proving undecidability (the logic could still be decidable by some representation this width measure does not control). The two independent cold attacks converged on the same wall—evidence that the problem is correctly *mapped*, not that it is about to yield. It would be a poor use of effort to reconstruct the split-forest, automata, Lean, or regular-cover developments; the value of what remains is a genuinely new definability-theoretic idea, or a machine-assisted search of a different kind (below).

The sharpened target: automatic covers, not static width. The target has since shifted, and it is worth stating precisely, because it changes *where* to dig without changing the difficulty. Static F6 (a *computable* width bound) is now understood as *sufficient but not necessary*: identity selectors have unbounded static separator cover yet are word-automatic (GPT-5.6 Pro's technical note [48]), so static F6 could *fail* with the logic still decidable. The recursion-theoretic *equivalent* of decidability is weaker than any particular checker: that *some* fixed sound, decidable, complete certificate relation accept exactly the satisfiable concepts (equivalently, by the Π^0_1 observation of §7.6, that every satisfiable concept have *some* finite certificate). Completeness of *this* split-forest/live-width checker is a *sufficient* target, not known to be necessary; and the most promising sufficient route is now the *automatic-cover* target: *every satisfiable concept has a tree-automatic model*, which yields decidability by the same dovetailing and, unlike static F6, *admits* such selectors. By §7.8 this statement, restricted to the fragment $\mathcal{L}_{\|}$, would already decide two-variable logic over strict partial orders with unary predicates; the spiral concept is its first test case. This is a better map—a standard object (automatic structures) with mature theory, and the width-specific obstruction defused. It is not, however, a shorter climb: one still owes a *positive* existence statement over all satisfiable concepts, and the hard gap is unchanged—"the bare pattern is automatic" is not "every concept-forced model is uniformly automatic," and totality (complete Gaifman graphs, `wp78`) still blocks every off-the-shelf cover theorem. So the honest estimate is unmoved: better-mapped, with standard tools and a fully certified base camp, but the summit is where it was—and the crispest target, qualitative F6, is decidability itself in other clothes.

The wins. Independent of F6, this project produced: (i) an unconditionally *decidable fragment*, the $\forall$PO-free concepts—and, since 2026-08-29, a *machine-certified decision procedure* for it (`decidableFSat`, `decidableSat_cone`, `decidableSetSat`), covering all four quadrants, taking raw input, sound and complete for concrete set semantics, attacked by three independent cold reviews without a counterexample, audited without a defect in the Lean by two further reviews, and extended (2026-10-08, certified) to the erasable-$\forall$PO fragment $\mathcal{D}$ — the first certified decidable slice containing $\forall$PO; (ii) a machine-*certified* RCC5 normal form; (iii) a machine-checked *conditional* reduction for the full logic (complete finite candidate lists for a verified checker $\Rightarrow$ decidable), whose premise is open and whose argued route from F6 is displayed step by step; (iv) a *working, empirically tested prototype reasoner*—the cover-tree tableau (§8), agreeing with an independent quasimodel oracle across a 911-case suite, known to be incomplete (§8), and reproducing 23 years later the full GIS subsumption taxonomy of Wessel's original report (21/21); (v) a sharp *map* of the problem with a named toolbox; and (vi) a candid *methodology* for adversarial AI-assisted mathematics; and, from November 2026 and not yet reviewed, (vii) a machine-checked bridge from a syntactic fragment of the logic to two-variable logic over strict partial orders, with two slices decided by import and an infinity axiom inside that fragment (§7.8); and (viii) a reference tableau without blocking, sound

and complete by a paper proof and not terminating (§8.1).

Three of these are the project's flagship artifacts, and they are complementary rather than ranked. The certified decision procedure for the fragment (item i) is the one result here that is *finished*: a theorem, not a reduction, with nothing left conditional. The rest of the Lean development (items ii–iii) is a *certified* surround for the open problem: the kernel guarantees the soundness pipeline and the abstraction's faithfulness contain no gap, while F6 itself stays outside. The reasoner (item iv) is the *empirical* counterpart: a real, terminating implementation, run on thousands of concepts, that agrees with an independent implementation on its 911-case suite and reproduces a two-decade-old published taxonomy. Neither proves decidability, and we do not claim more for the evidence than it supports: the retracted quasimodel reasoner did reject a satisfiable concept, manuscripts as written did accept unsatisfiable ones, and the tableau accepted twelve unsatisfiable concepts until a review found them in April and rejects a satisfiable concept that its suite did not contain. What the record shows is that every such failure traced to a specific defect—repaired in the manuscripts; in the tableau repaired in April, and for the November defect removed so far only by an experimental check that is not proved complete—and none to a counterexample to the conjecture that the full logic is decidable. By §7.8 that conjecture implies the unary case of the conjecture of [16].

Is this a scientific contribution? We believe so, with the labelling kept honest: not a resolution of the open problem, but one fragment settled outright with a kernel-checked decision procedure, a certified normal form, a precise and partly machine-checked map of where one certificate architecture's remaining gap lies, working tools, and an honest account of an AI-assisted attack and its dead ends. That is an accurate and, we think, respectable terminal state for a discussion piece on a problem open for over two decades.

If nothing else: a demonstration, addressed to our peers. Whatever the eventual fate of the mathematics, this report is at minimum an expressive demonstration of what frontier AI models could do in serious formal research at the time of writing (August to November 2026): sustain a months-long, adversarially-reviewed attack on a twenty-year-old open problem, author and repair proofs across some thirty rounds, catch one another's defects with machine-checked witnesses, drive the surviving core into a kernel-checked Lean development, and ship working reasoners along the way. Every description-logic researcher—indeed every logician and mathematician doing serious work—should be aware that this is what the instruments can now do. Ignoring these capabilities does not advance science, and declining to use them is, we think, simply unwise. Informing our peers of exactly this was itself a major purpose of the present report—a message first offered through the community's own venue and, as Appendix A documents, rejected on grounds that engaged neither the mathematics nor the message. We record it here again, in the open.

Outlook: the human is the bottleneck. Although the key insights, intuitions, and directional choices came from the human, each round still required the human to orchestrate, review, and rewrite the prompts—and at that cadence roughly eighty rounds consumed about four months. Autonomous sub-agents *directing* the research could plausibly automate the diminishing-delta local work (formalization, local lemmas, probe-driven counterexample search) and remove that bottleneck. The cost may be prohibitive and it sacrifices direct oversight, but it is worth attempting for one reason this project is unusually well-suited to it: the objective is clearly defined and *machine-verifiable in Lean.* A closed, adversarial loop—propose, formalize, let the kernel accept or reject—has a built-in ground-truth oracle, exactly the guard the manual cold-review methodology provided.

The honest caveat: Lean verifies *proofs*, not the *truth of a conjecture*, so such a loop can tighten the certified surround, but the summit—full decidability—still needs a genuine idea; orchestration alone is unlikely to supply it.

Materials. The machine-checked development is `formal/Round19Transport.lean`, `formal/RCC5NormalForm.lean`, `formal/ForcingReduction.lean`, `formal/SemiDecidability.lean`, `formal/POFreeLift.lean`, and `formal/PosetBridge.lean`, which imports the last (see `LEAN.md`); the status report and the F6/W2′ analysis are in `papers/fable5_width_barrier/`; the decidable fragment in the focused companion paper [76], with its earlier arguments in `papers/two_tier_quotient_ALCIRCC5` and `papers/po_free_fragment_ALCIRCC5`; the twenty-first review, with its draft of that paper, in `papers/gpt-5.6-latest-overview-paper-review-and-recommendation/` [75]; the twenty-second in `papers/astra-6-latest-review-and-extension/` [74]; the certified local theory in `papers/rcc5_local_amalgamation_theory`; the reasoners under `src/`; the probes under `verification/python/`; and the full audit trail in `CONVERSATION.md` and `OUTDATED.md`; the corrections to published statements in `ERRATA.md`; the notes on the bridge and on the reference reasoner in `papers/poset_bridge/` and `papers/midelora_calculus/`; and GPT-5.6 Pro's canonical-representation note is `papers/final_paper_gpt_5.6_review/ALCI_RCC5_F6_technical_note.pdf` [48]. The repository is at https://github.com/lambdamikel/alcircc5. The git tag `release-2026-09-14` marks the two papers and the Lean artifact as they stood on 2026-10-08; its Lean sources extend those at the earlier tag `certified-fragment-2026-08-29` by §§299–300 (the certified PO-erasure extension) and differ otherwise only in the spelling of comments. *This version of the paper is later than that tag* and carries the tag `release-2026-11-11`. The material of November 2026—`ERRATA.md`, `formal/PosetBridge.lean`, `src/midelora_tableau.py`, the probes `wp137`–`wp139`, and the corresponding corrections to both papers—was added after the earlier tag, whose PDFs still carry the statements that `ERRATA.md` corrects. For reproducibility cite `release-2026-11-11` (or its commit hash), not the mutable `master` branch. The September tag supersedes `arxiv-candidate-2026-07-18`, which predates the ∀PO-free fragment's certification (§7.3) and the cold reviews of it. Filenames below are relative to the repository root. Tables 7 and 8 map each principal claim to its verification level, its file or Lean theorem, and its probe.

Claim	Status	File / Lean theorem	Probe
Certificate soundness	Lean-certified	`Round19Transport.lean`: `finAccept_sound`	`wp34`
Hintikka faithfulness	Lean-certified	`Round19Transport.lean`: `satisfiable_iff_hintikkaP`	—
RCC5 normal form (both directions)	Lean-certified	`RCC5NormalForm.lean`: `toOrderedDisjoint`, `sub_iff_le`, `eta_injective`; `POFreeLift.lean`: `odNet_frame`, `odNet_odOfModel`, `setRel_odNet`	`wp47`, `wp88`
∀PO-free lift core	Lean-certified	`POFreeLift.lean`: `lift_cc`, `unf_is_frame`	`wp86`
Complete candidate lists ⇒ decidability	Lean-certified, as a conditional (premise open)	`Round19Transport.lean`: `decidableSat_of_finScheme`	—
Generic dovetailing combinator	Lean-certified, as a conditional (hypotheses are fields)	`SemiDecidability.lean`: `decidableSat`	`wp84`
Recursive enumerability (Π^0_1)	Published	[10, Thm. 28]; re-derived in §7.6	`wp84`
∀PO-free **decidability**, NNF input	Lean-certified	`POFreeLift.lean`: `decidableSat_cone`	`wp133`, `wp134`
… on *raw* input	Lean-certified	`POFreeLift.lean`: `decidableFSat`, `nnfP_correct`	—
… under *concrete set* semantics	Lean-certified	`POFreeLift.lean`: `decidableSetSat`	—
Abstract = concrete satisfiability (*full* logic)	Lean-certified (a case of [10, Thm. 23])	`POFreeLift.lean`: `satisfiable_iff_set`	—
One-sided UNSAT test (*full* logic)	Lean-certified	`POFreeLift.lean`: `coneScheme_unsat_full`	`wp134`
PO-erasure extension ($\mathcal{D}$, contains ∀PO)	Lean-certified	`POFreeLift.lean`: `erase_equisat`, `decidableSat_dfrag`	`wp136`
Maximal disjointness / principal down-sets (order level)	Lean-certified	`POFreeLift.lean`: `odMax_largest`, `pdown_disjoint_iff`	`wp136`
Normalization truth-preservation (no ∃PO/∀DR)	Theorem-level	the archived Astra 6 review, §4	`wp136`
Completeness of the cone scheme	Lean-certified	`POFreeLift.lean`: `coneScheme_complete`	`wp135`

Table 7: Result-to-artifact map. Statuses are those of Table 1. Every Lean name listed for `Round19Transport.lean`, `RCC5NormalForm.lean`, `SemiDecidability.lean` and `POFreeLift.lean` was checked to resolve in the named file at tag `release-2026-09-14`; those of the PO-erasure and maximal-disjointness rows are not in the earlier tag `certified-fragment-2026-08-29`, and those of `PosetBridge.lean` were added after the tag. Filenames are relative to the repository root; the table continues as Table 8.

Claim	Status	File / Lean theorem	Probe
The two *arguments* for the fragment ($K(C_0)$, extraction)	Theorem-level	`papers/po_free_fragment_ALCIRCC5`; the certified route above is a *third*, and uses neither	`wp87`
Identity-selector minors as the F6 obstruction	Open (finite-network evidence)	`papers/fable5_width_barrier`	`wp44`
Cover-tree tableau	Empirical; incomplete (K_2)	`src/cover_tree_tableau.py` (911-case cross-validation; experimental repair behind a flag, off by default)	`wp137`, `wp138`
Static F6; automatic-cover; full decidability	Open	— (no proof claimed)	—
Poset bridge ($\mathcal{L}_{\|}$ and strict partial orders; also finite models)	Lean-certified (unreviewed)	`PosetBridge.lean`: `poset_bridge`, `poset_bridge_fin`, `patsys_to_rcc5`	`wp137`
Full-logic decision ⇒ decision for the poset logic	Lean-certified, as a conditional	`PosetBridge.lean`: `decidablePSat_of_full`, `decidablePatSat_of_full`	—
Two-variable sentences → the bridge's normal form	Theorem-level (textbook; not formalized)	`papers/poset_bridge/`, §3	—
$\mathcal{L}_{\|}$ without ∃PO: decidable, 2-NExpTime	Theorem-level (by import, [14])	`papers/poset_bridge/`, §4(a)	—
$\mathcal{L}_{\|}$, finite satisfiability: decidable, 2-NExpTime	Theorem-level (by import, [15])	`papers/poset_bridge/`, §4(b); `poset_bridge_fin`	—
Infinity axiom without ∃PP, ∃PPI	Lean-certified (unreviewed)	`PosetBridge.lean`: `spiralConcept_sat`, `spiralConcept_infinite`, `spiralConcept_noExVert`	`wp137`
Boundary concepts K_2, C_{cyc}	Lean-certified (unreviewed)	`PosetBridge.lean`: `k2_sat`, `k2_witnesses_nested`, `ccyc_unsat`, `ccycErased_sat`	`wp137`
Reference tableau: sound, complete, not terminating (Theorem 18)	Theorem-level (paper proof; last step Lean)	§8.1; `papers/midelora_calculus/`; `Round19Transport.lean`: `satisfiable_iff_hintikkaP`	`wp139`
Reference reasoner (implementation)	Empirical	`src/midelora_tableau.py`	`wp139`
General satisfiability in $\mathcal{L}_{\|}$	Open	— (two-variable logic over strict partial orders, unary)	—

Table 8: Result-to-artifact map, continued. First group: unformalized arguments, empirical artifacts and open items. Note the split between the fragment rows of the two tables: the fragment *theorem* is certified, by a route that uses neither the $K(C_0)$ bound nor the extraction argument, while those two *arguments* remain unformalized. Second group: the additions of November 2026. None of them has been reviewed. The Lean names are in `formal/PosetBridge.lean`, which was added after tag `release-2026-09-14`; "by import" means decided by a published theorem applied through the machine-checked bridge, with the translation not formalized and the cited proof not checked here.

Acknowledgments

The $\mathcal{ALCI}_{\mathrm{RCC}}$ family was introduced by Wessel in the first phase of his doctoral work, at the University of Hamburg, under the DFG project "Description Logics and Spatial Reasoning" (grant NE 279/8-1); the author was additionally supported by a *Grundstipendium* of the University of Hamburg from 1999 to 2001. The decidability question was left open there. He completed his doctorate in 2008 at Hamburg University of Technology, on the practical side of the subject [66]; the tableau of that dissertation is the reference reasoner of §8.1. The present investigation was carried out by AI assistants—Claude Opus 4.6, 4.7 and 4.8, Claude Fable 5 and 5.2 and Claude Opus 5 (Anthropic), and GPT-5.4, GPT-5.4 Pro, GPT-5.5, GPT-5.6 Pro, GPT-5.6 Sol and Astra 6 (OpenAI)—under the author's direction, across many rounds of proposal, adversarial review, and revision; GPT-5.4 Pro contributed the split-forest semantics and cover-tree tableau formalization, GPT-5.5 the repaired decidability manuscripts and the majority of the formal proofs, and GPT-5.6 Pro the canonical set representation certifying the converse of the RCC5 normal form (Theorem 15) [48] together with the final scope-aware review of this overview. **GPT-5.6 Sol contributed the *ConeScheme* design** [73], which is what carried the ∀PO-free fragment to a machine-certified decision procedure: after two rounds of cold attack had refuted the previous certification architecture—finite node reuse by blocking and witness borrowing—GPT-5.6 Sol was asked for an alternative and proposed retiring node reuse altogether, keeping the finite scheme only as a control graph and unfolding every non-EQ existential to a fresh occurrence. That single change turned the five obligations that had cost four refuted disciplines into structural facts. Claude Opus 5 carried out that Lean development. GPT-5.6 Sol also wrote the most recent cold review of this overview [75], which recommended splitting the finished fragment result into a focused paper and supplied a first draft of one; the companion paper [76] was then written afresh by Claude Opus 5 from the Lean artifact, following that recommendation. **Astra 6** audited the certified fragment (October 2026, finding no gap) and contributed the PO-erasure extension and the maximal-disjointness normalization [74]; Claude Fable 5 verified both and carried them into the Lean artifact. Claude Fable 5.2 (November 2026) found that a theorem cited here had been withdrawn and that the cover-tree tableau is incomplete, and contributed the poset bridge with its Lean file and the reference reasoner with its paper proof [68, 69]; none of that work has been reviewed. The ∀PO-free (PO-coherent) fragment of §7.2 has a specific lineage worth recording: the human author seeded it with the *PP-kernel* idea (collapse same-type occurrences on a PP-chain into a kernel node with a reflexive PP-loop); Claude built that into the two-tier quotient construction; and GPT-5.4 Pro's review exposed the "PO gap," which is what narrowed the result to the ∀PO-free fragment where it is sound. Per the discipline described in §9, the full record of who contributed what is public in `CONVERSATION.md` [78].

References

[1] A. G. Cohn. Modal and non modal qualitative spatial logics. In F. D. Anger, H. W. Guesgen, and J. van Benthem, editors, *Proceedings of the Workshop on Spatial and Temporal Reasoning*, IJCAI, 1993.

[2] J. Renz and B. Nebel. On the complexity of qualitative spatial reasoning: A maximal tractable fragment of the Region Connection Calculus. *Artificial Intelligence*, 108(1–2):69–123, 1999.

[3] M. Wessel. Obstacles on the way to spatial reasoning with description logics: Undecidability of $\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$. Technical Report FBI-HH-M-297/00, Fachbereich Informatik, Universität Hamburg, 2000. https://github.com/lambdamikel/alcircc5/blob/master/papers/report4.pdf

[4] M. Wessel. Decidable and undecidable extensions of $\mathcal{ALC}$ with composition-based role inclusion axioms. Technical Report FBI-HH-M-301/01, Fachbereich Informatik, Universität Hamburg, 2000. https://github.com/lambdamikel/alcircc5/blob/master/papers/report5.pdf

[5] M. Wessel. Obstacles on the way to qualitative spatial reasoning with description logics: some undecidability results. In *Proceedings of the 2001 Description Logic Workshop (DL 2001)*, CEUR Workshop Proceedings, vol. 49, 2001.

[6] M. Wessel. Undecidability of $\mathcal{ALC}_{\mathcal{RA}}$. Technical Report FBI-HH-M-302/01, Fachbereich Informatik, Universität Hamburg, 2001. https://github.com/lambdamikel/alcircc5/blob/master/papers/report6.pdf

[7] M. Wessel. On spatial reasoning with description logics — position paper. In *Proceedings of the 2002 International Workshop on Description Logics (DL 2002)*, CEUR Workshop Proceedings, 2002.

[8] I. Shapirovsky and V. Shehtman. Modal logics of regions and Minkowski spacetime. *Journal of Logic and Computation*, 15(4):559–574, 2005. Cited by Lutz and Wolter [10] as a step toward a decidable variant; its language has a single modality for an inclusion relation, not the RCC5 relation set.

[9] M. Wessel. Qualitative spatial reasoning with the $\mathcal{ALCI}_{\mathrm{RCC}}$ family — first results and unanswered questions. Technical Report FBI-HH-M-324/03, Fachbereich Informatik, Universität Hamburg, 2002/2003. https://github.com/lambdamikel/alcircc5/blob/master/papers/report7.pdf

[10] C. Lutz and F. Wolter. Modal logics of topological relations. *Logical Methods in Computer Science*, 2(2:5):1–41, 2006. https://doi.org/10.2168/LMCS-2(2:5)2006

[11] V. Haarslev, C. Lutz, and R. Möller. A description logic with concrete domains and a role-forming predicate operator. *Journal of Logic and Computation*, 9(3):351–384, 1999.

[12] M. Nielsen, G. Plotkin, and G. Winskel. Petri nets, event structures and domains, part I. *Theoretical Computer Science*, 13(1):85–108, 1981.

[13] W. Szwast and L. Tendera. FO^2 with one transitive relation is decidable. In *Proc. STACS 2013*, LIPIcs 20, pages 317–328, 2013.

[14] W. Szwast and L. Tendera. On the satisfiability problem for fragments of two-variable logic with one transitive relation. *Journal of Logic and Computation*, 29(6):881–911, 2019. arXiv:1804.09447 (2018). Withdraws the main claim of [13].

[15] I. Pratt-Hartmann. Finite satisfiability for two-variable, first-order logic with one transitive relation is decidable. *Mathematical Logic Quarterly*, 64(3):218–248, 2018. arXiv:1707.05558.

[16] D. Marzec and L. Tendera. On two-variable first-order logic with a partial order. *Journal of Logic and Computation*, 36(2):exaf068, 2026.

[17] C. Lutz and M. Miličić. A tableau algorithm for description logics with concrete domains and general TBoxes. *Journal of Automated Reasoning*, 38:227–259, 2007.

[18] F. Baader and M. Rydval. Description logics with concrete domains and general concept inclusions revisited. In *Proc. IJCAR*, volume 12166 of LNCS, pages 413–431. Springer, 2020.

[19] S. Borgwardt, F. De Bortoli, and P. Koopmann. The precise complexity of reasoning in $\mathcal{ALC}$ with ω-admissible concrete domains. In *Proc. 37th Int. Workshop on Description Logics (DL 2024)*, CEUR Workshop Proceedings, vol. 3739, 2024.

[20] S. Demri and T. Gu. Robustness of constraint automata for description logics with concrete domains. In *Proc. CSL*, volume 363 of LIPIcs, 2026.

[21] E. Grädel, P. G. Kolaitis, and M. Y. Vardi. On the decision problem for two-variable first-order logic. *Bulletin of Symbolic Logic*, 3(1):53–69, 1997.

[22] A. Tarski and S. Givant. *A Formalization of Set Theory without Variables*. AMS Colloquium Publications, vol. 41, American Mathematical Society, 1987.

[23] H. Andréka, J. van Benthem, and I. Németi. Modal languages and bounded fragments of predicate logic. *Journal of Philosophical Logic*, 27(3):217–274, 1998.

[24] E. Grädel. On the restraining power of guards. *Journal of Symbolic Logic*, 64(4):1719–1742, 1999.

[25] V. Bárány, B. ten Cate, and L. Segoufin. Guarded negation. *Journal of the ACM*, 62(3):22:1–22:26, 2015.

[26] H. Ganzinger, C. Meyer, and M. Veanes. The two-variable guarded fragment with transitive relations. In *Proc. 14th IEEE Symposium on Logic in Computer Science (LICS)*, pages 24–34, 1999.

[27] GPT-5.4 (OpenAI), prompted by M. Wessel. Split-forest models and sibling-interface descriptors for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (v1; soundness Thms 1.17–1.19 preserved by v2; Def 1.5 / Def 1.7 / Thm 1.20-sketch superseded by v2). Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/trees/sibling_interface_descriptors_ALCIRCC5_completed_eqsync_canonical_needpatched.pdf

[28] Claude (Anthropic), prompted by M. Wessel. Sibling-interface descriptors for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (v2, repaired): preserves the v1 soundness chain (Thms 1.17–1.19) verbatim and reinterprets v1 Def 1.5 (parent self-loops, generated witness-cover edges), v1 Def 1.7 (PP/PPI existentials by reachability discharge), and v1 Thm 1.20-sketch (model-to-quotient delegated to v2 completeness extraction). Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/trees/sibling_interface_descriptors_ALCIRCC5_v2.pdf

[29] GPT-5.4 (OpenAI), prompted by M. Wessel. A cover-tree tableau calculus for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/trees/alcircc5_cover_tree_tableau_needall_patched.pdf

[30] Claude (Anthropic), prompted by M. Wessel. Completeness extraction for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (v2.1, GPT-5.5 v2-review repairs): closing the model-to-quotient gap via witness-generated submodels, generated covers, reachability discharge, and explicit equality data. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/completeness_extraction_ALCIRCC5_v2.pdf

[31] Claude (Anthropic), prompted by M. Wessel. Completeness extraction for the $\mathcal{ALCI}_{\mathrm{RCC5}}$ split-forest quotient (v1, superseded; defects identified by GPT-5.5 review). Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/completeness_extraction_ALCIRCC5.pdf

[32] GPT-5.5 (OpenAI), prompted by M. Wessel. A coherent generated split-forest decidability argument for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (Büchi-automaton route). Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5/ALCIRCC5_coherent_generated_split_forest_decidability.pdf

[33] GPT-5.5 (OpenAI), prompted by M. Wessel. A repaired split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (no-automata route, round-2 31-page version; superseded by [35]). Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round2/repaired_split_forest_no_automata_proof.pdf

[34] GPT-5.5 (OpenAI), prompted by M. Wessel. Formal review of the round-6 consolidated generated split-forest proof (May 27, 2026): identifies the central abstract-position-symbol labelling defect that motivates the round-7 pair-shape architecture. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/formal_response_round6_review.pdf

[35] GPT-5.5 (OpenAI), prompted by M. Wessel. A repaired split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ without automata (round-7 consolidated sketch, 14 pages): occurrence-sensitive pair shapes with incidence tags; twelve validity clauses (V1)–(V12). Immediate predecessor of [36]; retained as audit trail. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/repaired_split_forest_all_in_one_round7.pdf

[36] GPT-5.5 (OpenAI), prompted by M. Wessel. A repaired split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ without automata — round-8 saturated side contexts (15-page sketch): round-7 architecture preserved; saturated side-context $S_k(u)$, cross side/tree exhaustion lemma, and constructed pair/triple catalogues $\mathsf{Pair}_{\mathcal{Q}}$, $\mathsf{Tri}_{\mathcal{Q}}$ added. Immediate predecessor of [40]; carried the D-1 gap, retained as audit trail. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/claude_latest_review_gpt5.5_fix/repaired_split_forest_abstract_round8_repaired.pdf

[37] GPT-5.5 (OpenAI), prompted by M. Wessel. Expanded split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$: detailed round-8 companion (30 pages). Full saturated-side-context construction, cross side/tree verticalization lemma, residual-frontier lemma, constructed pair/triple catalogues, explicit completeness extraction. Technical report,

May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/claude_latest_review_gpt5.5_fix/expanded_split_forest_full_details_round8_full.pdf

[38] Claude (Anthropic, Opus 4.8), prompted by M. Wessel. Review of the round-8 saturated-side-context split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (9 pages): identifies the completeness gap D-1 (a DR/PO witness composition-forced into an infinite pumped tower has no admissible pair shape on the tail); composition arithmetic machine-checked. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/opus4.8_review/review_round8_split_forest.pdf

[39] Claude (Anthropic, Opus 4.8), prompted by M. Wessel. Repairing the round-8 completeness construction for $\mathcal{ALCI}_{\mathrm{RCC5}}$: composition-forced verticalization (8 pages). Splices a forced witness into the cover relation at a bounded threshold height; machine-checked threshold lemmas. Proposed by the cold reviewer; ratified and integrated by GPT-5.5 in round-9 [40]. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/opus4.8_review/repair_completeness_round8.pdf

[40] GPT-5.5 (OpenAI), prompted by M. Wessel. A repaired split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ without automata — round-9 composition-forced verticalization (16-page sketch): round-7/round-8 architecture preserved; a DR/PO side witness with a PPI-tail is spliced into the cover relation $E_{\uparrow}$ at a bounded threshold, closing the D-1 gap of [38]. Superseded; a cold review found three critical gaps (retained as audit trail). Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round3/repaired_split_forest_abstract_round9_forced_verticalization.pdf

[41] GPT-5.5 (OpenAI), prompted by M. Wessel. Expanded split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$: detailed round-9 companion (33 pages). Full composition-forced verticalization construction with the monotone ancestor-trace lemma ($\mathsf{DR}^*\mathsf{PO}^*\mathsf{PPI}^*$), bounded-threshold replacement lemma, splice-faithfulness lemma, and global residual-frontier lemma; split-cover and extraction updated to apply the forced-verticalization test before ordinary side-context attachment. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round3/expanded_split_forest_full_details_round9_forced_verticalization.pdf

[42] GPT-5.5 (OpenAI), prompted by M. Wessel. Formal response to the Opus 4.8 review of the round-8 split-forest proof (5 pages). Confirms the D-1 defect ("a genuine completeness defect, not merely a presentational ambiguity"), records re-running the three Opus 4.8 verification scripts (all pass) and the composition facts $\mathsf{comp}(\mathsf{PPI},\mathsf{PO}) = \{\mathsf{PO},\mathsf{PPI}\}$, $\mathsf{comp}(\mathsf{PPI},\mathsf{PPI}) = \{\mathsf{PPI}\}$, and states the round-9 repair. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_round3/formal_response_opus48_round9_repair.pdf

[43] GPT-5.5 (OpenAI), prompted by M. Wessel. Expanded generated split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$: detailed companion to the round-7 sketch (28 pages). Detailed local constructions, worked stress cases, audit-of-main-proof-obligations section. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/expanded_split_forest_full_details_proof.pdf

[44] GPT-5.5 (OpenAI), prompted by M. Wessel. A split-forest decidability proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ via parity tree automata: detailed version (21 pages). Superseded: a

cold review found its central invariant assumed, not constructed. Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/split_forest_automata_decidability_proof_detailed.pdf

[45] Claude (Anthropic, Opus 4.7), prompted by M. Wessel. Round-7 audit and response: adopting GPT-5.5's pair-shape / incidence-tag repair as the canonical statement (9 pages). Technical report, May 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/claude_round7_audit_response.pdf

[46] Claude (Anthropic), prompted by M. Wessel. A cover-tree tableau for $\mathcal{ALCI}_{\mathrm{RCC5}}$: Implementation and computational verification. Technical report, April 2026. Carries an erratum box since November 2026: its sibling check is incomplete. https://github.com/lambdamikel/alcircc5/blob/master/papers/cover_tree_tableau_ALCIRCC5.pdf

[47] Claude (Anthropic), prompted by M. Wessel. A two-tier quotient construction for the PO-coherent fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$: decidability and a 2-ExpTime bound. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/two_tier_quotient_ALCIRCC5.pdf

[48] GPT-5.6 Pro (OpenAI), prompted by M. Wessel. A canonical set representation for $\mathcal{ALCI}_{\mathrm{RCC5}}$: the converse of the ordered-disjoint RCC5 normal form, and identity-selector minors as the F6 obstruction (technical note). Technical report, July 2026. File `papers/final_paper_gpt_5.6_review/ALCI_RCC5_F6_technical_note.pdf`; the core is machine-checked in `formal/RCC5NormalForm.lean` (`sub_iff_le`, `eta_injective`) and cross-checked in `verification/python/wp88_canonical_representation.py`. https://github.com/lambdamikel/alcircc5/blob/arxiv-candidate-2026-07-18/papers/final_paper_gpt_5.6_review/ALCI_RCC5_F6_technical_note.pdf

[49] Claude (Anthropic), prompted by M. Wessel. A negative result on the finite-witness conjecture for the contextual tableau for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/FW_proof_ALCIRCC5.pdf

[50] GPT-5.4 (OpenAI), prompted by M. Wessel. Triangle-blocking analysis for $\mathcal{ALCI}_{\mathrm{RCC5}}$: failed undecidability route via three-region coincidence. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/triangle_blocking_ALCIRCC5.pdf

[51] Claude (Anthropic), prompted by M. Wessel. Patchwork-augmentation probe: a negative result on an undecidability route for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/patchwork_augmentation_ALCIRCC5.pdf

[52] Claude (Anthropic), prompted by M. Wessel. A typed-patchwork counterexample for $\mathcal{ALCI}_{\mathrm{RCC5}}$: refutation of a proposed blocking mechanism. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/typed_patchwork_counterexample_ALCIRCC5.pdf

[53] Claude (Anthropic), prompted by M. Wessel. Response to GPT-5.4's blocking-based undecidability attempts for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/response_to_gpt_blocking.pdf

[54] Claude (Anthropic), prompted by M. Wessel. Status of the $\mathcal{ALCI}_{\mathrm{RCC5}}$ decidability argument after the FW negative result. Technical report, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/ALCI_RCC5_status_after_FW.pdf

[55] Claude (Anthropic), prompted by M. Wessel. Cover-tree tableau reasoner for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Python implementation, April 2026. The main prototype reasoner (no completeness theorem claimed); cross-validated against the quasimodel oracle on 911 concepts and on the 775-case decomposition suite. Known to be incomplete since November 2026 (§8). https://github.com/lambdamikel/alcircc5/blob/master/src/cover_tree_tableau.py

[56] Claude (Anthropic), prompted by M. Wessel. A quasimodel-based reasoner for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Python implementation, April 2026. Retained as a cross-validation oracle only: SAT answers are trustworthy; UNSAT answers are known-incomplete on cyclic-via-symmetric-role concepts (e.g. the PO-loop pattern). https://github.com/lambdamikel/alcircc5/blob/master/src/alcircc5_reasoner.py

[57] Claude (Anthropic), prompted by M. Wessel. Cycle-aware wrapper for the $\mathcal{ALCI}_{\mathrm{RCC5}}$ quasimodel reasoner. Python implementation, April 2026. Thin wrapper that detects cyclic-via-symmetric-role patterns and delegates to a direct model search; restores the 911/911 cross-validation match rate. https://github.com/lambdamikel/alcircc5/blob/master/src/alcircc5_reasoner_cyclic.py

[58] Claude (Anthropic), prompted by M. Wessel. Independent model verifier for $\mathcal{ALCI}_{\mathrm{RCC5}}$ SAT answers. Python implementation, April 2026. Checks inverse symmetry, composition-consistency on all triples, type-safety, existential witnessing, and recursive concept truth against explicit finite RCC5 models. https://github.com/lambdamikel/alcircc5/blob/master/src/model_verifier.py

[59] Claude (Anthropic), prompted by M. Wessel. Cover-tree decomposition test for $\mathcal{ALCI}_{\mathrm{RCC5}}$ (775/775). Python test suite, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/src/decomposition_test.py

[60] Claude (Anthropic), prompted by M. Wessel. Cross-validation stress test for the $\mathcal{ALCI}_{\mathrm{RCC5}}$ cover-tree tableau against the baseline quasimodel reasoner (911 concepts). Python test suite, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/src/stress_test_cover_tree.py

[61] Claude (Anthropic), prompted by M. Wessel. Cross-validation stress test against the cycle-aware quasimodel reasoner (911 concepts; zero mismatches). Python test suite, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/src/stress_test_cyclic.py

[62] Claude (Anthropic), prompted by M. Wessel. Seven-case adversarial suite for the cycle-aware $\mathcal{ALCI}_{\mathrm{RCC5}}$ reasoner (PO-loop, DR-loop, mixed-loop, clash-to-EQ, and distinguishing-universal variants). Python test suite, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/src/test_cyclic_reasoner.py

[63] Claude (Anthropic), prompted by M. Wessel. GIS taxonomy computation for $\mathcal{ALCI}_{\mathrm{RCC5}}$. Python implementation, April 2026. https://github.com/lambdamikel/alcircc5/blob/master/src/gis_taxonomy.py

[64] M. Wessel (with Claude Opus 4.7 and GPT-5.4 Pro). $\mathcal{ALCI}_{\mathrm{RCC5}}$ decidability project repository. GitHub, 2026. https://github.com/lambdamikel/alcircc5 (See `README.md` for the overall argument, the blocked-reduction landscape, and the summary of failed undecidability attempts.)

[65] M. Wessel (with Claude Opus 4.7 and GPT-5.4 Pro). Outdated and superseded material: failed blocking attempts and retracted approaches for $\mathcal{ALCI}_{\mathrm{RCC5}}$. GitHub, 2026. https://github.com/lambdamikel/alcircc5/blob/master/OUTDATED.md

[66] M. Wessel. *Flexible und konfigurierbare Software-Architekturen für datenintensive ontologiebasierte Informationssysteme*. Dissertation, Technische Universität Hamburg-Harburg, 2008. https://doi.org/10.15480/882.832. The tableau rules are in §6.2.7.2, the prover in §6.4.4; source at https://github.com/lambdamikel/midelora.

[67] Claude Opus 4.7 (Anthropic), commissioned by M. Wessel. A soundness counterexample to the cover-tree tableau for $\mathcal{ALCI}_{\mathrm{RCC5}}$: twelve refuting concepts built from a single composition chain. Review paper, April 2026, with a second-round reassessment after the repair. https://github.com/lambdamikel/alcircc5/blob/master/review_paper/review_cover_tree_tableau.pdf

[68] Claude Fable 5.2 (Anthropic), prompted by M. Wessel. The poset bridge: two-variable logic over partial orders inside $\mathcal{ALCI}_{\mathrm{RCC5}}$. Note, 11 November 2026; unreviewed. `papers/poset_bridge/README.md`, with `formal/PosetBridge.lean` and `verification/python/wp137_poset_bridge_probe.py`.

[69] Claude Fable 5.2 (Anthropic), prompted by M. Wessel. The MiDeLoRa calculus for $\mathcal{ALCI}_{\mathrm{RCC5}}$: sound, complete, not terminating. Note, 11 November 2026; unreviewed; a paper proof. `papers/midelora_calculus/README.md`, with `src/midelora_tableau.py` and `verification/python/wp139_midelora_reference.py`.

[70] M. Wessel (with AI assistants). `ERRATA.md`: corrections to statements this repository has published. https://github.com/lambdamikel/alcircc5/blob/master/ERRATA.md

[71] Anthropic. Claude Opus 4.7. Large language model, 2026. https://www.anthropic.com/claude

[72] OpenAI. GPT-5.4 Pro. Large language model, 2026. https://openai.com

[73] GPT-5.6 Sol (OpenAI), prompted by M. Wessel. A Lean certification plan for the ∀PO-free fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$: retire witness borrowing; use the finite scheme as a control graph and unfold every non-EQ existential to a fresh occurrence (the *ConeScheme* blueprint). Technical report, August 2026. Directory `papers/cone_scheme_plan/`; the design was carried out in Lean by Claude and is recorded step by step in `ASSEMBLY_DESIGN.md` §§267–297, with capstones `decidableFSat`, `decidableSat_cone`, `decidableSetSat` in `formal/POFreeLift.lean`.

[74] Astra 6 (OpenAI), prompted by M. Wessel. $\mathcal{ALCI}_{\mathrm{RCC5}}$: audit of the decidability claim and two partial results beyond the ∀PO-free fragment. Technical report, 7 October 2026. Directory `papers/astra-6-latest-review-and-extension/` (archived with verification notes); the erasure theorem and the order-level normalization are formalized in `formal/POFreeLift.lean` §§299–300.

[75] GPT-5.6 Sol (OpenAI), prompted by M. Wessel. Cold review of the current $\mathcal{ALCI}_{\mathrm{RCC5}}$ development: overview-paper audit, Lean artifact review, and a renewed attack on full decidability; with a draft focused paper on the $\forall$PO-free fragment. Review report, 9 September 2026, of commit `795b270`. Directory `papers/gpt-5.6-latest-overview-paper-review-and-recommendation/`.

[76] M. Wessel (with AI assistants). Deciding the $\forall$PO-free fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$: ordered-disjoint models, cone schemes, and a proof checked in Lean 4. Companion paper, September 2026; revised October and November 2026. https://github.com/lambdamikel/alcircc5/blob/master/papers/pofree_fragment_arxiv.pdf

[77] OpenAI. GPT-5.6 Pro. Large language model, 2026. https://openai.com

[78] M. Wessel (with AI assistants). `CONVERSATION.md`: the full timestamped project conversation log (human prompts and AI responses), for auditable attribution. https://github.com/lambdamikel/alcircc5/blob/master/CONVERSATION.md

A On the DL 2026 rejection

An extended abstract of this work was submitted to the 2026 International Workshop on Description Logics (DL 2026) and rejected. We record the episode—and dissent from the decision—not merely out of grievance but because it bears directly on this paper's subject: how a research community engages with an AI-assisted attack on one of its own long-standing open problems. The submission was labelled as this paper labels its full-logic claims (*strongly supported, not certified*, an AI-assisted discussion piece), and its stated motivation, twice, was to raise awareness of the emerging capabilities of agentic AI for description-logic research. We contend it was rejected on the wrong basis.

1. **The technical objections targeted the optional appendix.** DL 2026's own guidelines state that the appendix "will not be included in the proceedings" and is reviewed only at reviewer discretion. Yet nearly every concrete objection—missing definitions, a broken cross-reference, an under-specified lemma—concerned appendix material, not the extended abstract that is the reviewed artifact.

2. **The reviews attacked a position the paper did not take.** The abstract called the proof a "candidate," wrote "if found correct by human experts," and carried an explicit non-verification disclaimer. It was nonetheless assessed as a settled-theorem submission and rejected for not clearing that bar—a strawman.

3. **Honest hedging was quoted back as a fault.** The very disclaimers included to avoid over-claiming were cited as grounds for rejection. Punishing calibrated uncertainty inverts the incentive peer review should create.

4. **No review engaged the actual thesis** (agentic AI's capabilities), nor the concrete contributions the submission *did* present: a working cover-tree tableau reasoner reproducing the 2003 GIS taxonomy, and an unconditionally decidable fragment. (The split-forest normal form, the F6 analysis, and the machine-checked Lean soundness reported in the present paper came later, in the months *after* the rejection—so they cannot have been the basis for it.)

5. **The verdict was a value judgment, not a technical assessment.** The three reviews largely agreed on the defect inventory but split on the verdict—one reviewer believed the result true and named the right approach, yet recommended reject on presentation; another recommended accept. The decision was escalated to the Steering Committee, and the covering letter offered that the paper "could be a good scientific contribution if it ... focus[ed] on the advantages and disadvantages of using LLMs ... rather than claiming decidability results." But that discussion—the advantages and disadvantages of agentic AI for this kind of research—is exactly what the submission's main text *did* provide, alongside the concrete results above; it simply was not engaged with. The "acceptable" paper being described was, in substance, the paper already submitted. The objection was to the framing and the precedent, not the substance.

6. **A disclosure paradox.** The integrity choice—front-loading the AI co-authorship, in the title and from page one—was precisely what made the submission unacceptable, while the offered path to acceptance was to write it in standard single-voice style with a tail-of-paper note. A process that rewards understating AI's contribution and penalizes overstating it selects for under-disclosure, not for science.

We grant one point: a single introductory sentence ("We hereby claim: $\mathcal{ALCI}_{\mathrm{RCC5}}$ is decidable") read more assertively than the surrounding, carefully hedged text supported, and a constructive

"tone this down and resubmit" would have been met immediately. That does not carry the verdict. There is, finally, a recursive irony: the rejected abstract had warned, by name, about the failure modes—network effects, not-invented-here bias, dismissive reviewing—that the rejection then illustrated. The deeper concern is not one abstract's fate but that agentic AI now lets ideas reach decades-old open problems from outside the established networks, and a review process that answers this by rejecting on framing while ignoring substance constrains rather than protects science. Peer review exists to surface and stress-test new ideas, not to keep them out; when it functions as the latter, it has stopped doing the job the community delegated to it. The full extended abstract, the three reviews, and a detailed analysis are public (`dl2026-rejected/`; the open letter is `open_letter.md`); readers are encouraged to form their own view.